Biochemistry Exam #2

Complete Comprehensive Guide with Analogies

Covering: Enzymes, Kinetics, Metabolism, Glycolysis, Gluconeogenesis, Diabetes, and the Citric Acid Cycle

Exam Experts

Contents

1 Enzymes & Regulation

1.1 Introduction to Enzymes

Definition

An **enzyme** is a biological catalyst, typically a protein, that accelerates the rate of a chemical reaction without being consumed in the process. Enzymes lower the **activation energy** (E_a) required for a reaction to proceed, thereby increasing reaction rates by factors of 10^6 to 10^{12}.

Enzymes are the molecular workhorses of all living cells. Without enzymes, the chemical reactions necessary for life would occur far too slowly to sustain biological processes. Every metabolic pathway, from digestion to DNA replication, depends on specific enzymes working in precise coordination.

The study of enzymes—**enzymology**—is fundamental to understanding metabolism, drug design, and disease mechanisms. Enzymes exhibit remarkable specificity, typically catalyzing only one reaction or a small set of closely related reactions.

Exam Experts Analogy

Think of an enzyme like a teacher helping students solve math problems. The students (substrates) could eventually figure out the problem on their own, but it would take a very long time. The teacher (enzyme) helps them understand the problem faster and more efficiently. After helping, the teacher isn't used up—she's ready to help the next group of students!

1.2 Enzyme Nomenclature and Classification

Enzymes are classified into six major classes based on the type of reaction they catalyze:

1. **Oxidoreductases:** Catalyze oxidation-reduction (redox) reactions. Transfer electrons between molecules.
   - Example: Lactate dehydrogenase (converts lactate to pyruvate)
   - Common names end in "-dehydrogenase" or "-oxidase"
2. **Transferases:** Transfer functional groups (amino, phosphate, methyl, etc.) from one molecule to another.
   - Example: Hexokinase (transfers phosphate from ATP to glucose)
   - Common names end in "-transferase" or "-kinase"
3. **Hydrolases:** Catalyze hydrolysis reactions (breaking bonds using water).
   - Example: Trypsin (breaks peptide bonds in proteins)
   - Common names end in "-ase" attached to substrate name

4. **Lyases:** Add or remove groups to/from double bonds, or form double bonds by removing groups.
   - Example: Aldolase (cleaves fructose-1,6-bisphosphate)
   - Common names include "-lyase," "-decarboxylase," "-dehydratase"
5. **Isomerases:** Catalyze structural rearrangements within a single molecule.
   - Example: Phosphoglucose isomerase (converts G6P to F6P)
   - Common names end in "-isomerase" or "-mutase"
6. **Ligases:** Join two molecules together using ATP energy (forming new bonds).
   - Example: DNA ligase (joins DNA fragments)
   - Common names end in "-ligase" or "-synthetase"

1.3 Enzyme Structure and the Active Site

Definition

The **active site** is a specific region on an enzyme where substrate binding and catalysis occur. It is typically a three-dimensional cleft or pocket formed by amino acid residues from different parts of the polypeptide chain.

Key features of the active site:

1. **Specificity:** The shape and chemical properties of the active site determine which substrates can bind
2. **Binding residues:** Amino acids that interact with the substrate through hydrogen bonds, ionic interactions, and hydrophobic contacts
3. **Catalytic residues:** Amino acids directly involved in the chemical transformation
4. **Microenvironment:** The active site often has different properties (pH, polarity) than the surrounding solution

Key Principle

Lock-and-Key Model vs. Induced Fit Model:

- **Lock-and-Key (Emil Fischer, 1894):** The enzyme active site has a rigid, pre-formed shape that exactly matches the substrate, like a key fitting into a lock.
- **Induced Fit (Daniel Koshland, 1958):** The enzyme active site is flexible and changes shape upon substrate binding to achieve optimal fit. This is the more accurate modern understanding.

Exam Experts Analogy

The lock-and-key model is like putting your hand into a rigid glove that's exactly your size. The induced fit model is like putting your hand into a flexible mitten—the mitten adjusts and wraps around your hand to create a snug fit. Most enzymes work more like the mitten!

1.4 Cofactors and Coenzymes

Many enzymes require non-protein molecules called **cofactors** for activity.

Definition

- **Cofactor:** A non-protein molecule required for enzyme activity. Can be inorganic (metal ions) or organic (coenzymes).
- **Coenzyme:** An organic cofactor, often derived from vitamins, that participates in the catalytic reaction.
- **Prosthetic group:** A cofactor that is tightly or covalently bound to the enzyme.
- **Apoenzyme:** The protein portion of an enzyme without its cofactor.
- **Holoenzyme:** The complete, catalytically active enzyme (apoenzyme + cofactor).

Cofactor Type	Examples	Function
Metal ions	Zn^{2+}, Mg^{2+}, Fe^{2+}, Cu^{2+}	Stabilize charges, participate in redox
NAD^+/NADH	From niacin (B3)	Electron carrier in redox reactions
FAD/$FADH_2$	From riboflavin (B2)	Electron carrier in redox reactions
Coenzyme A	From pantothenic acid (B5)	Acyl group carrier
Thiamine pyrophosphate	From thiamine (B1)	Decarboxylation reactions
Pyridoxal phosphate	From pyridoxine (B6)	Amino acid metabolism
Biotin	Vitamin B7	Carboxylation reactions
Tetrahydrofolate	From folic acid	One-carbon transfers

1.5 Mechanisms of Enzyme Catalysis

Enzymes accelerate reactions through several mechanisms:

1. **Proximity and Orientation Effects:** Enzymes bring substrates together in the correct orientation for reaction, increasing the effective concentration of reactants.
2. **Transition State Stabilization:** Enzymes preferentially bind and stabilize the transition state, lowering the activation energy. This is the most important catalytic mechanism.

3. **Acid-Base Catalysis:** Amino acid residues donate or accept protons during the reaction.
   - General acid catalysis: Proton donation
   - General base catalysis: Proton acceptance
4. **Covalent Catalysis:** A covalent bond forms transiently between the enzyme and substrate.
   - Example: Serine proteases form acyl-enzyme intermediates
5. **Metal Ion Catalysis:** Metal ions stabilize negative charges, participate in redox reactions, or activate water molecules.

Example

Chymotrypsin Mechanism:
Chymotrypsin is a serine protease that cleaves peptide bonds. It uses a **catalytic triad** consisting of Ser195, His57, and Asp102.
Step 1: His57 acts as a general base, abstracting a proton from Ser195, making it a powerful nucleophile.
Step 2: The activated Ser195 attacks the carbonyl carbon of the peptide bond (covalent catalysis), forming a tetrahedral intermediate.
Step 3: The tetrahedral intermediate collapses, releasing the first product (amine portion) and forming an acyl-enzyme intermediate.
Step 4: Water attacks the acyl-enzyme intermediate, forming another tetrahedral intermediate.
Step 5: The second tetrahedral intermediate collapses, releasing the carboxylic acid product and regenerating free enzyme.

1.6 Enzyme Regulation

Cells must precisely control enzyme activity to maintain homeostasis and respond to changing conditions. Several regulatory mechanisms exist:

1.6.1 Allosteric Regulation

Definition

Allosteric regulation occurs when a molecule binds to a site on the enzyme *other than* the active site (the allosteric site), causing a conformational change that affects enzyme activity.

- **Allosteric activators:** Increase enzyme activity by stabilizing the active (R) state
- **Allosteric inhibitors:** Decrease enzyme activity by stabilizing the inactive (T) state

- **Homotropic effectors:** The substrate itself acts as an allosteric modulator
- **Heterotropic effectors:** A molecule other than the substrate acts as the modulator

Exam Experts Analogy

Imagine a door with two handles—one on the front (active site) and one on the back (allosteric site). Normally, you open the door by turning the front handle. But if someone holds the back handle in a certain way, it can either make the front handle easier to turn (activator) or lock it so it won't turn at all (inhibitor). The person at the back controls how well the front handle works!

Key Principle

Concerted (MWC) Model of Allostery:

- Enzyme exists in equilibrium between T (tense, low affinity) and R (relaxed, high affinity) states
- All subunits change conformation simultaneously (concerted)
- Substrates preferentially bind R state, shifting equilibrium toward R
- Inhibitors stabilize T state; activators stabilize R state

1.6.2 Covalent Modification

Definition

Covalent modification involves the addition or removal of chemical groups to/from the enzyme, altering its activity. The most common modification is **phosphorylation**.

- **Phosphorylation:** Addition of phosphate group by kinases; removal by phosphatases
  - Adds negative charges, can activate or inhibit depending on enzyme
  - Example: Glycogen phosphorylase is activated by phosphorylation
  - Example: Glycogen synthase is inhibited by phosphorylation
- **Acetylation:** Addition of acetyl groups
- **Methylation:** Addition of methyl groups
- **Ubiquitination:** Addition of ubiquitin proteins (often targets for degradation)

1.6.3 Proteolytic Activation (Zymogens)

Definition

A **zymogen** (or proenzyme) is an inactive enzyme precursor that requires proteolytic cleavage to become active. This is an irreversible activation mechanism.

Example

Digestive Enzyme Zymogens:

- Pepsinogen $\xrightarrow{\text{HCl, pepsin}}$ Pepsin (in stomach)
- Trypsinogen $\xrightarrow{\text{enteropeptidase}}$ Trypsin (in small intestine)
- Chymotrypsinogen $\xrightarrow{\text{trypsin}}$ Chymotrypsin

Why zymogens? They prevent enz ymes from digesting the cells that produce them. Pancreatic cells would be destroyed if trypsin were active inside them!

1.6.4 Isozymes

Definition

Isozymes (or isoenzymes) are different molecular forms of the same enzyme that catalyze the same reaction but differ in kinetic properties, regulatory properties, or tissue distribution.

Example

Lactate Dehydrogenase (LDH) Isozymes:
LDH is a tetramer composed of H (heart) and M (muscle) subunits:

- LDH-1 (H4): Predominant in heart, high affinity for lactate
- LDH-2 (H3M1): Found in RBCs
- LDH-3 (H2M2): Found in brain, kidney
- LDH-4 (HM3): Found in kidney, placenta
- LDH-5 (M4): Predominant in liver and skeletal muscle, high affinity for pyruvate

Clinical Application: Elevated LDH-1 in blood indicates heart damage (myocardial infarction). Different isozyme patterns help diagnose tissue-specific damage.

1.7 Feedback Inhibition in Metabolic Pathways

Definition

Feedback inhibition occurs when the end product of a metabolic pathway inhibits an enzyme early in the pathway, preventing overproduction of the product.

Exam Experts Analogy

Feedback inhibition is like a thermostat. When your house gets warm enough, the thermostat tells the heater to turn off. The "product" (warm air) inhibits the "enzyme" (heater) to prevent overheating. Similarly, when a cell has enough of a product, that product tells the pathway to slow down.

Example

Isoleucine Biosynthesis:

$$\text{Threonine} \xrightarrow{\text{threonine deaminase}} \alpha\text{-ketobutyrate} \rightarrow \ldots \rightarrow \text{Isoleucine}$$

When isoleucine accumulates, it allosterically inhibits threonine deaminase (the first committed step), preventing unnecessary synthesis.

1.8 Regulation Summary Table

Mechanism	Speed	Reversibility	Example
Allosteric regulation	Fast (seconds)	Reversible	PFK-1, ATCase
Covalent modification	Moderate (minutes)	Reversible	Glycogen phosphorylase
Proteolytic cleavage	Slow (minutes-hours)	Irreversible	Trypsinogen $\rightarrow$ trypsin
Gene expression	Slow (hours-days)	Reversible	Enzyme induction

2 Enzyme Kinetics

2.1 Introduction to Enzyme Kinetics

Definition

Enzyme kinetics is the study of the rates of enzyme-catalyzed reactions and how these rates are affected by various factors including substrate concentration, enzyme concentration, pH, temperature, and the presence of inhibitors or activators.

Exam Experts Analogy

Think of enzyme kinetics like studying how fast a pizza restaurant can make pizzas. The "rate" is how many pizzas per hour. This depends on how many orders come in (substrate concentration), how many chefs are working (enzyme concentration), whether the ovens are at the right temperature, and whether anything is slowing down the kitchen (inhibitors).

2.2 Reaction Velocity and Substrate Concentration

2.2.1 The Michaelis-Menten Model

Definition

The **Michaelis-Menten model** describes the kinetics of many enzymes by relating reaction velocity (v) to substrate concentration ([S]).

Key Principle

Michaelis-Menten Equation:

$$v = \frac{V_{\max}[S]}{K_m + [S]}$$

Where:

- v = initial reaction velocity
- $V_{\max}$ = maximum velocity (when enzyme is saturated)
- $[S]$ = substrate concentration
- K_m = Michaelis constant (substrate concentration at $\frac{1}{2}V_{\max}$)

2.2.2 Derivation of the Michaelis-Menten Equation

The model assumes a simple reaction mechanism:

$$E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_2} E + P$$

Key Assumptions:

1. **Steady-state assumption:** The concentration of ES remains constant over time (rate of ES formation = rate of ES breakdown)
2. **Initial velocity conditions:** Measured before significant product accumulates (so reverse reaction $P \to S$ is negligible)

3. $[S] >> [E]_{\text{total}}$, so substrate concentration doesn't change significantly

Derivation Steps:
1. Rate of ES formation: $k_1[E][S]$
2. Rate of ES breakdown: $k_{-1}[ES] + k_2[ES] = (k_{-1} + k_2)[ES]$
3. At steady state: $k_1[E][S] = (k_{-1} + k_2)[ES]$
4. Define $K_m = \frac{k_{-1}+k_2}{k_1}$ (Michaelis constant)
5. Total enzyme: $[E]_{\text{total}} = [E] + [ES]$, so $[E] = [E]_{\text{total}} - [ES]$
6. Substituting and solving for [ES]:

$$[ES] = \frac{[E]_{\text{total}}[S]}{K_m + [S]}$$

7. Reaction velocity: $v = k_2[ES] = \frac{k_2[E]_{\text{total}}[S]}{K_m+[S]}$
8. Since $V_{\text{max}} = k_2[E]_{\text{total}}$:

$$\boxed{v = \frac{V_{\text{max}}[S]}{K_m + [S]}}$$

2.3 Understanding Kinetic Parameters

2.3.1 V_{max} (Maximum Velocity)

Definition

V_{max} is the maximum rate of the reaction when the enzyme is fully saturated with substrate (all enzyme molecules are in the ES complex).

- $V_{\text{max}} = k_2[E]_{\text{total}} = k_{\text{cat}}[E]_{\text{total}}$
- Directly proportional to enzyme concentration
- Units: M/s, μmol/min, etc.
- Increasing [E] increases V_{max}

2.3.2 K_m (Michaelis Constant)

Definition

K_m is the substrate concentration at which the reaction velocity is half of V_{max}. It is a measure of the enzyme's affinity for its substrate.

- **Low K_m:** High affinity (enzyme reaches half-maximal velocity at low [S])
- **High K_m:** Low affinity (enzyme needs high [S] to reach half-maximal velocity)
- Units: Concentration (M, mM, μM)

- K_m is independent of enzyme concentration
- K_m is characteristic of each enzyme-substrate pair

Exam Experts Analogy

K_m is like how hungry you are for pizza. If you have a low K_m for pizza, even the smell of one slice (low concentration) gets you halfway to full satisfaction. If you have a high K_m, you need a whole pizza shop's worth before you're even halfway interested. Low K_m = high affinity = easily satisfied!

2.3.3 k_{cat} (Turnover Number)

Definition

The **turnover number** (k_{cat}) is the number of substrate molecules converted to product per enzyme molecule per unit time when the enzyme is saturated.

$$k_{\text{cat}} = \frac{V_{\text{max}}}{[E]_{\text{total}}}$$

- Units: s^{-1} (per second)
- Represents the intrinsic catalytic efficiency of the enzyme
- Also called the "molecular activity"

Example

Turnover Numbers of Various Enzymes:

Enzyme	k_{cat} (s^{-1})
Carbonic anhydrase	6×10^5
Catalase	4×10^7
Chymotrypsin	100
Lysozyme	0.5

Carbonic anhydrase converts 600,000 CO_2 molecules to bicarbonate per second!

2.3.4 Catalytic Efficiency (k_{cat}/K_m)

Definition

Catalytic efficiency (k_{cat}/K_m) measures how efficiently an enzyme converts substrate to product when substrate is limiting. It accounts for both binding affinity and catalytic speed.

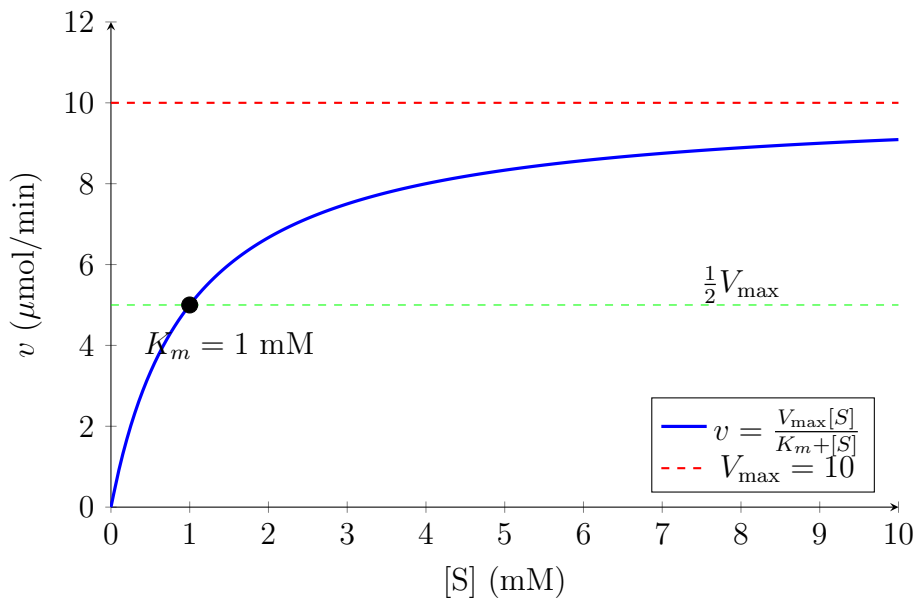
- Units: $M^{-1}s^{-1}$
- Upper limit is the diffusion-controlled limit: $\sim 10^8 - 10^9$ $M^{-1}s^{-1}$
- Enzymes approaching this limit are called **catalytically perfect**
- Useful for comparing different enzymes or same enzyme with different substrates

Example

Catalytically Perfect Enzymes:
Triose phosphate isomerase has $k_{\text{cat}}/K_m \approx 2 \times 10^8$ $M^{-1}s^{-1}$, approaching the diffusion limit. Every collision between enzyme and substrate results in product formation!

2.4 Graphical Analysis of Enzyme Kinetics

2.4.1 Michaelis-Menten Plot



Key Features:

- Hyperbolic curve approaching $V_{\max}$ asymptotically
- At low [S]: $v \approx \frac{V_{\max}}{K_m}[S]$ (first-order kinetics)
- At high [S]: $v \approx V_{\max}$ (zero-order kinetics)
- At [S] $= K_m$: $v = \frac{1}{2}V_{\max}$

2.4.2 Lineweaver-Burk Plot (Double Reciprocal Plot)

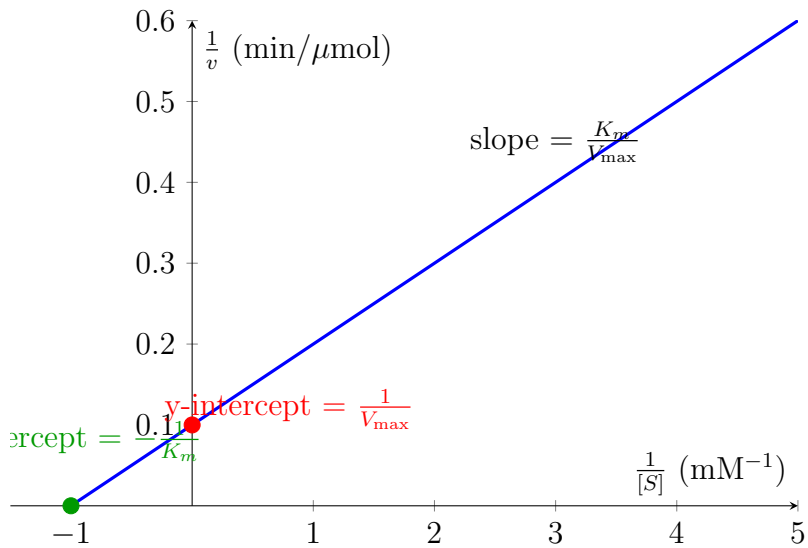
Taking the reciprocal of the Michaelis-Menten equation:

Key Principle

Lineweaver-Burk Equation:

$$\frac{1}{v} = \frac{K_m}{V_{\max}} \cdot \frac{1}{[S]} + \frac{1}{V_{\max}}$$

This is in the form $y = mx + b$, giving a straight line when $\frac{1}{v}$ is plotted against $\frac{1}{[S]}$.



Key Features:

- y-intercept = $\frac{1}{V_{\max}}$
- x-intercept = $-\frac{1}{K_m}$
- Slope = $\frac{K_m}{V_{\max}}$
- Useful for determining kinetic parameters and analyzing inhibition

2.5 Enzyme Inhibition

Definition

Enzyme inhibition is the decrease in enzyme activity caused by the binding of specific molecules (inhibitors) to the enzyme.

2.5.1 Reversible Inhibition

1. Competitive Inhibition

Definition

A **competitive inhibitor** binds to the active site of the enzyme, competing directly with the substrate for binding.

- Inhibitor resembles substrate structurally
- Effect can be overcome by increasing [S]
- K_m increases (apparent lower affinity)
- V_{max} unchanged (at infinite [S], all inhibitor is displaced)

$$v = \frac{V_{\text{max}}[S]}{K_m\left(1 + \frac{[I]}{K_i}\right) + [S]} = \frac{V_{\text{max}}[S]}{K_m^{\text{app}} + [S]}$$

where $K_m^{\text{app}} = K_m\left(1 + \frac{[I]}{K_i}\right)$

Example

Methotrexate is a competitive inhibitor of dihydrofolate reductase (DHFR). It resembles the substrate dihydrofolate and is used as a chemotherapy drug because rapidly dividing cancer cells require folate for DNA synthesis.

2. Uncompetitive Inhibition

Definition

An **uncompetitive inhibitor** binds only to the enzyme-substrate (ES) complex, not to the free enzyme.

- Inhibitor binds to a site created when substrate binds
- Both K_m and V_{max} decrease by the same factor
- Cannot be overcome by increasing [S]
- Lineweaver-Burk: parallel lines (same slope)

$$v = \frac{V_{\text{max}}^{\text{app}}[S]}{K_m^{\text{app}} + [S]}$$

where $V_{\text{max}}^{\text{app}} = \frac{V_{\text{max}}}{1+\frac{[I]}{K_i}}$ and $K_m^{\text{app}} = \frac{K_m}{1+\frac{[I]}{K_i}}$

3. Noncompetitive Inhibition (Mixed Inhibition)

Definition

A **noncompetitive inhibitor** binds to a site other than the active site and can bind to either the free enzyme (E) or the enzyme-substrate complex (ES).

- Pure noncompetitive: binds E and ES with equal affinity
- V_{max} decreases (fewer functional enzyme molecules)
- K_m unchanged (in pure noncompetitive)
- Cannot be overcome by increasing [S]

$$v = \frac{V_{\text{max}}^{\text{app}}[S]}{K_m + [S]}$$

where $V_{\text{max}}^{\text{app}} = \frac{V_{\text{max}}}{1+\frac{[I]}{K_i}}$

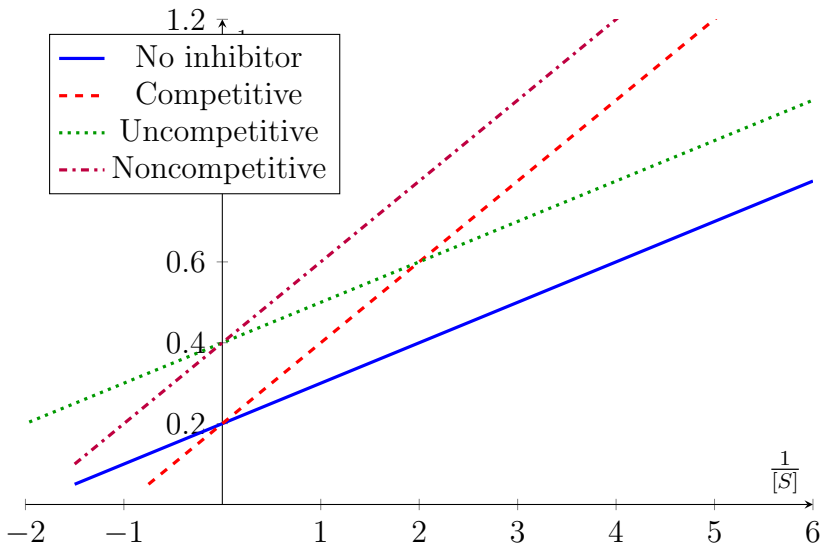
Exam Experts Analogy

Types of Inhibition - A Restaurant Analogy:

- **Competitive:** Another customer sits in your reserved seat. If you wait long enough (more customers = more substrate), you'll eventually get a seat.
- **Uncompetitive:** A health inspector closes tables only after customers sit down. Fewer tables available, but also fewer customers can even try to sit.
- **Noncompetitive:** The chef gets sick. It doesn't matter how many customers come—the kitchen can only work at half capacity.

2.5.2 Summary of Reversible Inhibition

Inhibition Type	Binds To	K_m	V_{max}
Competitive	E only	Increases	Unchanged
Uncompetitive	ES only	Decreases	Decreases
Noncompetitive (pure)	E and ES	Unchanged	Decreases



2.5.3 Irreversible Inhibition

Definition

Irreversible inhibitors form covalent bonds with the enzyme, permanently inactivating it. The enzyme must be replaced by new protein synthesis.

Example

Examples of Irreversible Inhibitors:

- **Aspirin:** Acetylates cyclooxygenase (COX), blocking prostaglandin synthesis
- **Penicillin:** Covalently modifies transpeptidase, preventing bacterial cell wall synthesis
- **Organophosphates** (nerve agents, pesticides): Phosphorylate acetylcholinesterase
- **DIPF** (diisopropyl fluorophosphate): Modifies serine in serine proteases

2.6 Catalytic Efficiency

Definition

The **catalytic efficiency** of an enzyme is measured by the ratio $\frac{k_{\text{cat}}}{K_m}$, which accounts for both the rate of catalysis and substrate binding.

- k_{cat} (turnover number) $= \frac{V_{\max}}{[E]_T}$ = molecules of substrate converted per enzyme per unit time

- Units of k_{cat}: s^{-1}
- Units of $\frac{k_{\text{cat}}}{K_m}$: $\text{M}^{-1}\text{s}^{-1}$
- Maximum theoretical limit: $10^8 - 10^9$ $\text{M}^{-1}\text{s}^{-1}$ (diffusion-controlled)

Example

Catalase has one of the highest k_{cat} values known:

- $k_{\text{cat}} \approx 4 \times 10^7$ s^{-1}
- This means each catalase molecule converts 40 million H_2O_2 molecules to H_2O and O_2 every second!

Key Principle

Specificity Constant:
The ratio $\frac{k_{\text{cat}}}{K_m}$ is called the specificity constant and is used to compare:

- The efficiency of different enzymes
- The preference of one enzyme for different substrates
- Higher $\frac{k_{\text{cat}}}{K_m}$ = better enzyme or preferred substrate

2.7 Enzyme Kinetics Problem-Solving Strategies

Laboratory/Clinical Skill

Calculating K_m and V_{max} from Data:
Method 1: Direct Inspection

1. Find V_{max} as the plateau value at high [S]
2. Find $\frac{V_{\text{max}}}{2}$
3. K_m = [S] at which $v = \frac{V_{\text{max}}}{2}$

Method 2: Lineweaver-Burk Plot

1. Calculate $\frac{1}{v}$ and $\frac{1}{[S]}$ for each data point
2. Plot $\frac{1}{v}$ vs $\frac{1}{[S]}$
3. y-intercept = $\frac{1}{V_{\text{max}}}$
4. x-intercept = $-\frac{1}{K_m}$

Example

Given the following kinetic data, determine K_m and V_{max}:

[S] (mM)	v (μmol/min)
0.5	25
1.0	40
2.0	57
4.0	73
8.0	84
16.0	92

Solution

Using Lineweaver-Burk Plot:
First, calculate reciprocals:

[S] (mM)	v (μmol/min)	$\frac{1}{[S]}$ (mM^{-1})	$\frac{1}{v}$ (min/μmol)
0.5	25	2.0	0.040
1.0	40	1.0	0.025
2.0	57	0.5	0.018
4.0	73	0.25	0.014
8.0	84	0.125	0.012
16.0	92	0.0625	0.011

From linear regression or graphical analysis:

- y-intercept ≈ 0.010 min/μmol
- Therefore: $V_{\text{max}} = \frac{1}{0.010} = 100$ μmol/min
- x-intercept ≈ -1.0 mM^{-1}
- Therefore: $K_m = -\frac{1}{-1.0} = 1.0$ mM

Verification: At [S] $= K_m = 1.0$ mM, v should equal $\frac{V_{\text{max}}}{2} = 50$ μmol/min.
The measured value is 40 μmol/min, which is close (experimental error accounts for difference).

3 Biochemical Energy, Metabolism, and Carbohydrates

3.1 Introduction to Bioenergetics

Definition

Bioenergetics is the study of energy transformations in living organisms, including how cells capture, store, and use energy to perform biological work.

Exam Experts Analogy

Think of your body as a city. The city needs electricity to power everything—lights, computers, factories. Your cells are like buildings in this city, and ATP is the electricity. Just as the power plant converts fuel (coal, natural gas) into electricity, your mitochondria convert food (glucose, fats) into ATP. Every activity in the city/cell requires this energy currency!

3.2 Thermodynamics in Biochemistry

3.2.1 Free Energy (Gibbs Free Energy)

Definition

Gibbs Free Energy (G) is the energy available to do useful work at constant temperature and pressure. The change in free energy (ΔG) determines whether a reaction is spontaneous.

Key Principle

Gibbs Free Energy Equation:

$$\Delta G = \Delta H - T\Delta S$$

where:

- ΔG = change in free energy (kJ/mol)
- ΔH = change in enthalpy (heat content)
- T = absolute temperature (Kelvin)
- ΔS = change in entropy (disorder)

Interpretation of ΔG:

- $\Delta G < 0$: **Exergonic** (spontaneous, releases energy)
- $\Delta G > 0$: **Endergonic** (non-spontaneous, requires energy input)

- $\Delta G = 0$: System is at **equilibrium**

3.2.2 Standard Free Energy Change

Definition

The **standard free energy change** ($\Delta G'$) is measured under biochemical standard conditions:

- Temperature: 298 K (25°C)
- Pressure: 1 atm
- Concentration of reactants and products: 1 M
- pH: 7.0 (this is the prime notation)
- Water concentration: 55.5 M (treated as constant)

Key Principle

Relationship Between ΔG and $\Delta G'$:

$$\Delta G = \Delta G' + RT \ln Q$$

where:

- R = gas constant (8.314 J/mol·K)
- T = temperature in Kelvin
- Q = reaction quotient = $\frac{[\text{products}]}{[\text{reactants}]}$

At equilibrium ($\Delta G = 0$):

$$\Delta G' = -RT \ln K_{eq}$$

Exam Experts Analogy

$\Delta G'$ is like the "official" difficulty rating of a hiking trail—it's measured under standard conditions. But the actual difficulty you experience (ΔG) depends on current conditions: Is it raining? Are you carrying a heavy pack? Similarly, the actual free energy change depends on the current concentrations in the cell.

3.3 ATP: The Energy Currency

Definition

Adenosine triphosphate (ATP) is the primary energy currency of the cell. It consists of:

- Adenine (nitrogenous base)
- Ribose (5-carbon sugar)
- Three phosphate groups (α, β, γ)

3.3.1 ATP Hydrolysis

$$\text{ATP} + \text{H}_2\text{O} \rightarrow \text{ADP} + \text{P}_i + \text{H}^+ \quad \Delta G' = -30.5 \text{ kJ/mol}$$

$$\text{ATP} + \text{H}_2\text{O} \rightarrow \text{AMP} + \text{PP}_i + \text{H}^+ \quad \Delta G' = -45.6 \text{ kJ/mol}$$

Why is ATP hydrolysis so exergonic?

1. **Electrostatic repulsion:** Three negative phosphate groups repel each other
2. **Resonance stabilization:** Products (ADP + P_i) have more resonance forms than ATP
3. **Hydration:** Products are better solvated by water
4. **Ionization:** Products ionize, releasing H^+ and driving equilibrium forward

Exam Experts Analogy

ATP is like a compressed spring. The three phosphate groups are pushed together, creating tension (electrostatic repulsion). When you release the spring (hydrolysis), energy is released as the "spring" relaxes. The products (ADP + P_i) are in a more relaxed, stable state.

3.3.2 Energy Coupling

Definition

Energy coupling is the use of an exergonic reaction to drive an endergonic reaction. The two reactions are linked, often through a common intermediate.

Example

Glutamine Synthesis:
The synthesis of glutamine from glutamate and ammonia is endergonic:

$$\text{Glutamate} + \text{NH}_3 \rightarrow \text{Glutamine} + \text{H}_2\text{O} \quad \Delta G' = +14.2 \text{ kJ/mol}$$

Coupling with ATP hydrolysis:

$$\text{ATP} + \text{H}_2\text{O} \rightarrow \text{ADP} + \text{P}_i \quad \Delta G' = -30.5 \text{ kJ/mol}$$

Overall coupled reaction:

$$\text{Glutamate} + \text{NH}_3 + \text{ATP} \rightarrow \text{Glutamine} + \text{ADP} + \text{P}_i$$

$$\Delta G'_{\text{overall}} = +14.2 + (-30.5) = -16.3 \text{ kJ/mol}$$

The unfavorable reaction becomes favorable when coupled to ATP hydrolysis!

3.4 Other High-Energy Compounds

Compound	**$\Delta G'$ of Hydrolysis (kJ/mol)**
Phosphoenolpyruvate (PEP)	-61.9
1,3-Bisphosphoglycerate	-49.4
Phosphocreatine	-43.1
ATP $\rightarrow$ AMP + PP_i	-45.6
ATP $\rightarrow$ ADP + P_i	-30.5
Glucose-6-phosphate	-13.8

Key Principle

Phosphoryl Group Transfer Potential:
Compounds with more negative $\Delta G'$ of hydrolysis can transfer their phosphate groups to compounds with less negative values. ATP occupies an intermediate position, allowing it to:

- Accept phosphate from high-energy compounds (PEP, 1,3-BPG)
- Donate phosphate to lower-energy compounds (glucose, glycerol)

Exam Experts Analogy

Think of phosphate groups like water flowing downhill. High-energy compounds like PEP are at the top of the hill—they can easily "pour" their phosphate down to ATP. ATP is at a middle elevation, so it can receive phosphate from above and donate it to compounds below (like glucose-6-phosphate at the bottom). Water (phosphate) always flows from high to low energy!

3.5 Introduction to Carbohydrates

Definition

Carbohydrates are polyhydroxy aldehydes or ketones, or substances that yield such compounds upon hydrolysis. The general formula is $(CH_2O)_n$, hence "hydrates of carbon."

3.5.1 Classification of Carbohydrates

1. **Monosaccharides:** Simple sugars that cannot be hydrolyzed further
   - Trioses (3C): Glyceraldehyde, dihydroxyacetone
   - Tetroses (4C): Erythrose
   - Pentoses (5C): Ribose, deoxyribose, ribulose
   - Hexoses (6C): Glucose, fructose, galactose, mannose
2. **Disaccharides:** Two monosaccharides joined by glycosidic bond
   - Maltose: Glucose + Glucose (α-1,4)
   - Lactose: Galactose + Glucose (β-1,4)
   - Sucrose: Glucose + Fructose (α-1,β-2)
3. **Oligosaccharides:** 3-10 monosaccharide units
4. **Polysaccharides:** Many monosaccharide units
   - Storage: Starch (plants), Glycogen (animals)
   - Structural: Cellulose (plants), Chitin (arthropods)

3.5.2 Monosaccharide Structure

Definition

Aldoses contain an aldehyde group; **ketoses** contain a ketone group. The position of the carbonyl determines the classification.

D and L Configuration:

- Based on the configuration of the chiral carbon farthest from the carbonyl
- D-sugars: OH on the right in Fischer projection (most biological sugars)
- L-sugars: OH on the left in Fischer projection

Epimers:

Definition

Epimers are sugars that differ in configuration at only one chiral carbon.

Example

- D-Glucose and D-Mannose are C-2 epimers
- D-Glucose and D-Galactose are C-4 epimers

3.5.3 Cyclic Forms of Monosaccharides

Definition

In aqueous solution, monosaccharides with 5+ carbons form cyclic structures through intramolecular hemiacetal or hemiketal formation:

- **Pyranose:** 6-membered ring (like glucose)
- **Furanose:** 5-membered ring (like fructose)

Anomers and Mutarotation:

Definition

Anomers are cyclic sugars that differ in configuration at the anomeric carbon (C-1 for aldoses, C-2 for ketoses):

- α-anomer: OH on anomeric carbon is axial (down in Haworth projection for D-sugars)
- β-anomer: OH on anomeric carbon is equatorial (up in Haworth projection for D-sugars)

Mutarotation is the interconversion between α and β anomers through the open-chain form.

Exam Experts Analogy

Imagine a revolving door (the open-chain form). People can enter as "α-visitors" or "β-visitors" depending on which side they exit. In solution, sugars constantly spin through this revolving door, switching between α and β forms until they reach equilibrium!

3.5.4 Glycosidic Bonds

Definition

A **glycosidic bond** forms when the anomeric carbon of one sugar reacts with a hydroxyl group of another molecule (sugar, amino acid, lipid, etc.), releasing water.

- α-glycosidic bond: Formed by α-anomer
- β-glycosidic bond: Formed by β-anomer
- Notation: $\alpha(1\rightarrow4)$ means α-linkage from C-1 to C-4

Example

Important Glycosidic Bonds:

- **Maltose:** $\alpha(1\rightarrow4)$ glucose-glucose (digestible)
- **Cellobiose:** $\beta(1\rightarrow4)$ glucose-glucose (humans cannot digest)
- **Lactose:** $\beta(1\rightarrow4)$ galactose-glucose
- **Sucrose:** $\alpha(1\rightarrow2\beta)$ glucose-fructose

3.5.5 Polysaccharides

Storage Polysaccharides:

- **Starch** (plants):
  - Amylose: Linear $\alpha(1\rightarrow4)$ glucose chains
  - Amylopectin: Branched with $\alpha(1\rightarrow6)$ branch points every 24-30 residues
- **Glycogen** (animals):
  - Similar to amylopectin but more highly branched
  - $\alpha(1\rightarrow6)$ branch points every 8-12 residues
  - Stored primarily in liver and muscle
  - More branches = more ends = faster mobilization

Exam Experts Analogy

Glycogen is like a tree with many branches. When you need energy quickly (like running from danger), you can pick fruit from many branch ends simultaneously. Amylose is like a single vine—you can only pick from one end at a time. That's why animals store glycogen (highly branched) for quick energy access!

Structural Polysaccharides:

- **Cellulose:** Linear $\beta(1\rightarrow4)$ glucose chains
  - Forms rigid, fibrous structures
  - Humans lack enzymes to cleave $\beta(1\rightarrow4)$ bonds
  - Dietary fiber

4 Glycolysis

4.1 Introduction to Glycolysis

Definition

Glycolysis (from Greek: *glykys* = sweet, *lysis* = splitting) is the metabolic pathway that converts one molecule of glucose (6 carbons) into two molecules of pyruvate (3 carbons each), with the net production of ATP and NADH.

- Location: Cytoplasm (cytosol)
- Oxygen requirement: None (can occur aerobically or anaerobically)
- Universal pathway: Found in virtually all living organisms
- 10 enzyme-catalyzed reactions

Exam Experts Analogy

Glycolysis is like breaking a $20 bill into smaller bills so you can use them in vending machines. You start with one big glucose molecule (the $20 bill) and end up with two pyruvate molecules (like two $10 bills). Along the way, you have to pay a small fee (ATP investment), but you get more back than you spent (ATP payoff). The "change" you get (NADH) can be "deposited" later for even more energy!

4.2 Overview: Two Phases of Glycolysis

Key Principle

Glycolysis consists of two phases:
Phase 1: Energy Investment Phase (Steps 1-5)

- Glucose is phosphorylated and converted to two 3-carbon molecules
- Consumes 2 ATP
- "Priming" the glucose for energy extraction

Phase 2: Energy Payoff Phase (Steps 6-10)

- Two 3-carbon molecules are oxidized and converted to pyruvate
- Produces 4 ATP and 2 NADH
- Net yield: 2 ATP and 2 NADH per glucose

4.3 The Ten Reactions of Glycolysis

4.3.1 Phase 1: Energy Investment Phase

Reaction 1: Hexokinase

$$\text{Glucose} + \text{ATP} \xrightarrow{\text{Hexokinase}} \text{Glucose-6-phosphate} + \text{ADP}$$

- $\Delta G' = -16.7$ kJ/mol (irreversible)
- **Enzyme:** Hexokinase (or Glucokinase in liver)
- **Purpose:**
  - Traps glucose in the cell (G6P cannot cross membrane)
  - Destabilizes glucose for subsequent reactions
- **Regulation:** Hexokinase inhibited by G6P (product inhibition)
- Requires Mg^{2+}

Definition

Hexokinase vs. Glucokinase:

- **Hexokinase:** Found in most tissues, low K_m (high affinity), inhibited by G6P
- **Glucokinase:** Found in liver and pancreatic β-cells, high K_m (low affinity), NOT inhibited by G6P

Glucokinase acts as a "glucose sensor"—it only phosphorylates glucose when blood glucose is high.

Reaction 2: Phosphoglucose Isomerase

$$\text{Glucose-6-phosphate} \xleftrightarrow{\text{Phosphoglucose Isomerase}} \text{Fructose-6-phosphate}$$

- $\Delta G' = +1.7$ kJ/mol (reversible, near equilibrium)
- **Type:** Isomerization (aldose $\rightarrow$ ketose)
- **Purpose:** Prepares for phosphorylation at C-1

Reaction 3: Phosphofructokinase-1 (PFK-1) [MAJOR REGULATORY STEP]

$$\text{Fructose-6-phosphate} + \text{ATP} \xrightarrow{\text{PFK-1}} \text{Fructose-1,6-bisphosphate} + \text{ADP}$$

- $\Delta G' = -14.2$ kJ/mol (irreversible)
- **COMMITTED STEP** of glycolysis
- Most important regulatory enzyme in glycolysis
- Requires Mg^{2+}

Key Principle

Regulation of PFK-1:
Activators (stimulate glycolysis):

- AMP (signals low energy)
- Fructose-2,6-bisphosphate (F-2,6-BP) — most potent activator
- ADP
- P_i

Inhibitors (slow glycolysis):

- ATP (signals high energy) — despite being a substrate!
- Citrate (signals abundant biosynthetic precursors)
- H^+ (low pH) — prevents excessive lactate production

Exam Experts Analogy

PFK-1 is like the "gatekeeper" of glycolysis. Imagine a bouncer at a club who decides how many people can enter. When the club is empty (low ATP, high AMP), the bouncer lets everyone in quickly. When the club is packed (high ATP, citrate), the bouncer slows down entry. Fructose-2,6-bisphosphate is like the club owner telling the bouncer "Let more people in!"

Reaction 4: Aldolase

$$\text{Fructose-1,6-bisphosphate} \xleftrightarrow{\text{Aldolase}} \text{DHAP} + \text{Glyceraldehyde-3-phosphate}$$

- $\Delta G' = +23.8$ kJ/mol (unfavorable, but pulled forward)
- **Type:** Aldol cleavage
- Splits 6-carbon sugar into two 3-carbon pieces:
  - Dihydroxyacetone phosphate (DHAP) — ketose
  - Glyceraldehyde-3-phosphate (G3P) — aldose

Reaction 5: Triose Phosphate Isomerase

$$\text{Dihydroxyacetone phosphate} \xleftrightarrow{\text{TIM}} \text{Glyceraldehyde-3-phosphate}$$

- $\Delta G' = +7.5$ kJ/mol (reversible)
- Only G3P continues in glycolysis

- Effectively: 1 glucose → 2 G3P
- TIM is a "kinetically perfect" enzyme (diffusion-limited)

End of Phase 1: 2 ATP consumed, 2 G3P molecules ready for Phase 2

4.3.2 Phase 2: Energy Payoff Phase

Note: Each reaction occurs TWICE per glucose (once for each G3P)

Reaction 6: Glyceraldehyde-3-phosphate Dehydrogenase (GAPDH)

$$\text{G3P} + \text{NAD}^+ + \text{P}_i \xleftrightarrow{\text{GAPDH}} \text{1,3-Bisphosphoglycerate} + \text{NADH} + \text{H}^+$$

- $\Delta G' = +6.3$ kJ/mol (reversible)
- **Type:** Oxidation and phosphorylation
- Two coupled processes:
  1. Oxidation of aldehyde to carboxylic acid ($\text{NAD}^+ \rightarrow \text{NADH}$)
  2. Attachment of inorganic phosphate (substrate-level phosphorylation setup)
- Creates high-energy acyl phosphate bond in 1,3-BPG
- **Produces: 2 NADH per glucose**

Exam Experts Analogy

This reaction is like a two-for-one deal. The enzyme oxidizes G3P (removing electrons to NAD^+) AND attaches a phosphate group at the same time. It's like a car wash that also fills your gas tank—two services in one stop. The energy from oxidation is captured in the high-energy phosphate bond.

Reaction 7: Phosphoglycerate Kinase

$$\text{1,3-Bisphosphoglycerate} + \text{ADP} \xleftrightarrow{\text{PGK}} \text{3-Phosphoglycerate} + \text{ATP}$$

- $\Delta G' = -18.5$ kJ/mol (favorable)
- **Type:** Substrate-level phosphorylation
- First ATP-generating step in glycolysis
- **Produces: 2 ATP per glucose** (one per G3P)
- Named "kinase" because reverse reaction was discovered first

Definition

Substrate-level phosphorylation is the direct transfer of a phosphate group from a high-energy substrate to ADP, forming ATP. This occurs in the cytoplasm and does not require oxygen or an electron transport chain.

Reaction 8: Phosphoglycerate Mutase

$$\text{3-Phosphoglycerate} \xleftrightarrow{\text{PGM}} \text{2-Phosphoglycerate}$$

- $\Delta G' = +4.4$ kJ/mol (reversible)
- **Type:** Isomerization (mutase = intramolecular phosphate transfer)
- Moves phosphate from C3 to C2
- Prepares molecule for dehydration

Reaction 9: Enolase

$$\text{2-Phosphoglycerate} \xleftrightarrow{\text{Enolase}} \text{Phosphoenolpyruvate} + H_2O$$

- $\Delta G' = +1.7$ kJ/mol (reversible)
- **Type:** Dehydration
- Removes water, creating double bond
- Creates phosphoenolpyruvate (PEP)—highest energy phosphate compound in metabolism!
- Requires Mg^{2+}
- Inhibited by fluoride (used in blood collection tubes)

Reaction 10: Pyruvate Kinase

$$\text{Phosphoenolpyruvate} + \text{ADP} \xrightarrow{\text{Pyruvate Kinase}} \text{Pyruvate} + \text{ATP}$$

- $\Delta G' = -31.4$ kJ/mol (**highly favorable, irreversible**)
- **Type:** Substrate-level phosphorylation
- Second ATP-generating step
- **Produces: 2 ATP per glucose**
- Third regulatory point of glycolysis
- Requires K^+ and Mg^{2+}

Key Principle

Regulation of Pyruvate Kinase:
Activators:

- Fructose-1,6-bisphosphate (feedforward activation)
- High [PEP]

Inhibitors:

- ATP (signals high energy)
- Alanine (signals abundant amino acids)
- Acetyl-CoA
- Phosphorylation by protein kinase A (in liver)

4.4 Overall Glycolysis Summary

Net Reaction of Glycolysis

$$\text{Glucose} + 2\text{NAD}^+ + 2\text{ADP} + 2\text{P}_i \rightarrow 2\text{Pyruvate} + 2\text{NADH} + 2\text{H}^+ + 2\text{ATP} + 2\text{H}_2\text{O}$$

Component	Consumed	Produced
Glucose	1	—
ATP	2 (investment)	4 (payoff)
Net ATP	—	**2**
NAD^+	2	—
NADH	—	2
Pyruvate	—	2

4.5 Fates of Pyruvate

Definition

The fate of pyruvate depends on the oxygen availability and the metabolic needs of the cell.

4.5.1 Aerobic Conditions (Oxygen Present)

$$\text{Pyruvate} \xrightarrow{\text{Pyruvate Dehydrogenase}} \text{Acetyl-CoA} \rightarrow \text{Citric Acid Cycle} \rightarrow \text{Electron Transport Chain}$$

- Pyruvate enters mitochondria

- Oxidized to acetyl-CoA (releases CO_2, generates NADH)
- Complete oxidation yields 30-32 ATP per glucose

4.5.2 Anaerobic Conditions (No Oxygen)

Lactic Acid Fermentation (Animals, some bacteria):

$$\text{Pyruvate} + \text{NADH} + \text{H}^+ \xrightarrow{\text{Lactate Dehydrogenase}} \text{Lactate} + \text{NAD}^+$$

- Regenerates NAD^+ for continued glycolysis
- Occurs in exercising muscle, red blood cells
- Lactate can be recycled to glucose in liver (Cori cycle)

Alcoholic Fermentation (Yeast, some bacteria):

$$\text{Pyruvate} \xrightarrow{\text{Pyruvate Decarboxylase}} \text{Acetaldehyde} + \text{CO}_2$$

$$\text{Acetaldehyde} + \text{NADH} + \text{H}^+ \xrightarrow{\text{Alcohol Dehydrogenase}} \text{Ethanol} + \text{NAD}^+$$

- Used in brewing and baking
- CO_2 makes bread rise and beer fizzy

Exam Experts Analogy

The fate of pyruvate is like what happens to a traveler at a crossroads. If the highway is open (oxygen available), they continue the journey to the big city (mitochondria) for maximum reward. If the highway is closed (no oxygen), they must take a detour (fermentation) that lets them keep moving but doesn't get them as far.

4.6 Regulation of Glycolysis

Key Principle

Three Irreversible Steps = Three Control Points:

1. **Hexokinase** (Reaction 1): Inhibited by G6P (product inhibition)
2. **Phosphofructokinase-1** (Reaction 3): Main control point
3. **Pyruvate kinase** (Reaction 10): Fine-tuning at the end

4.6.1 The Role of Fructose-2,6-Bisphosphate

Definition

Fructose-2,6-bisphosphate (F-2,6-BP) is the most potent activator of PFK-1 and inhibitor of fructose-1,6-bisphosphatase (gluconeogenesis enzyme). It is NOT a glycolytic intermediate but a regulatory molecule.

- Produced by phosphofructokinase-2 (PFK-2)
- Degraded by fructose-2,6-bisphosphatase (FBPase-2)
- PFK-2 and FBPase-2 are the same protein (bifunctional enzyme)!
- Hormonal control:
  - **Insulin** → activates PFK-2 → increases F-2,6-BP → stimulates glycolysis
  - **Glucagon** → activates FBPase-2 → decreases F-2,6-BP → inhibits glycolysis

5 Gluconeogenesis and Diabetes

5.1 Introduction to Gluconeogenesis

Definition

Gluconeogenesis is the metabolic pathway that synthesizes glucose from non-carbohydrate precursors. It occurs primarily in the **liver** (90%) and to a lesser extent in the **kidney** (10%).

Exam Experts Analogy

If glycolysis is like spending money (breaking down glucose for energy), gluconeogenesis is like earning money (building glucose from scratch). When your "glucose bank account" runs low (during fasting or exercise), your liver acts like a glucose factory, manufacturing new glucose from raw materials to keep your blood sugar stable.

5.2 Why Gluconeogenesis Matters

- **Brain** requires 120 g glucose/day (cannot use fatty acids)
- **Red blood cells** have no mitochondria—rely entirely on glycolysis
- **Glycogen stores** last only 12-24 hours during fasting
- **Exercise** depletes muscle glycogen rapidly

5.3 Precursors for Gluconeogenesis

Precursor	Source	Entry Point
Lactate	Muscle, RBCs	Pyruvate
Alanine	Muscle protein	Pyruvate
Glycerol	Adipose (fat breakdown)	DHAP
Glucogenic amino acids	Protein breakdown	Various TCA intermediates
Propionate	Odd-chain fatty acids	Succinyl-CoA

Key Principle

Important: Fatty acids (even-chain) CANNOT be converted to glucose in animals because:

- Acetyl-CoA cannot be converted to pyruvate (PDH is irreversible)
- Acetyl-CoA enters TCA cycle but carbons are lost as CO_2
- No net synthesis of oxaloacetate from acetyl-CoA

5.4 Gluconeogenesis Pathway

Gluconeogenesis is largely the reverse of glycolysis, but three irreversible glycolytic reactions must be bypassed with different enzymes.

5.4.1 Bypass 1: Pyruvate to Phosphoenolpyruvate

The pyruvate kinase reaction ($\Delta G' = -31.4$ kJ/mol) is too energetically unfavorable to reverse. Two enzymes are required:

Step 1a: Pyruvate Carboxylase

$$\text{Pyruvate} + CO_2 + \text{ATP} + H_2O \xrightarrow{\text{Pyruvate Carboxylase}} \text{Oxaloacetate} + \text{ADP} + P_i$$

- Location: **Mitochondrial matrix**
- Requires **biotin** as cofactor
- Requires **acetyl-CoA** as allosteric activator
- Adds CO_2 to pyruvate (carboxylation)

Step 1b: Phosphoenolpyruvate Carboxykinase (PEPCK)

$$\text{Oxaloacetate} + \text{GTP} \xrightarrow{\text{PEPCK}} \text{Phosphoenolpyruvate} + CO_2 + \text{GDP}$$

- Location: **Cytosol** (in humans) or mitochondria (varies by species)

- Removes CO_2 and adds phosphate from GTP
- OAA must be transported out of mitochondria (as malate or aspartate)

Exam Experts Analogy

Bypassing pyruvate kinase is like taking a detour around a one-way street. You can't go backward on the highway (pyruvate kinase), so you take two side roads (pyruvate carboxylase and PEPCK) to get around it. It costs more fuel (2 high-energy phosphates instead of 1), but it gets you where you need to go.

5.4.2 Bypass 2: Fructose-1,6-bisphosphate to Fructose-6-phosphate

$$\text{Fructose-1,6-bisphosphate} + H_2O \xrightarrow{\text{Fructose-1,6-bisphosphatase}} \text{Fructose-6-phosphate} + P_i$$

- Hydrolysis reaction (no ATP produced)
- **Key regulatory enzyme** of gluconeogenesis
- Inhibited by AMP and F-2,6-BP
- Activated by ATP and citrate

5.4.3 Bypass 3: Glucose-6-phosphate to Glucose

$$\text{Glucose-6-phosphate} + H_2O \xrightarrow{\text{Glucose-6-phosphatase}} \text{Glucose} + P_i$$

- Located in **endoplasmic reticulum** membrane
- Found only in **liver and kidney** (not muscle!)
- This is why muscle cannot release free glucose into blood
- G6P must be transported into ER lumen for hydrolysis

5.5 Energy Cost of Gluconeogenesis

Energy Investment for Gluconeogenesis

Per glucose synthesized from 2 pyruvate:

- 4 ATP (2 for pyruvate carboxylase, 2 for phosphoglycerate kinase reversal)
- 2 GTP (for PEPCK)
- 2 NADH (for GAPDH reversal)

Total: 6 high-energy phosphate bonds
Compare to glycolysis yield: 2 ATP + 2 NADH
Gluconeogenesis costs more than glycolysis produces!

5.6 Reciprocal Regulation of Glycolysis and Gluconeogenesis

Key Principle

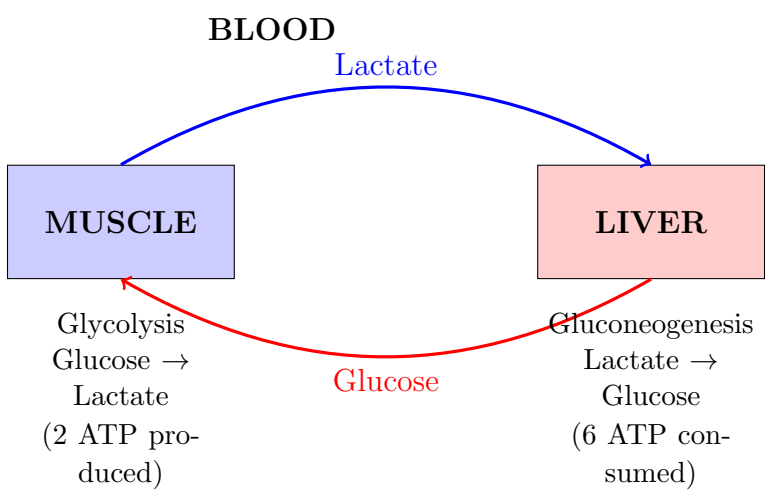
Glycolysis and gluconeogenesis are reciprocally regulated to prevent a futile cycle (simultaneous operation would waste ATP).

Regulator	Glycolysis	Gluconeogenesis
ATP (high)	Inhibits PFK-1	Activates FBPase-1
AMP (low energy)	Activates PFK-1	Inhibits FBPase-1
Citrate	Inhibits PFK-1	Activates FBPase-1
F-2,6-BP	Activates PFK-1	Inhibits FBPase-1
Insulin	Stimulates glycolysis	Inhibits gluconeogenesis
Glucagon	Inhibits glycolysis	Stimulates gluconeogenesis

5.7 The Cori Cycle

Definition

The **Cori cycle** is the metabolic pathway in which lactate produced by anaerobic glycolysis in muscle is transported to the liver, converted to glucose via gluconeogenesis, and returned to muscle.



- Allows muscle to function anaerobically during intense exercise
- Shifts metabolic burden (ATP cost) to the liver
- Net cost: 4 ATP per cycle (liver pays 6, muscle gains 2)
- Important during exercise, trauma, and in cancer (Warburg effect)

Clinical Correlation

Clinical Correlation: Lactic Acidosis
When lactate production exceeds the liver's capacity for gluconeogenesis, lactate accumulates in blood, causing **lactic acidosis**. Causes include:

- Tissue hypoxia (shock, cardiac arrest)
- Liver failure (cannot perform gluconeogenesis)
- Metformin toxicity (inhibits gluconeogenesis)
- Thiamine deficiency (pyruvate cannot enter TCA cycle)

5.8 Practice Problems: Gluconeogenesis

1. Which of the following enzymes is unique to gluconeogenesis?

   A. Hexokinase
   B. Phosphofructokinase-1
   C. Fructose-1,6-bisphosphatase
   D. Pyruvate kinase

   Answer: C. Fructose-1,6-bisphosphatase is one of the four enzymes unique to gluconeogenesis, bypassing the irreversible PFK-1 reaction of glycolysis.

2. A patient with liver cirrhosis develops hypoglycemia after fasting. Which metabolic pathway is most likely impaired?

   A. Glycolysis
   B. Glycogenolysis in muscle
   C. Gluconeogenesis
   D. Pentose phosphate pathway

   Answer: C. Gluconeogenesis occurs primarily in the liver. During fasting, the liver synthesizes glucose to maintain blood glucose levels. Liver damage impairs this critical function.

3. Fructose-2,6-bisphosphate (F-2,6-BP) has what effect on gluconeogenesis?

   A. Activates fructose-1,6-bisphosphatase
   B. Inhibits fructose-1,6-bisphosphatase
   C. Activates pyruvate carboxylase
   D. Has no effect on gluconeogenesis

   Answer: B. F-2,6-BP inhibits FBPase-1 (gluconeogenesis) while activating PFK-1 (glycolysis), ensuring reciprocal regulation of these opposing pathways.

6 Citric Acid Cycle (TCA/Krebs Cycle)

Learning Objectives

After completing this section, you should be able to:

1. Describe the location and overall function of the citric acid cycle
2. List all eight reactions, enzymes, and products of the TCA cycle
3. Identify the regulatory enzymes and their allosteric modulators
4. Calculate the ATP yield from one acetyl-CoA through the TCA cycle
5. Explain the amphibolic nature of the TCA cycle
6. Describe anaplerotic reactions that replenish TCA intermediates

6.1 Overview and Location

Definition

The **citric acid cycle** (also called the **tricarboxylic acid (TCA) cycle** or **Krebs cycle**) is a series of eight enzymatic reactions that oxidize acetyl-CoA to CO_2, generating reduced electron carriers (NADH, $FADH_2$) and GTP.

- **Location:** Mitochondrial matrix
- **Input:** Acetyl-CoA (2 carbons)
- **Output:** 2 CO_2, 3 NADH, 1 $FADH_2$, 1 GTP per acetyl-CoA
- **Function:** Final common pathway for oxidation of carbohydrates, fats, and amino acids

Exam Experts Analogy

The TCA cycle is like a recycling center. Oxaloacetate (OAA) is the "container" that picks up acetyl-CoA (the "trash"). As the container moves through the cycle, the trash is processed (oxidized), energy is extracted (NADH, $FADH_2$, GTP), and waste products are released (CO_2). At the end, the empty container (OAA) is ready to pick up more trash!

6.2 Pyruvate Dehydrogenase Complex (PDC)

Before entering the TCA cycle, pyruvate must be converted to acetyl-CoA.

Key Principle

Pyruvate Dehydrogenase Reaction:

$$\text{Pyruvate} + \text{CoA} + \text{NAD}^+ \xrightarrow{\text{PDC}} \text{Acetyl-CoA} + \text{CO}_2 + \text{NADH}$$

- $\Delta G' = -33.4$ kJ/mol (irreversible)
- Location: Mitochondrial matrix
- Links glycolysis to TCA cycle

6.2.1 PDC Structure and Mechanism

The PDC is a massive multi-enzyme complex with three catalytic enzymes:

Component	Enzyme	Coenzyme	Vitamin Source
E1	Pyruvate dehydrogenase	TPP	Thiamine (B_1)
E2	Dihydrolipoyl transacetylase	Lipoic acid, CoA	Pantothenic acid (B_5)
E3	Dihydrolipoyl dehydrogenase	FAD, NAD^+	Riboflavin (B_2), Niacin (B_3)

Memory Aid: PDC Coenzymes

"Tender Loving Care For Nancy"

- **T**PP (Thiamine)
- **L**ipoic acid
- **C**oA (pantothenic acid)
- **F**AD (riboflavin)
- **N**AD^+ (niacin)

6.2.2 Regulation of PDC

Key Principle

PDC Regulation by Covalent Modification:

Active (dephosphorylated)	Inactive (phosphorylated)
PDC phosphatase activated by:	PDC kinase activated by:
Ca^{2+} (muscle contraction)	ATP (high energy)
Insulin	Acetyl-CoA (product)
Pyruvate (substrate)	NADH (product)
	Fatty acids (alternative fuel)

Clinical Correlation

Clinical Correlation: Pyruvate Dehydrogenase Deficiency
PDC deficiency is an X-linked disorder causing:

- Lactic acidosis (pyruvate converted to lactate)
- Neurological defects (brain depends on glucose oxidation)
- Treatment: Ketogenic diet (provides acetyl-CoA from fats, bypassing PDC)

6.3 Reactions of the TCA Cycle

TCA Cycle Overview

Net Reaction (per acetyl-CoA):

$$\text{Acetyl-CoA} + 3\text{NAD}^+ + \text{FAD} + \text{GDP} + \text{P}_i + 2\text{H}_2\text{O}$$
$$\downarrow$$
$$2\text{CO}_2 + 3\text{NADH} + \text{FADH}_2 + \text{GTP} + \text{CoA} + 3\text{H}^+$$

6.3.1 The Eight Reactions

Reaction 1: Citrate Synthase

$$\text{Oxaloacetate} + \text{Acetyl-CoA} + \text{H}_2\text{O} \rightarrow \text{Citrate} + \text{CoA-SH}$$

- $\Delta G' = -31.4$ kJ/mol (**irreversible**)
- **Type:** Condensation (aldol condensation)
- 4C + 2C → 6C

- **Regulatory enzyme:** Inhibited by ATP, NADH, succinyl-CoA, citrate

Reaction 2: Aconitase

$$\text{Citrate} \xleftrightarrow{\text{Aconitase}} \text{Isocitrate}$$

- $\Delta G' = +6.3$ kJ/mol (reversible)
- **Type:** Isomerization (via cis-aconitate intermediate)
- Moves hydroxyl group from C3 to C2
- Contains iron-sulfur cluster (Fe-S)

Clinical Correlation

Clinical Note: Fluoroacetate (rat poison) is converted to fluorocitrate, which inhibits aconitase, blocking the TCA cycle.

Reaction 3: Isocitrate Dehydrogenase (IDH)

$$\text{Isocitrate} + \text{NAD}^+ \rightarrow \alpha\text{-Ketoglutarate} + \text{CO}_2 + \text{NADH}$$

- $\Delta G' = -20.9$ kJ/mol (**irreversible**)
- **Type:** Oxidative decarboxylation
- 6C $\rightarrow$ 5C (first CO_2 released)
- **First NADH produced**
- **Rate-limiting step of TCA cycle**
- **Regulation:**
  - Activated by: ADP, Ca^{2+}
  - Inhibited by: ATP, NADH

Reaction 4: α-Ketoglutarate Dehydrogenase Complex

$$\alpha\text{-Ketoglutarate} + \text{NAD}^+ + \text{CoA} \rightarrow \text{Succinyl-CoA} + \text{CO}_2 + \text{NADH}$$

- $\Delta G' = -33.5$ kJ/mol (**irreversible**)
- **Type:** Oxidative decarboxylation
- 5C $\rightarrow$ 4C (second CO_2 released)
- **Second NADH produced**

- Same coenzymes as PDC: TPP, lipoic acid, CoA, FAD, NAD^+
- **Regulation:**
  - Activated by: Ca^{2+}
  - Inhibited by: Succinyl-CoA, NADH, ATP

Reaction 5: Succinyl-CoA Synthetase

$$\text{Succinyl-CoA} + \text{GDP} + \text{P}_i \leftrightarrow \text{Succinate} + \text{CoA} + \text{GTP}$$

- $\Delta G' = -2.9$ kJ/mol (reversible)
- **Type:** Substrate-level phosphorylation
- **Only GTP (or ATP) produced directly in TCA cycle**
- High-energy thioester bond drives GTP synthesis
- GTP can be converted to ATP: GTP + ADP $\rightleftharpoons$ GDP + ATP

Reaction 6: Succinate Dehydrogenase (Complex II)

$$\text{Succinate} + \text{FAD} \leftrightarrow \text{Fumarate} + \text{FADH}_2$$

- $\Delta G' = 0$ kJ/mol (reversible)
- **Type:** Oxidation (dehydrogenation)
- **Only $FADH_2$ produced in TCA cycle**
- **Only enzyme embedded in inner mitochondrial membrane**
- Part of electron transport chain (Complex II)
- FAD used because reaction not energetic enough to reduce NAD^+
- **Competitively inhibited by malonate** (succinate analog)

Reaction 7: Fumarase

$$\text{Fumarate} + \text{H}_2\text{O} \leftrightarrow \text{L-Malate}$$

- $\Delta G' = -3.8$ kJ/mol (reversible)
- **Type:** Hydration
- Stereospecific: produces only L-malate

Reaction 8: Malate Dehydrogenase

$$\text{L-Malate} + \text{NAD}^+ \leftrightarrow \text{Oxaloacetate} + \text{NADH} + \text{H}^+$$

- $\Delta G' = +29.7$ kJ/mol (highly unfavorable!)
- **Type:** Oxidation
- **Third NADH produced**
- Reaction pulled forward by:
  - Low [OAA] — rapidly consumed by citrate synthase
  - Removal of NADH by ETC
- Regenerates oxaloacetate to continue the cycle

Memory Aid: TCA Cycle Intermediates

"Citrate Is Krebs' Starting Substrate For Making Oxaloacetate"

- **C**itrate (6C)
- **I**socitrate (6C)
- α-**K**etoglutarate (5C)
- **S**uccinyl-CoA (4C)
- **S**uccinate (4C)
- **F**umarate (4C)
- **M**alate (4C)
- **O**xaloacetate (4C)

6.4 Energy Yield of the TCA Cycle

Product	Per Acetyl-CoA	ATP Equivalent
NADH	3	$3 \times 2.5 = 7.5$ ATP
$FADH_2$	1	$1 \times 1.5 = 1.5$ ATP
GTP	1	1 ATP
Total		**10 ATP**

Per glucose molecule (2 acetyl-CoA):

- 6 NADH $\rightarrow$ 15 ATP

- 2 $FADH_2 \rightarrow$ 3 ATP
- 2 GTP $\rightarrow$ 2 ATP
- **Total from TCA: 20 ATP per glucose**

6.5 Regulation of the TCA Cycle

Key Principle

Three Regulatory Enzymes:

1. **Citrate synthase** (entry point)
2. **Isocitrate dehydrogenase** (rate-limiting)
3. **α-Ketoglutarate dehydrogenase**

General principles:

- High ATP, NADH, succinyl-CoA, citrate $\rightarrow$ **inhibit** cycle
- High ADP, NAD^+, Ca^{2+} $\rightarrow$ **activate** cycle

Regulator	Citrate Synthase	Isocitrate DH	α-KG DH
ATP	Inhibits	Inhibits	—
ADP	—	Activates	—
NADH	Inhibits	Inhibits	Inhibits
NAD^+	—	Activates	Activates
Succinyl-CoA	—	—	Inhibits
Ca^{2+}	—	Activates	Activates
Citrate	Inhibits	—	—

Exam Experts Analogy

The TCA cycle is like a factory production line. When the warehouse is full of products (high ATP, NADH), the line slows down. When orders are piling up (high ADP, NAD^+), production speeds up. Calcium is like an "urgent order" signal—during muscle contraction, Ca^{2+} increases, signaling the need for more ATP.

6.6 Anaplerotic Reactions

Definition

Anaplerotic reactions are reactions that replenish TCA cycle intermediates when they are removed for biosynthesis. "Anaplerotic" means "filling up."

Key Anaplerotic Reactions:

1. **Pyruvate carboxylase** (most important):

$$\text{Pyruvate} + CO_2 + \text{ATP} \xrightarrow{\text{Biotin}} \text{Oxaloacetate} + \text{ADP} + P_i$$

   - Activated by acetyl-CoA
   - Located in mitochondria
   - Also used in gluconeogenesis

2. **PEP carboxykinase** (in some tissues):

$$\text{PEP} + CO_2 \rightarrow \text{Oxaloacetate} + P_i$$

3. **Amino acid degradation**:

   - Glutamate $\rightarrow$ α-ketoglutarate
   - Aspartate $\rightarrow$ oxaloacetate

6.7 TCA Cycle as a Metabolic Hub

Key Principle

The TCA cycle intermediates serve as precursors for many biosynthetic pathways:

- **Citrate** $\rightarrow$ Fatty acid synthesis (exported to cytoplasm)
- **α-Ketoglutarate** $\rightarrow$ Glutamate $\rightarrow$ other amino acids, purines
- **Succinyl-CoA** $\rightarrow$ Heme synthesis, odd-chain fatty acid metabolism
- **Oxaloacetate** $\rightarrow$ Aspartate $\rightarrow$ pyrimidines, other amino acids
- **Oxaloacetate** $\rightarrow$ Gluconeogenesis (via PEP)

Clinical Correlation: TCA Cycle Defects

Mutations in TCA cycle enzymes are associated with various cancers:

- **Succinate dehydrogenase (SDH) mutations**: Paragangliomas, pheochromocytomas
- **Fumarase mutations**: Renal cell carcinoma, leiomyomas
- **Isocitrate dehydrogenase (IDH) mutations**: Gliomas, acute myeloid leukemia

These mutations lead to accumulation of "oncometabolites" that promote tumor growth.

6.8 Practice Problems: TCA Cycle

1. A patient has a thiamine deficiency. Which TCA cycle reactions would be directly affected?
2. Calculate the total ATP yield from one molecule of acetyl-CoA through the TCA cycle and oxidative phosphorylation.
3. Why does the TCA cycle stop when oxygen is absent, even though oxygen doesn't directly participate in any TCA cycle reaction?
4. A researcher adds fluoroacetate (converted to fluorocitrate, an aconitase inhibitor) to cells. What would happen to:
   (a) Citrate levels?
   (b) Isocitrate levels?
   (c) ATP production?

Solutions:

1. Thiamine (vitamin B_1) is a component of TPP, required for:
   - Pyruvate dehydrogenase (before TCA)
   - α-Ketoglutarate dehydrogenase (in TCA)

   Both reactions would be impaired.
2. Per acetyl-CoA:
   - 3 NADH $\times$ 2.5 ATP = 7.5 ATP
   - 1 $FADH_2$ $\times$ 1.5 ATP = 1.5 ATP
   - 1 GTP = 1 ATP
   - **Total: 10 ATP**
3. Without oxygen, the electron transport chain cannot function, so NADH and $FADH_2$ cannot be reoxidized to NAD^+ and FAD. Without NAD^+ and FAD, the dehydrogenase reactions of the TCA cycle cannot proceed.
4. (a) Citrate levels would **increase** (accumulate before block)
   (b) Isocitrate levels would **decrease** (cannot be formed)
   (c) ATP production would **decrease dramatically** (cycle blocked)

7 Electron Transport Chain and Oxidative Phosphorylation

Learning Objectives

After completing this section, you should be able to:

- Describe the location and components of the electron transport chain
- Explain the function of each complex (I-IV) and mobile electron carriers
- Describe the chemiosmotic theory and proton motive force
- Explain ATP synthesis by ATP synthase
- Calculate ATP yield from glucose oxidation
- Identify inhibitors and uncouplers of oxidative phosphorylation

7.1 Overview of Oxidative Phosphorylation

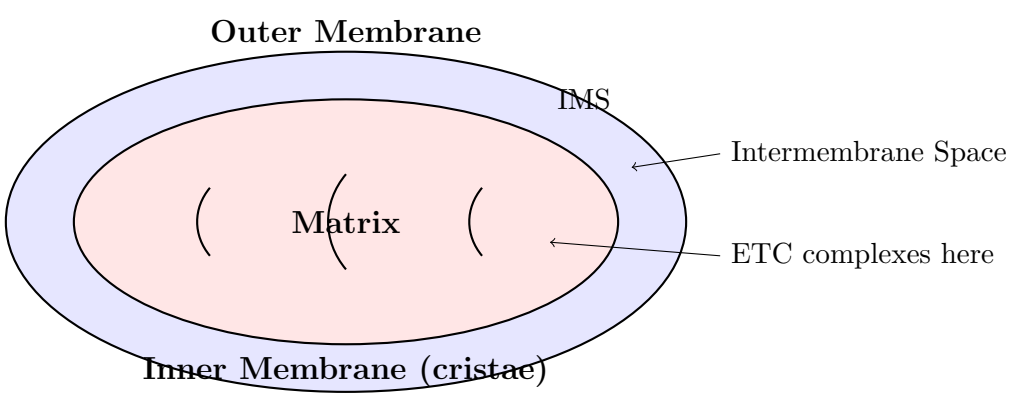
Definition

Oxidative phosphorylation is the process by which ATP is synthesized using energy released from the transfer of electrons from NADH and $FADH_2$ to oxygen through a series of electron carriers in the inner mitochondrial membrane.

Two coupled processes:

1. **Electron transport chain (ETC)**: Electrons flow from NADH/$FADH_2$ to O_2, releasing energy used to pump H^+
2. **Chemiosmosis**: H^+ flows back through ATP synthase, driving ATP synthesis

7.2 Mitochondrial Structure



Key locations:

- **Outer membrane**: Permeable to small molecules (contains porins)
- **Intermembrane space (IMS)**: Site of H^+ accumulation; similar composition to cytosol
- **Inner membrane**: Impermeable to most molecules; contains ETC complexes and ATP synthase
- **Matrix**: Contains TCA cycle enzymes, pyruvate dehydrogenase, mtDNA

7.3 Components of the Electron Transport Chain

7.3.1 Complex I: NADH-Ubiquinone Oxidoreductase

- **Also called**: NADH dehydrogenase
- **Prosthetic groups**: FMN, Fe-S clusters
- **Function**: Transfers 2 electrons from NADH to ubiquinone (CoQ)
- **H^+ pumped**: 4 H^+ per NADH

$$\text{NADH} + \text{H}^+ + \text{Q} \rightarrow \text{NAD}^+ + \text{QH}_2$$

7.3.2 Complex II: Succinate-Ubiquinone Oxidoreductase

- **Also called**: Succinate dehydrogenase (same enzyme as TCA cycle!)
- **Prosthetic groups**: FAD, Fe-S clusters
- **Function**: Transfers electrons from succinate (via $FADH_2$) to ubiquinone
- **H^+ pumped**: 0 (no proton pumping!)

$$\text{Succinate} + \text{Q} \rightarrow \text{Fumarate} + \text{QH}_2$$

Exam Experts Analogy

Complex II is like a side entrance to the ETC. Electrons from $FADH_2$ enter here, bypassing Complex I. Because they skip the first proton pump, $FADH_2$ produces less ATP than NADH—like getting on a bus at a later stop and paying a reduced fare.

7.3.3 Ubiquinone (Coenzyme Q)

- **Mobile electron carrier** in the inner membrane
- Lipid-soluble (moves freely in membrane)
- Accepts electrons from Complex I AND Complex II
- Can carry 1 or 2 electrons (semiquinone intermediate)
- Delivers electrons to Complex III

7.3.4 Complex III: Ubiquinol-Cytochrome c Oxidoreductase

- **Also called**: Cytochrome bc_1 complex
- **Prosthetic groups**: Heme b, Heme c_1, Fe-S cluster (Rieske protein)
- **Function**: Transfers electrons from QH_2 to cytochrome c via Q cycle
- **H^+ pumped**: 4 H^+ per pair of electrons

7.3.5 Cytochrome c

- **Mobile electron carrier** in intermembrane space
- Water-soluble peripheral membrane protein
- Contains heme c prosthetic group
- Carries only ONE electron at a time
- Also plays role in apoptosis when released to cytoplasm

7.3.6 Complex IV: Cytochrome c Oxidase

- **Function**: Transfers electrons from cytochrome c to O_2 (final electron acceptor)
- **Prosthetic groups**: Heme a, Heme a_3, Cu_A, Cu_B
- **H^+ pumped**: 2 H^+ per pair of electrons
- Requires 4 electrons to fully reduce O_2 to 2 H_2O

$$4 \text{ Cyt c (reduced)} + 4H^+ + O_2 \rightarrow 4 \text{ Cyt c (oxidized)} + 2H_2O$$

Summary: Electron Flow Through ETC

NADH pathway:

$$\text{NADH} \xrightarrow{\text{Complex I}} \text{Q} \xrightarrow{\text{Complex III}} \text{Cyt c} \xrightarrow{\text{Complex IV}} \text{O}_2$$

$FADH_2$ pathway:

$$\text{FADH}_2 \xrightarrow{\text{Complex II}} \text{Q} \xrightarrow{\text{Complex III}} \text{Cyt c} \xrightarrow{\text{Complex IV}} \text{O}_2$$

Total H^+ pumped:

- NADH: 4 (CI) + 4 (CIII) + 2 (CIV) = **10 H^+**
- $FADH_2$: 0 (CII) + 4 (CIII) + 2 (CIV) = **6 H^+**

7.4 Chemiosmotic Theory

Key Principle

Peter Mitchell's Chemiosmotic Theory (Nobel Prize, 1978):
The energy released during electron transport is used to pump H^+ from the matrix to the intermembrane space, creating an **electrochemical gradient** (proton motive force). The flow of H^+ back into the matrix through ATP synthase drives ATP synthesis.

Proton Motive Force (Δp) has two components:

1. **Chemical gradient** (ΔpH): Higher $[H^+]$ in IMS than matrix
   - IMS: pH $\approx$ 7.0
   - Matrix: pH $\approx$ 8.0
2. **Electrical gradient** ($\Delta\psi$): Positive charge in IMS, negative in matrix
   - Membrane potential $\approx$ 140 mV (inside negative)

$$\Delta p = \Delta\psi - \frac{2.3RT}{F}\Delta\text{pH}$$

Exam Experts Analogy

The proton motive force is like water behind a dam. The ETC pumps water (H^+) uphill to create a reservoir. The pressure of this water (proton motive force) can then be used to turn a turbine (ATP synthase) and generate electricity (ATP).

7.5 ATP Synthase (Complex V)

Definition

ATP synthase (F_0F_1 ATPase) is a molecular machine that uses the proton motive force to synthesize ATP from ADP and P_i.

Structure of ATP Synthase:

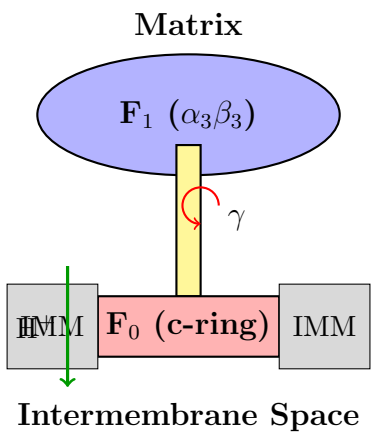
1. **F_0 portion** (membrane-embedded):
   - Contains the **c-ring** (proton channel)
   - Subunit a: provides proton half-channels
   - c-ring rotation driven by H^+ flow
   - Number of c subunits varies (8-15 depending on species)
2. **F_1 portion** (matrix-facing):
   - $\alpha_3\beta_3$ hexamer: catalytic head
   - β subunits contain catalytic sites for ATP synthesis
   - γ subunit: central rotating shaft
   - δ and ϵ subunits: connect F_0 to F_1

Key Principle

Boyer's Binding Change Mechanism (Nobel Prize, 1997):
The three β subunits cycle through three conformational states:

1. **O (Open)**: Low affinity — releases ATP
2. **L (Loose)**: Binds ADP + P_i loosely
3. **T (Tight)**: High affinity — catalyzes ATP formation

Rotation of the γ subunit (driven by H^+ flow through F_0) causes sequential conformational changes in each β subunit.
One complete rotation (360°) produces 3 ATP molecules.



7.6 ATP Yield from Complete Glucose Oxidation

Complete ATP Accounting per Glucose

Stage	ATP	NADH	$FADH_2$	ATP Equivalent
Glycolysis	2	2	—	2
Pyruvate → Acetyl-CoA	—	2	—	—
TCA Cycle (×2)	2 (GTP)	6	2	2
Subtotal	4	10	2	4
Oxidative Phosphorylation:				
10 NADH × 2.5 ATP				25
2 $FADH_2$ × 1.5 ATP				3
TOTAL				**30-32 ATP**

Note: The range (30-32) depends on which shuttle transports cytosolic NADH:

- Malate-aspartate shuttle: 2.5 ATP per NADH (total = 32)
- Glycerol-3-phosphate shuttle: 1.5 ATP per NADH (total = 30)

7.7 Shuttle Systems for Cytosolic NADH

The inner mitochondrial membrane is impermeable to NADH. Two shuttle systems transfer reducing equivalents from cytosolic NADH into the mitochondrial matrix:

7.7.1 Malate-Aspartate Shuttle

- Predominant in **liver, heart, and kidney**
- Transfers electrons to mitochondrial NAD^+
- Produces **2.5 ATP per cytosolic NADH**

- Reversible; can work in either direction

Mechanism:

1. Cytosolic OAA + NADH → Malate + NAD^+ (cytosolic malate DH)
2. Malate enters matrix via malate-αKG antiporter
3. Matrix malate + NAD^+ → OAA + NADH (mitochondrial malate DH)
4. OAA converted to aspartate (transamination)
5. Aspartate exits via glutamate-aspartate antiporter
6. Aspartate converted back to OAA in cytosol

7.7.2 Glycerol-3-Phosphate Shuttle

- Predominant in **brain and skeletal muscle**
- Transfers electrons to FAD (via mitochondrial G3P DH)
- Produces **1.5 ATP per cytosolic NADH**
- Irreversible; simpler but less efficient

Mechanism:

1. DHAP + NADH → Glycerol-3-P + NAD^+ (cytosolic G3P DH)
2. Glycerol-3-P oxidized by mitochondrial G3P DH (FAD-linked)
3. $FADH_2$ transfers electrons directly to CoQ
4. DHAP regenerated and returns to cytosol

7.8 Inhibitors of Oxidative Phosphorylation

Target	Inhibitor	Effect
Complex I	Rotenone, Amytal, MPP^+	Blocks NADH oxidation
Complex II	Malonate	Competitive inhibitor (succinate analog)
Complex III	Antimycin A	Blocks cytochrome b_{566}
Complex IV	CN^-, CO, H_2S, Azide	Binds to Fe^{3+} in cytochrome a_3
ATP Synthase	Oligomycin	Blocks F_0 proton channel

Example

Cyanide Poisoning:
Cyanide (CN^-) binds tightly to Fe^{3+} in cytochrome c oxidase (Complex IV), preventing electron transfer to O_2.
Consequences:

- ETC stops completely
- No proton gradient formed
- No ATP synthesis via oxidative phosphorylation
- Cells must rely on glycolysis alone (insufficient for high-energy tissues)
- Brain and heart fail rapidly

Treatment: Nitrites (convert Hb to MetHb, which binds CN^-) followed by thiosulfate (converts CN^- to thiocyanate for excretion)

7.9 Uncouplers of Oxidative Phosphorylation

Definition

Uncouplers are lipid-soluble weak acids that carry H^+ across the inner mitochondrial membrane, dissipating the proton gradient without ATP synthesis. Electron transport continues (even accelerates), but energy is released as **heat**.

Examples of Uncouplers:

- **2,4-Dinitrophenol (DNP)**: Historical diet pill (dangerous!)
- **FCCP/CCCP**: Laboratory uncouplers
- **Aspirin** (high doses): Weak uncoupling effect
- **Thermogenin (UCP1)**: Natural uncoupler in brown fat

Key Principle

Brown Adipose Tissue and Thermogenesis:
Brown fat contains **uncoupling protein 1 (UCP1/thermogenin)**, which allows H^+ to flow back into the matrix without passing through ATP synthase.
Function: Generate heat for:

- Thermoregulation in newborns
- Non-shivering thermogenesis in cold exposure
- Hibernating animals

Regulation: Activated by norepinephrine (sympathetic nervous system) and free fatty acids

Exam Experts Analogy

Uncouplers are like holes in a dam. The water (H^+) flows through the holes instead of the turbine (ATP synthase). The potential energy is wasted as the water crashes down (heat) rather than doing useful work (ATP).

7.10 Regulation of Oxidative Phosphorylation

Key Principle

Respiratory Control:
The rate of oxidative phosphorylation is tightly coupled to ATP demand through:

1. **Availability of ADP**: Primary regulator
   - High ADP → stimulates ATP synthase → increases ETC rate
   - Low ADP → ATP synthase slows → ETC backs up
2. **Availability of substrates**: NADH, $FADH_2$, O_2
3. **ATP/ADP ratio**: Reflects cellular energy status

Respiratory States:

- **State 3**: Active respiration (ADP present, substrate present)
- **State 4**: Resting respiration (ADP depleted, substrate present)
- **Respiratory Control Ratio (RCR)**: State 3 / State 4 rate
  - Normal: 5-10 (tightly coupled)
  - Uncoupled: approaches 1

8 Pentose Phosphate Pathway

8.1 Learning Objectives

- Describe the two phases of the pentose phosphate pathway
- Explain the production and importance of NADPH
- Understand the synthesis of ribose-5-phosphate for nucleotide biosynthesis
- Describe the regulation of the pathway
- Explain the clinical significance of G6PD deficiency

8.2 Overview and Functions

Definition

The **pentose phosphate pathway** (PPP), also called the **hexose monophosphate shunt**, is an alternative route for glucose-6-phosphate oxidation that produces:

1. **NADPH**: Reducing power for biosynthesis and antioxidant defense
2. **Ribose-5-phosphate**: Precursor for nucleotide synthesis

Location: Cytosol (same as glycolysis)
Tissues with high PPP activity:

- **Liver**: Fatty acid and cholesterol synthesis (NADPH)
- **Adipose tissue**: Fatty acid synthesis (NADPH)
- **Adrenal cortex**: Steroid hormone synthesis (NADPH)
- **Red blood cells**: Glutathione reduction (NADPH)
- **Rapidly dividing cells**: Nucleotide synthesis (ribose-5-P)

8.3 Oxidative Phase (Irreversible)

The oxidative phase produces NADPH and converts glucose-6-phosphate to ribulose-5-phosphate.

Reaction 1: Glucose-6-phosphate Dehydrogenase (G6PD)

$$\text{Glucose-6-P} + \text{NADP}^+ \xrightarrow{\text{G6PD}} \text{6-Phosphogluconolactone} + \text{NADPH} + \text{H}^+$$

- **Rate-limiting step** of the PPP
- **First NADPH produced**
- Irreversible; commits G6P to the PPP

- **Regulated by $NADP^+$/NADPH ratio**

Reaction 2: Lactonase

$$\text{6-Phosphogluconolactone} + H_2O \rightarrow \text{6-Phosphogluconate}$$

- Hydrolysis of the lactone ring
- Spontaneous but enzyme-catalyzed for speed

Reaction 3: 6-Phosphogluconate Dehydrogenase

$$\text{6-Phosphogluconate} + NADP^+ \rightarrow \text{Ribulose-5-P} + CO_2 + NADPH$$

- Oxidative decarboxylation
- **Second NADPH produced**
- Converts 6C sugar to 5C sugar (releases CO_2)

Oxidative Phase Summary

Input: Glucose-6-phosphate + 2 $NADP^+$ + H_2O
Output: Ribulose-5-phosphate + 2 NADPH + 2 H^+ + CO_2
Key points:

- Irreversible (committed pathway)
- Generates reducing power (NADPH)
- Loss of one carbon as CO_2

8.4 Non-Oxidative Phase (Reversible)

The non-oxidative phase interconverts sugars of different carbon lengths through a series of reversible reactions.

Key Enzymes:

1. **Phosphopentose Isomerase**

$$\text{Ribulose-5-P} \leftrightarrow \text{Ribose-5-P}$$

   - Ketose to aldose isomerization
   - Produces ribose-5-P for nucleotide synthesis

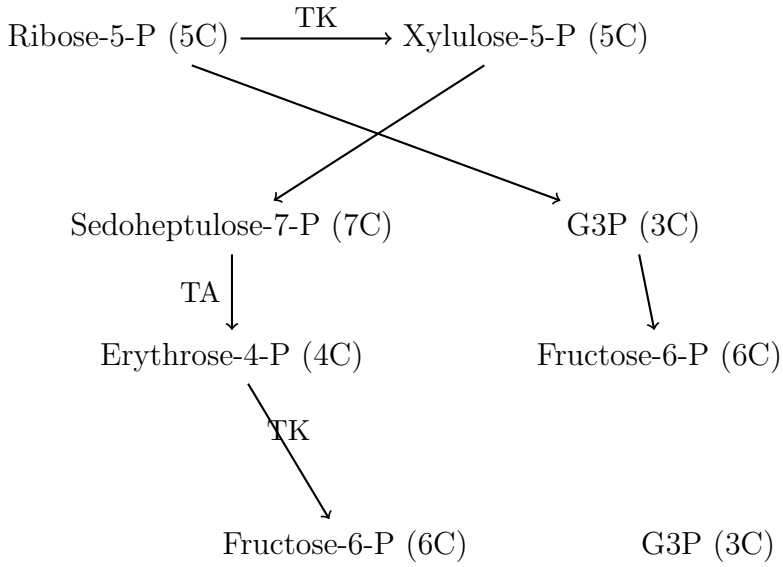
2. **Phosphopentose Epimerase**

$$\text{Ribulose-5-P} \leftrightarrow \text{Xylulose-5-P}$$

- Epimerization at C3
- Generates substrate for transketolase

3. **Transketolase** (requires TPP)
   - Transfers 2-carbon units
   - Thiamine pyrophosphate (vitamin B_1) is cofactor
4. **Transaldolase**
   - Transfers 3-carbon units
   - No cofactor required

Carbon Shuffling Summary:



Net reaction of non-oxidative phase:

$$3 \text{ Ribulose-5-P (5C)} \leftrightarrow 2 \text{ Fructose-6-P (6C)} + 1 \text{ G3P (3C)}$$

- 15 carbons in = 15 carbons out (12 + 3)
- F6P and G3P can enter glycolysis or gluconeogenesis
- Reversibility allows pathway to meet cellular needs

8.5 Modes of the Pentose Phosphate Pathway

Key Principle

The PPP operates in different modes depending on cellular needs:

Mode 1: Need NADPH and Ribose-5-P (balanced)

- Oxidative phase runs
- Ribulose-5-P → Ribose-5-P (isomerase only)
- Example: Rapidly dividing cells

Mode 2: Need NADPH only (excess ribose)

- Oxidative phase runs maximally
- Non-oxidative phase converts R5P → F6P + G3P
- F6P and G3P enter glycolysis
- Example: Adipose tissue during lipogenesis

Mode 3: Need Ribose-5-P only (no NADPH need)

- Oxidative phase bypassed
- Non-oxidative phase runs in reverse
- F6P + G3P → Ribose-5-P
- Example: Rapidly dividing cells with adequate NADPH

Mode 4: Need NADPH and ATP

- Oxidative phase produces NADPH
- R5P converted to F6P and G3P
- These enter glycolysis for ATP production
- Example: RBCs (need NADPH for glutathione, ATP for ion pumps)

8.6 Regulation of the Pentose Phosphate Pathway

- **Primary control:** G6PD (rate-limiting enzyme)
- **Activated by:** High $NADP^+$/NADPH ratio (need for NADPH)
- **Inhibited by:** High NADPH (product inhibition)

Clinical Correlation: G6PD Deficiency

G6PD deficiency is the most common enzyme deficiency worldwide, affecting over 400 million people.
Pathophysiology:

- Decreased NADPH production in RBCs
- Impaired glutathione reduction (GSH maintenance)
- RBCs vulnerable to oxidative stress
- Hemolysis upon exposure to oxidants

Triggers of hemolytic crisis:

- Fava beans (favism)
- Drugs: Primaquine, sulfonamides, dapsone, nitrofurantoin
- Infections (increased oxidative stress)
- Diabetic ketoacidosis

Clinical features:

- Acute hemolytic anemia
- Jaundice, dark urine (hemoglobinuria)
- "Bite cells" and Heinz bodies on blood smear

Inheritance: X-linked recessive (males predominantly affected)
Protective effect: Confers resistance to *Plasmodium falciparum* malaria

8.7 Functions of NADPH

1. **Reductive biosynthesis:**
   - Fatty acid synthesis
   - Cholesterol synthesis
   - Steroid hormone synthesis
   - Neurotransmitter synthesis
2. **Antioxidant defense:**
   - Regeneration of reduced glutathione (GSH)
   - Thioredoxin reduction

- Protection against reactive oxygen species

3. **Detoxification:**
   - Cytochrome P450 reactions
   - Drug metabolism in liver
4. **Immune function:**
   - NADPH oxidase in phagocytes
   - Respiratory burst (superoxide production)

Example

Glutathione System in RBCs
Red blood cells lack mitochondria and rely entirely on the PPP for NADPH production.
The glutathione cycle:

1. Glutathione peroxidase uses GSH to reduce H_2O_2:

$$2\text{GSH} + \text{H}_2\text{O}_2 \xrightarrow{\text{GPx}} \text{GSSG} + 2\text{H}_2\text{O}$$

2. Glutathione reductase regenerates GSH using NADPH:

$$\text{GSSG} + \text{NADPH} + \text{H}^+ \xrightarrow{\text{GR}} 2\text{GSH} + \text{NADP}^+$$

Without adequate NADPH (as in G6PD deficiency), GSH cannot be regenerated, and oxidative damage accumulates.

9 Glycogen Metabolism

9.1 Overview of Glycogen

Definition

Glycogen is a highly branched polymer of glucose that serves as the primary storage form of glucose in animals. It is stored mainly in:

- **Liver** (∼100g): Maintains blood glucose levels
- **Skeletal muscle** (∼400g): Fuel for muscle contraction

Structure of Glycogen:

- Linear chains: $\alpha(1\rightarrow4)$ glycosidic bonds

- Branch points: $\alpha(1\rightarrow6)$ glycosidic bonds (every 8-12 residues)
- Highly branched structure allows rapid mobilization
- Each glycogen granule contains up to 55,000 glucose residues
- **Glycogenin** protein at the core (primer for synthesis)

Exam Experts Analogy

Glycogen is like a tree with many branches. The trunk represents the core protein (glycogenin), and the branches (α-1,6 linkages) allow many "leaves" (glucose residues) to be added or removed simultaneously, enabling rapid storage or release of glucose.

9.2 Glycogenesis (Glycogen Synthesis)

Definition

Glycogenesis is the process of synthesizing glycogen from glucose. It occurs primarily in liver and muscle when glucose is abundant (fed state).

Steps of Glycogenesis:
Step 1: Glucose Phosphorylation

$$\text{Glucose} + \text{ATP} \xrightarrow{\text{Hexokinase/Glucokinase}} \text{Glucose-6-P} + \text{ADP}$$

Step 2: Isomerization

$$\text{Glucose-6-P} \xleftrightarrow{\text{Phosphoglucomutase}} \text{Glucose-1-P}$$

Step 3: Activation (UDP-glucose formation)

$$\text{Glucose-1-P} + \text{UTP} \xrightarrow{\text{UDP-glucose pyrophosphorylase}} \text{UDP-glucose} + \text{PP}_i$$

- UDP-glucose is the "activated" form of glucose
- Pyrophosphate (PP_i) hydrolysis makes reaction irreversible
- $\text{PP}_i \rightarrow 2\ \text{P}_i$ (by pyrophosphatase)

Step 4: Chain Elongation

$$\text{UDP-glucose} + \text{Glycogen}_{(n)} \xrightarrow{\text{Glycogen synthase}} \text{Glycogen}_{(n+1)} + \text{UDP}$$

- **Glycogen synthase**: Key regulatory enzyme
- Adds glucose to non-reducing end via $\alpha(1\rightarrow4)$ bond
- Requires pre-existing primer (at least 4 glucose residues)

Step 5: Branch Formation

$$\text{Linear chain} \xrightarrow{\text{Branching enzyme}} \text{Branched chain}$$

- **Branching enzyme** (glycosyl 4:6 transferase)
- Transfers 6-7 glucose block from end of chain
- Creates new $\alpha(1\rightarrow6)$ branch point
- Branch must be at least 4 residues from existing branch

Primer Synthesis (Glycogenin):

- Glycogenin is a protein that self-glucosylates
- Attaches first glucose to tyrosine residue
- Extends chain to $\sim$8 glucose residues
- Then glycogen synthase takes over

Energy Cost of Glycogenesis

Per glucose added to glycogen:

- 1 ATP (hexokinase)
- 1 UTP (UDP-glucose pyrophosphorylase)

Total: 2 ATP equivalents per glucose stored

9.3 Glycogenolysis (Glycogen Breakdown)

Definition

Glycogenolysis is the process of breaking down glycogen to release glucose (liver) or glucose-6-phosphate (muscle) for energy production.

Steps of Glycogenolysis:
Step 1: Chain Shortening

$$\text{Glycogen}_{(n)} + \text{P}_i \xrightarrow{\text{Glycogen phosphorylase}} \text{Glycogen}_{(n-1)} + \text{Glucose-1-P}$$

- **Glycogen phosphorylase**: Key regulatory enzyme
- Cleaves $\alpha(1\rightarrow4)$ bonds by phosphorolysis
- Releases glucose-1-phosphate (not free glucose)

- Stops 4 residues from branch point
- Requires pyridoxal phosphate (PLP, vitamin B_6) cofactor

Step 2: Branch Removal (Debranching Enzyme)
The debranching enzyme has two activities:

1. **Transferase activity:**
   - Moves 3 of the 4 remaining glucose residues to another chain
   - Creates new $\alpha(1\rightarrow4)$ linkage
2. **α-1,6-Glucosidase activity:**
   - Hydrolyzes the $\alpha(1\rightarrow6)$ bond
   - Releases single free glucose molecule
   - This is the only free glucose released in glycogenolysis

Step 3: Conversion to Glucose-6-P

$$\text{Glucose-1-P} \xleftrightarrow{\text{Phosphoglucomutase}} \text{Glucose-6-P}$$

Step 4: Fate of Glucose-6-P

- **In muscle:** G6P enters glycolysis (muscle lacks G6Pase)
- **In liver:** G6P $\rightarrow$ Glucose (via glucose-6-phosphatase)

$$\text{Glucose-6-P} + \ textH_2\text{O} \xrightarrow{\text{Glucose-6-phosphatase}} \text{Glucose} + \text{P}_i$$

  - Located in ER membrane
  - Allows liver to release glucose into blood

Glycogenolysis Products

Approximately 90%: Glucose-1-phosphate (from α-1,4 cleavage)
Approximately 10%: Free glucose (from α-1,6 branch points)

9.4 Regulation of Glycogen Metabolism

Glycogen metabolism is regulated by reciprocal control of glycogen synthase and glycogen phosphorylase through:

- Hormonal signals (insulin, glucagon, epinephrine)
- Allosteric effectors
- Covalent modification (phosphorylation)

Hormonal Regulation:

Hormone	Target Tissue	Glycogen Synthase	Glycogen Phosphorylase
Insulin	Liver, Muscle	Activates	Inhibits
Glucagon	Liver only	Inhibits	Activates
Epinephrine	Liver, Muscle	Inhibits	Activates

Phosphorylation States:

- **Glycogen synthase:**
  - Dephosphorylated = Active (GSa)
  - Phosphorylated = Inactive (GSb)
- **Glycogen phosphorylase:**
  - Dephosphorylated = Inactive (GPb)
  - Phosphorylated = Active (GPa)

Clinical Correlation: Glycogen Storage Diseases

Von Gierke Disease (Type I):

- Deficiency: Glucose-6-phosphatase
- Features: Severe fasting hypoglycemia, hepatomegaly, lactic acidosis
- Liver cannot release free glucose

Pompe Disease (Type II):

- Deficiency: Lysosomal α-1,4-glucosidase (acid maltase)
- Features: Cardiomegaly, muscle weakness, early death
- Glycogen accumulates in lysosomes

McArdle Disease (Type V):

- Deficiency: Muscle glycogen phosphorylase
- Features: Exercise intolerance, muscle cramps, myoglobinuria
- Cannot mobilize muscle glycogen during exercise

Cori Disease (Type III):

- Deficiency: Debranching enzyme
- Features: Milder hypoglycemia, limit dextrin accumulation
- Abnormal glycogen structure with short outer branches

10 Overview of Lipid Metabolism

Learning Objectives

After completing this chapter, students should be able to:

1. Describe the process of fatty acid oxidation (β-oxidation)
2. Calculate ATP yield from fatty acid oxidation
3. Explain fatty acid synthesis and its regulation
4. Describe ketone body metabolism and its clinical significance
5. Understand cholesterol synthesis and its regulation

Definition

Lipids serve as:

- Major energy storage molecules (triacylglycerols)
- Structural components of membranes (phospholipids)
- Signaling molecules (steroid hormones, eicosanoids)
- Insulation and protection

11 Fatty Acid Oxidation (β-Oxidation)

11.1 Overview and Location

Definition

β-Oxidation is the process of breaking down fatty acids into acetyl-CoA units for energy production. The name refers to oxidation at the β-carbon (C3) of the fatty acid.

Location: Mitochondrial matrix
When does it occur?

- Fasting state
- Prolonged exercise
- Low carbohydrate availability
- Diabetes mellitus (uncontrolled)

11.2 Activation of Fatty Acids

Before oxidation, fatty acids must be activated to acyl-CoA in the cytosol:

$$\text{Fatty acid} + \text{CoA} + \text{ATP} \xrightarrow{\text{Acyl-CoA synthetase}} \text{Acyl-CoA} + \text{AMP} + \text{PP}_i$$

- Also called fatty acid thiokinase
- Located on outer mitochondrial membrane
- Costs 2 ATP equivalents (ATP $\rightarrow$ AMP + PP_i)
- PP_i is hydrolyzed to 2P_i (makes reaction irreversible)

11.3 Carnitine Shuttle

Long-chain acyl-CoA cannot cross the inner mitochondrial membrane. The **carnitine shuttle** transports fatty acids into the matrix.

Step 1: Formation of Acylcarnitine

$$\text{Acyl-CoA} + \text{Carnitine} \xrightarrow{\text{CPT-I}} \text{Acylcarnitine} + \text{CoA}$$

- **CPT-I** (Carnitine palmitoyltransferase I)
- Located on outer mitochondrial membrane
- **Rate-limiting step** of fatty acid oxidation
- **Inhibited by malonyl-CoA** (key regulatory point)

Step 2: Transport Across Inner Membrane

$$\text{Acylcarnitine}_{(\text{cytosol})} \xleftrightarrow{\text{Translocase}} \text{Acylcarnitine}_{(\text{matrix})}$$

- Carnitine-acylcarnitine translocase
- Antiport: acylcarnitine in, carnitine out

Step 3: Regeneration of Acyl-CoA

$$\text{Acylcarnitine} + \text{CoA} \xrightarrow{\text{CPT-II}} \text{Acyl-CoA} + \text{Carnitine}$$

- **CPT-II** on inner membrane (matrix side)
- Regenerates acyl-CoA for β-oxidation
- Carnitine returns to cytosol via translocase

Clinical Correlation: Carnitine Deficiency

Primary carnitine deficiency:

- Defect in carnitine transporter
- Features: Hypoketotic hypoglycemia, cardiomyopathy, muscle weakness
- Treatment: Carnitine supplementation

CPT-I deficiency:

- Cannot transport fatty acids into mitochondria
- Presents with fasting hypoglycemia without ketosis
- Hepatomegaly common

CPT-II deficiency:

- Most common disorder of fatty acid oxidation
- Adult form: Exercise-induced muscle pain and rhabdomyolysis
- Triggered by prolonged exercise, fasting, cold exposure

11.4 The β-Oxidation Spiral

Each round of β-oxidation consists of four reactions that remove a 2-carbon unit as acetyl-CoA.

Reaction 1: Oxidation (Acyl-CoA Dehydrogenase)

$$\text{Acyl-CoA} + \text{FAD} \rightarrow \text{trans-}\Delta^2\text{-Enoyl-CoA} + \text{FADH}_2$$

- Creates double bond between C2 and C3 (trans configuration)
- **$FADH_2$ produced** $\rightarrow$ 1.5 ATP
- Multiple isozymes for different chain lengths:
  - VLCAD: Very long chain (C12-C18)
  - MCAD: Medium chain (C6-C12)
  - SCAD: Short chain (C4-C6)

Reaction 2: Hydration (Enoyl-CoA Hydratase)

$$\text{trans-}\Delta^2\text{-Enoyl-CoA} + \text{H}_2\text{O} \rightarrow \text{L-3-Hydroxyacyl-CoA}$$

- Adds water across the double bond

- Produces L-stereoisomer (important for next step)

Reaction 3: Oxidation (3-Hydroxyacyl-CoA Dehydrogenase)

$$\text{L-3-Hydroxyacyl-CoA} + \text{NAD}^+ \rightarrow \text{3-Ketoacyl-CoA} + \text{NADH} + \text{H}^+$$

- Oxidizes hydroxyl group to ketone
- **NADH produced** $\rightarrow$ 2.5 ATP

Reaction 4: Thiolysis (β-Ketothiolase)

$$\text{3-Ketoacyl-CoA} + \text{CoA-SH} \rightarrow \text{Acetyl-CoA} + \text{Acyl-CoA}_{(n-2)}$$

- Cleaves between C2 and C3 (α and β carbons)
- Releases acetyl-CoA
- Shortened acyl-CoA re-enters the spiral

Memory Aid: β-Oxidation Steps

"Oh Holy Day, That's Christmas!"

- **O**xidation (FAD $\rightarrow$ $FADH_2$)
- **H**ydration (add H_2O)
- **D**ehydrogenation ($NAD^+ \rightarrow$ NADH)
- **T**hiolysis (cleavage with CoA)

11.5 Energy Yield from Fatty Acid Oxidation

Example

Calculate the ATP yield from complete oxidation of palmitate (16:0).
Step 1: Determine number of β-oxidation cycles

- Palmitate has 16 carbons
- Number of cycles = (n/2) - 1 = (16/2) - 1 = **7 cycles**

Step 2: Products of β-oxidation

- Acetyl-CoA: 16/2 = **8 acetyl-CoA**
- $FADH_2$: 7 (one per cycle)
- NADH: 7 (one per cycle)

Step 3: Calculate ATP from each source

Source	Amount	ATP
Acetyl-CoA (TCA cycle)	8 × 10 ATP	80 ATP
$FADH_2$ (β-oxidation)	7 × 1.5 ATP	10.5 ATP
NADH (β-oxidation)	7 × 2.5 ATP	17.5 ATP
Gross total		108 ATP
Activation cost		−2 ATP
Net ATP yield		**106 ATP**

Key Principle

General Formula for Saturated Fatty Acid Oxidation:
For a saturated fatty acid with n carbons:

- Number of β-oxidation cycles = (n/2) - 1
- Acetyl-CoA produced = n/2
- $FADH_2$ produced = (n/2) - 1
- NADH from β-oxidation = (n/2) - 1

Total ATP = [n/2 × 10] + [(n/2 - 1) × 1.5] + [(n/2 - 1) × 2.5] - 2

11.6 Oxidation of Unsaturated Fatty Acids

Unsaturated fatty acids require additional enzymes:

1. Enoyl-CoA Isomerase

- Converts *cis*-Δ^3 double bonds to *trans*-Δ^2
- Required for monounsaturated fatty acids (e.g., oleate)
- No ATP or reducing equivalents involved

2. 2,4-Dienoyl-CoA Reductase

- Required for polyunsaturated fatty acids
- Uses NADPH as reducing agent
- Reduces conjugated double bond system

ATP Yield from Unsaturated Fatty Acids

Each double bond in a fatty acid results in:

- Loss of one $FADH_2$ (1.5 ATP less)
- The oxidation step that would create that double bond is bypassed

Example: Oleate (18:1Δ^9)

- Saturated 18-carbon would yield: 120 - 2 = 118 ATP
- Oleate yields: 118 - 1.5 = **116.5 ATP**

11.7 Oxidation of Odd-Chain Fatty Acids

Odd-chain fatty acids (from bacterial sources, some plants) produce propionyl-CoA in the final thiolysis step.

Conversion of Propionyl-CoA to Succinyl-CoA:

Step 1: Carboxylation

$$\text{Propionyl-CoA} + CO_2 + \text{ATP} \xrightarrow{\text{Propionyl-CoA carboxylase}} \text{D-Methylmalonyl-CoA}$$

- Requires **biotin** as cofactor
- Consumes 1 ATP

Step 2: Racemization

$$\text{D-Methylmalonyl-CoA} \xleftrightarrow{\text{Racemase}} \text{L-Methylmalonyl-CoA}$$

Step 3: Isomerization

$$\text{L-Methylmalonyl-CoA} \xrightarrow{\text{Methylmalonyl-CoA mutase}} \text{Succinyl-CoA}$$

- Requires **vitamin B_{12}** (adenosylcobalamin)
- Succinyl-CoA enters TCA cycle
- This is an **anaplerotic reaction**

Clinical Correlation: Methylmalonic Acidemia

Causes:

- Methylmalonyl-CoA mutase deficiency
- Vitamin B_{12} deficiency

Features:

- Elevated methylmalonic acid in blood and urine
- Metabolic acidosis
- Developmental delay, lethargy
- Distinguishes B_{12} deficiency from folate deficiency

12 Fatty Acid Synthesis

12.1 Overview

Definition

Fatty acid synthesis (lipogenesis) is the process of building fatty acids from acetyl-CoA units. It is essentially the reverse of β-oxidation but uses different enzymes and occurs in a different cellular location.

Feature	β-Oxidation	Fatty Acid Synthesis
Location	Mitochondrial matrix	Cytosol
Acyl carrier	CoA	ACP (acyl carrier protein)
Reducing agent	FAD, NAD^+	NADPH
2-Carbon donor	—	Malonyl-CoA
Key enzyme	Multiple enzymes	Fatty acid synthase (FAS)
Direction	C $\rightarrow$ N-terminus	N-terminus $\rightarrow$ C-terminus
Energy yield	Produces ATP	Consumes ATP

Primary sites of fatty acid synthesis:

- **Liver**: Major site; exports fat as VLDL
- **Adipose tissue**: Storage of synthesized fat

- **Lactating mammary gland**: Milk fat production
- **Brain**: Limited synthesis for myelin

12.2 Transport of Acetyl-CoA to Cytosol

Acetyl-CoA is produced in mitochondria but fatty acid synthesis occurs in the cytosol. The **citrate shuttle** transports acetyl-CoA equivalents across the inner mitochondrial membrane.

Steps of the Citrate Shuttle:

Step 1: Citrate Formation (Mitochondria)

$$\text{Acetyl-CoA} + \text{OAA} \xrightarrow{\text{Citrate synthase}} \text{Citrate}$$

Step 2: Transport

- Citrate exits mitochondria via tricarboxylate transporter
- Exchange for malate

Step 3: Citrate Cleavage (Cytosol)

$$\text{Citrate} + \text{ATP} + \text{CoA} \xrightarrow{\text{ATP-citrate lyase}} \text{Acetyl-CoA} + \text{OAA} + \text{ADP} + \text{P}_i$$

- Key regulatory enzyme
- Activated by insulin
- Inhibited by glucagon

Step 4: OAA Return to Mitochondria

Two possible routes:

Route A (generates cytosolic NADPH):

$$\text{OAA} \xrightarrow{\text{Malate DH}} \text{Malate} \xrightarrow{\text{Malic enzyme}} \text{Pyruvate} + \text{CO}_2 + \text{NADPH}$$

Route B:

$$\text{OAA} \xrightarrow{\text{Malate DH}} \text{Malate} \rightarrow \text{(enters mitochondria)}$$

NADPH Sources for Fatty Acid Synthesis

Each palmitate synthesis requires **14 NADPH**:

1. **Pentose phosphate pathway**: Primary source (oxidative phase)
2. **Malic enzyme**: Cytosolic; from citrate shuttle
3. **Cytosolic isocitrate dehydrogenase**: Minor contribution

12.3 Synthesis of Malonyl-CoA

Definition

The committed step of fatty acid synthesis is the carboxylation of acetyl-CoA to form **malonyl-CoA**, catalyzed by **acetyl-CoA carboxylase (ACC)**.

$$\text{Acetyl-CoA} + CO_2 + \text{ATP} \xrightarrow{\text{ACC}} \text{Malonyl-CoA} + \text{ADP} + P_i$$

Properties of ACC:

- Requires **biotin** as cofactor
- Two-step reaction:
  1. Biotin carboxylation: Biotin + CO_2 + ATP → Carboxybiotin
  2. Carboxyl transfer: Carboxybiotin + Acetyl-CoA → Malonyl-CoA
- **Rate-limiting enzyme** of fatty acid synthesis

Regulation of ACC:

Activators	Inhibitors
Citrate (allosteric)	Palmitoyl-CoA (product inhibition)
Insulin (dephosphorylation)	Glucagon (phosphorylation via PKA)
High carbohydrate diet	Epinephrine (phosphorylation)
	AMP (via AMPK phosphorylation)

- **Active form**: Dephosphorylated polymer
- **Inactive form**: Phosphorylated monomer
- Citrate promotes polymerization (activation)
- Palmitoyl-CoA promotes depolymerization (inhibition)

12.4 Fatty Acid Synthase (FAS)

Definition

Fatty acid synthase is a large, multifunctional enzyme complex that catalyzes all remaining steps of palmitate synthesis. In mammals, it is a homodimer with each monomer containing all seven enzymatic activities.

Domain Organization of FAS:

1. **Acetyl-CoA-ACP transacylase (AT)**

2. **Malonyl-CoA-ACP transacylase (MT)**
3. **β-Ketoacyl-ACP synthase (KS)** — condensing enzyme
4. **β-Ketoacyl-ACP reductase (KR)**
5. **β-Hydroxyacyl-ACP dehydratase (DH)**
6. **Enoyl-ACP reductase (ER)**
7. **Thioesterase (TE)**

Acyl Carrier Protein (ACP):

- Contains phosphopantetheine prosthetic group (from vitamin B_5)
- Same prosthetic group as CoA
- Holds growing fatty acid chain during synthesis

12.5 Reactions of Fatty Acid Synthesis

Loading Reactions:

Step 1: Acetyl Loading

$$\text{Acetyl-CoA} + \text{ACP-SH} \xrightarrow{\text{AT}} \text{Acetyl-ACP} + \text{CoA-SH}$$

Then transferred to KS domain:

$$\text{Acetyl-ACP} + \text{KS-SH} \rightarrow \text{Acetyl-KS} + \text{ACP-SH}$$

Step 2: Malonyl Loading

$$\text{Malonyl-CoA} + \text{ACP-SH} \xrightarrow{\text{MT}} \text{Malonyl-ACP} + \text{CoA-SH}$$

Elongation Cycle (repeated 7 times):

Reaction 1: Condensation (β-Ketoacyl-ACP Synthase)

$$\text{Acetyl-KS} + \text{Malonyl-ACP} \rightarrow \text{Acetoacetyl-ACP} + CO_2 + \text{KS-SH}$$

- Releases CO_2 (same CO_2 added by ACC)
- Decarboxylation drives reaction forward
- Forms β-ketoacyl intermediate

Reaction 2: Reduction (β-Ketoacyl-ACP Reductase)

$$\text{Acetoacetyl-ACP} + \text{NADPH} + H^+ \rightarrow \text{D-}\beta\text{-Hydroxybutyryl-ACP} + \text{NADP}^+$$

- Reduces ketone to hydroxyl

- Uses NADPH (not NADH)
- Produces D-isomer (opposite of β-oxidation)

Reaction 3: Dehydration (β-Hydroxyacyl-ACP Dehydratase)

$$\text{D-}\beta\text{-Hydroxybutyryl-ACP} \rightarrow \text{Crotonyl-ACP} + H_2O$$

- Removes water
- Forms trans double bond (enoyl intermediate)

Reaction 4: Reduction (Enoyl-ACP Reductase)

$$\text{Crotonyl-ACP} + \text{NADPH} + H^+ \rightarrow \text{Butyryl-ACP} + \text{NADP}^+$$

- Reduces double bond
- Second NADPH used per cycle
- Product is saturated acyl-ACP (4 carbons)

Cycle Continuation:

- Butyryl group transferred to KS domain
- New malonyl-CoA loaded onto ACP
- Cycle repeats, adding 2 carbons each time
- After 7 cycles: 16-carbon palmitoyl-ACP

Termination: Thioesterase

$$\text{Palmitoyl-ACP} + H_2O \xrightarrow{\text{TE}} \text{Palmitate} + \text{ACP-SH}$$

- Hydrolyzes thioester bond
- Releases free palmitate (16:0)
- Thioesterase is specific for 16-carbon chains

12.6 Stoichiometry of Palmitate Synthesis

Overall Reaction for Palmitate Synthesis

$$8 \text{ Acetyl-CoA} + 7 \text{ ATP} + 14 \text{ NADPH} + 14 \text{ H}^+ \rightarrow$$
$$\text{Palmitate} + 8 \text{ CoA} + 7 \text{ ADP} + 7 \text{ P}_i + 14 \text{ NADP}^+ + 6 \text{ H}_2\text{O}$$

Accounting:

- 8 acetyl-CoA provides 16 carbons
- 7 ATP used to make 7 malonyl-CoA
- 14 NADPH: 2 per elongation cycle $\times$ 7 cycles
- 7 CO_2 added then removed (net zero)

12.7 Elongation and Desaturation

Fatty Acid Elongation:

Palmitate can be elongated to longer fatty acids:

- **Endoplasmic reticulum**: Primary site
  - Uses malonyl-CoA as 2-carbon donor
  - Uses NADPH as reductant
  - Similar reactions to FAS but separate enzymes
- **Mitochondria**: Minor pathway
  - Uses acetyl-CoA directly
  - Reverse of β-oxidation

Fatty Acid Desaturation:

Definition

Desaturases introduce double bonds into fatty acid chains. Mammals have $\Delta 9$, $\Delta 6$, $\Delta 5$, and $\Delta 4$ desaturases but **cannot** introduce double bonds beyond C-9.

Δ9-Desaturase (Stearoyl-CoA Desaturase):

$$\text{Stearoyl-CoA} + \text{NADH} + \text{H}^+ + \text{O}_2 \rightarrow \text{Oleoyl-CoA} + \text{NAD}^+ + 2\text{H}_2\text{O}$$

- Located in ER membrane
- Requires cytochrome b_5 and cytochrome b_5 reductase

- Mixed-function oxidase (uses O_2)
- Creates cis double bond at $\Delta 9$ position
- Converts stearate (18:0) to oleate (18:1Δ9)

Essential Fatty Acids

Because humans lack $\Delta 12$ and $\Delta 15$ desaturases, we cannot synthesize:
ω-6 family:

- Linoleic acid (18:2Δ9,12) — essential
- Arachidonic acid (20:4Δ5,8,11,14) — conditionally essential

ω-3 family:

- α-Linolenic acid (18:3Δ9,12,15) — essential
- EPA (20:5) and DHA (22:6) — conditionally essential

These must be obtained from the diet!

12.8 Regulation of Fatty Acid Synthesis

Short-term Regulation (minutes):

1. **Allosteric regulation of ACC:**
   - Citrate activates (indicates abundant acetyl-CoA)
   - Palmitoyl-CoA inhibits (product feedback)
2. **Covalent modification of ACC:**
   - **Insulin**: Activates phosphatase → dephosphorylates ACC → active
   - **Glucagon/Epinephrine**: Activates PKA → phosphorylates ACC → inactive
   - **AMPK**: Phosphorylates ACC when ATP is low → inactive

Long-term Regulation (hours to days):

1. **Transcriptional control:**
   - **SREBP-1c**: Activated by insulin; induces ACC, FAS, ATP-citrate lyase
   - **ChREBP**: Activated by glucose; induces lipogenic enzymes
2. **Nutritional state:**
   - High carbohydrate diet: Increases lipogenic enzyme synthesis
   - Fasting/starvation: Decreases lipogenic enzyme synthesis
   - High fat diet: Decreases lipogenic enzymes

13 Ketone Body Metabolism

13.1 Overview of Ketogenesis

Definition

Ketone bodies are water-soluble molecules produced in the liver from acetyl-CoA during periods of low carbohydrate availability. They serve as an alternative fuel source for extrahepatic tissues, especially the brain.

The three ketone bodies:

1. **Acetoacetate** — primary ketone body
2. **β-Hydroxybutyrate** (D-3-hydroxybutyrate) — most abundant in blood
3. **Acetone** — produced by spontaneous decarboxylation; exhaled

Conditions favoring ketogenesis:

- Prolonged fasting/starvation
- Uncontrolled diabetes mellitus
- Very low carbohydrate (ketogenic) diets
- Alcoholism
- High-intensity prolonged exercise

Why Ketone Bodies?

Problem: During fasting, the brain needs fuel but:

- Cannot use fatty acids (don't cross blood-brain barrier efficiently)
- Glucose supply is limited (gluconeogenesis is slow)

Solution: Liver converts fatty acids to ketone bodies, which:

- Are water-soluble (cross blood-brain barrier)
- Spare glucose for tissues that absolutely require it (RBCs)
- Reduce protein breakdown for gluconeogenesis

13.2 Pathway of Ketogenesis

Location: Mitochondrial matrix of **hepatocytes only**

Substrate: Acetyl-CoA from β-oxidation of fatty acids

Step 1: Formation of Acetoacetyl-CoA

$$2 \text{ Acetyl-CoA} \xrightarrow{\text{Thiolase}} \text{Acetoacetyl-CoA} + \text{CoA-SH}$$

- Reverse of the last step of β-oxidation
- Condensation of two acetyl-CoA molecules

Step 2: Formation of HMG-CoA

$$\text{Acetoacetyl-CoA} + \text{Acetyl-CoA} + H_2O \xrightarrow{\text{HMG-CoA synthase}} \text{HMG-CoA} + \text{CoA-SH}$$

- **HMG-CoA synthase**: Rate-limiting enzyme of ketogenesis
- HMG-CoA = 3-hydroxy-3-methylglutaryl-CoA
- This is the **mitochondrial** HMG-CoA synthase (distinct from cytosolic enzyme in cholesterol synthesis)

Step 3: Cleavage to Acetoacetate

$$\text{HMG-CoA} \xrightarrow{\text{HMG-CoA lyase}} \text{Acetoacetate} + \text{Acetyl-CoA}$$

- Releases the first ketone body (acetoacetate)
- Acetyl-CoA is recycled

Step 4a: Reduction to β-Hydroxybutyrate

$$\text{Acetoacetate} + \text{NADH} + H^+ \xleftrightarrow{\beta\text{-hydroxybutyrate DH}} \beta\text{-Hydroxybutyrate} + NAD^+$$

- Reversible reaction
- Ratio of β-hydroxybutyrate to acetoacetate reflects mitochondrial NADH/NAD^+ ratio
- Typically 3:1 under normal conditions; higher in severe ketosis

Step 4b: Spontaneous Decarboxylation to Acetone

$$\text{Acetoacetate} \xrightarrow{\text{spontaneous}} \text{Acetone} + CO_2$$

- Non-enzymatic reaction
- Acetone is volatile; exhaled through lungs
- Causes "fruity breath" odor in diabetic ketoacidosis

Summary of Ketogenesis Pathway:

$$\text{Fatty Acids} \xrightarrow{\beta\text{-oxidation}} \text{Acetyl-CoA} \xrightarrow{\text{Thiolase}} \text{Acetoacetyl-CoA}$$

$$\downarrow \text{ HMG-CoA synthase}$$

$$\text{HMG-CoA} \xrightarrow{\text{HMG-CoA lyase}} \text{Acetoacetate}$$

$$\swarrow \qquad \searrow$$

$$\beta\text{-Hydroxybutyrate} \qquad \text{Acetone}$$

13.3 Ketolysis (Ketone Body Utilization)

Definition

Ketolysis is the process by which extrahepatic tissues convert ketone bodies back to acetyl-CoA for energy production via the TCA cycle.

Tissues that use ketone bodies:

- **Brain**: Major consumer during prolonged fasting (up to 75% of energy needs)
- **Heart**: Prefers ketone bodies over glucose
- **Skeletal muscle**: Uses ketones during fasting
- **Kidney cortex**: Active ketone consumer

Tissues that CANNOT use ketone bodies:

- **Liver**: Lacks succinyl-CoA:acetoacetate CoA transferase (SCOT)
- **Red blood cells**: No mitochondria

Steps of Ketolysis:
Step 1: Oxidation of β-Hydroxybutyrate

$$\beta\text{-Hydroxybutyrate} + \text{NAD}^+ \xrightarrow{\beta\text{-hydroxybutyrate DH}} \text{Acetoacetate} + \text{NADH}$$

Step 2: Activation of Acetoacetate

$$\text{Acetoacetate} + \text{Succinyl-CoA} \xrightarrow{\text{SCOT}} \text{Acetoacetyl-CoA} + \text{Succinate}$$

- **SCOT** = Succinyl-CoA:acetoacetate CoA transferase (thiophorase)
- Key enzyme absent in liver (prevents futile cycle)
- Uses succinyl-CoA from TCA cycle as CoA donor

Step 3: Thiolytic Cleavage

$$\text{Acetoacetyl-CoA} + \text{CoA-SH} \xrightarrow{\text{Thiolase}} 2\ \text{Acetyl-CoA}$$

Step 4: Entry into TCA Cycle

$$2\ \text{Acetyl-CoA} \xrightarrow{\text{TCA cycle}} \text{ATP production}$$

ATP Yield from Ketone Bodies

From β-Hydroxybutyrate:

- 1 NADH (from β-HB dehydrogenase) = 2.5 ATP
- 2 Acetyl-CoA $\times$ 10 ATP = 20 ATP
- Minus 1 GTP equivalent (succinyl-CoA used for activation) = -1 ATP
- **Total: 21.5 ATP**

From Acetoacetate:

- 2 Acetyl-CoA $\times$ 10 ATP = 20 ATP
- Minus 1 GTP equivalent = -1 ATP
- **Total: 19 ATP**

13.4 Regulation of Ketogenesis

Factors promoting ketogenesis:

1. **Increased fatty acid supply to liver:**
   - Low insulin $\rightarrow$ increased lipolysis in adipose tissue
   - High glucagon/epinephrine $\rightarrow$ activated hormone-sensitive lipase
2. **Increased β-oxidation:**
   - Low malonyl-CoA (glucagon effect) $\rightarrow$ active CPT-I
   - More fatty acids enter mitochondria
3. **Decreased TCA cycle activity:**
   - Oxaloacetate diverted to gluconeogenesis
   - Acetyl-CoA accumulates and enters ketogenesis
4. **High NADH/NAD$^+$ ratio:**
   - From excessive β-oxidation

- Inhibits isocitrate dehydrogenase and α-ketoglutarate dehydrogenase
- Slows TCA cycle, promoting ketone body formation

Clinical Correlation: Diabetic Ketoacidosis (DKA)

Pathophysiology:

- Insulin deficiency (Type 1 DM) or severe insulin resistance
- Unopposed glucagon action
- Massive lipolysis $\rightarrow$ excess fatty acids to liver
- Overwhelming ketone body production

Clinical Features:

- Hyperglycemia (>250 mg/dL)
- Metabolic acidosis (pH $<$ 7.3, bicarbonate $<$ 18 mEq/L)
- Ketonemia and ketonuria
- Fruity breath (acetone)
- Kussmaul breathing (deep, rapid respirations)
- Dehydration, altered mental status

Treatment:

- IV fluids (correct dehydration)
- IV insulin (stop ketogenesis)
- Potassium replacement (insulin drives K^+ into cells)
- Monitor and correct electrolytes

13.5 Starvation Ketosis vs. Diabetic Ketoacidosis

Feature	Starvation Ketosis	DKA
Insulin level	Low but present	Absent or very low
Glucagon level	Elevated	Very elevated
Blood glucose	Low to normal	High (>250 mg/dL)
Ketone levels	Moderate (2-5 mM)	Very high (>10 mM)
pH	Mildly decreased	Severely decreased (<7.3)
Severity	Self-limited	Life-threatening

14 Cholesterol Metabolism

14.1 Overview of Cholesterol

Definition

Cholesterol is a 27-carbon sterol that serves as:

- Structural component of cell membranes (modulates fluidity)
- Precursor for steroid hormones, bile acids, and vitamin D
- Component of lipoproteins

Sources of cholesterol:

- **Endogenous synthesis**: ∼800-1000 mg/day (primarily liver)
- **Dietary intake**: ∼300-500 mg/day

Fates of cholesterol:

- Membrane synthesis
- Bile acid synthesis (primary excretion route)
- Steroid hormone synthesis
- Vitamin D synthesis
- Esterification for storage/transport

14.2 Cholesterol Biosynthesis

Location: Cytosol and endoplasmic reticulum
Primary tissue: Liver (major site); also intestine, adrenal cortex, gonads
Precursor: Acetyl-CoA (from carbohydrates, amino acids, or fatty acids)
The pathway can be divided into four stages:

14.2.1 Stage 1: Synthesis of Mevalonate

Step 1: Formation of Acetoacetyl-CoA

$$2\ \text{Acetyl-CoA} \xrightarrow{\text{Thiolase}} \text{Acetoacetyl-CoA} + \text{CoA-SH}$$

Step 2: Formation of HMG-CoA

$$\text{Acetoacetyl-CoA} + \text{Acetyl-CoA} \xrightarrow{\text{HMG-CoA synthase}} \text{HMG-CoA}$$

- This is the **cytosolic** HMG-CoA synthase

- Different from mitochondrial enzyme (ketogenesis)

Step 3: Formation of Mevalonate (RATE-LIMITING STEP)

$$\text{HMG-CoA} + 2\text{NADPH} + 2\text{H}^+ \xrightarrow{\textbf{HMG-CoA reductase}} \text{Mevalonate} + 2\text{NADP}^+ + \text{CoA-SH}$$

HMG-CoA Reductase: Key Regulatory Enzyme

- **Rate-limiting enzyme** of cholesterol synthesis
- Located in ER membrane
- Target of **statin drugs** (competitive inhibitors)
- Regulated by multiple mechanisms (see regulation section)

14.2.2 Stage 2: Conversion of Mevalonate to Isopentenyl Pyrophosphate

Three ATP-dependent reactions convert mevalonate (C6) to isopentenyl pyrophosphate (C5), the basic 5-carbon isoprenoid unit.

$$\text{Mevalonate} \xrightarrow{\text{3 ATP}} \text{Isopentenyl pyrophosphate (IPP)} + \text{CO}_2$$

IPP can isomerize to dimethylallyl pyrophosphate (DMAPP).

14.2.3 Stage 3: Condensation of Isoprene Units

Formation of Geranyl Pyrophosphate (C10):

$$\text{IPP (C5)} + \text{DMAPP (C5)} \rightarrow \text{Geranyl-PP (C10)}$$

Formation of Farnesyl Pyrophosphate (C15):

$$\text{Geranyl-PP (C10)} + \text{IPP (C5)} \rightarrow \text{Farnesyl-PP (C15)}$$

Formation of Squalene (C30):

$$2\ \text{Farnesyl-PP (C15)} \xrightarrow{\text{Squalene synthase}} \text{Squalene (C30)} + 2\text{PP}_i$$

- Head-to-head condensation
- Requires NADPH
- First committed step to cholesterol (farnesyl-PP has other fates)

14.2.4 Stage 4: Cyclization of Squalene to Cholesterol

Squalene Epoxidation:

$$\text{Squalene} + O_2 + \text{NADPH} \xrightarrow{\text{Squalene monooxygenase}} \text{Squalene-2,3-epoxide}$$

Cyclization:

$$\text{Squalene-2,3-epoxide} \xrightarrow{\text{Oxidos qualene cyclase}} \text{Lanosterol}$$

Conversion to Cholesterol:

$$\text{Lanosterol} \xrightarrow{\text{19 reactions}} \text{Cholesterol}$$

- Removal of 3 methyl groups
- Reduction of double bond in side chain
- Migration of double bond from C8 to C5

14.3 Regulation of Cholesterol Synthesis

Cholesterol synthesis is tightly regulated at multiple levels, primarily targeting HMG-CoA reductase.

1. Transcriptional Regulation (SREBP System):

SREBP-2 Pathway

When cholesterol is LOW:

1. SREBP-2 (Sterol Regulatory Element Binding Protein) is in ER membrane
2. SCAP (SREBP Cleavage-Activating Protein) escorts SREBP to Golgi
3. Site-1 and Site-2 proteases cleave SREBP
4. Active SREBP fragment enters nucleus
5. Activates transcription of HMG-CoA reductase and LDL receptor genes

When cholesterol is HIGH:

1. Cholesterol binds to SCAP
2. SCAP conformation changes; binds to Insig (ER retention protein)
3. SREBP-SCAP complex retained in ER
4. No transcription of cholesterol synthesis genes

2. Translational Regulation:

- High cholesterol decreases HMG-CoA reductase mRNA translation
- Mediated by sequences in the 5' UTR of mRNA

3. Protein Degradation:

- High cholesterol/oxysterols accelerate degradation of HMG-CoA reductase
- Insig-mediated ubiquitination and proteasomal degradation
- Half-life can decrease from ¿12 hours to ¡1 hour

4. Covalent Modification:

- **AMPK** phosphorylates HMG-CoA reductase → **inactive**
- Occurs when cellular energy is low
- Prevents ATP consumption for cholesterol synthesis during energy deficit

5. Hormonal Regulation:

- **Insulin**: Activates phosphatase → dephosphorylates enzyme → active
- **Glucagon**: Activates AMPK → phosphorylates enzyme → inactive
- **Thyroid hormone**: Increases LDL receptor expression

14.4 Cholesterol Transport: Lipoproteins

Definition

Lipoproteins are spherical particles that transport hydrophobic lipids (cholesterol, triglycerides) through the aqueous bloodstream. They consist of a hydrophobic core surrounded by an amphipathic shell.

Structure of Lipoproteins:

- **Core**: Triglycerides and cholesteryl esters (hydrophobic)
- **Shell**: Phospholipids, free cholesterol, apolipoproteins (amphipathic)

Lipoprotein	Density	Main Lipid	Source	Key Apo	Function
Chylomicron	Lowest	TG (dietary)	Intestine	B-48, C-II, E	Dietary lipid transport
VLDL	Very low	TG (hepatic)	Liver	B-100, C-II, E	Hepatic TG export
IDL	Intermediate	TG/Chol	VLDL remnant	B-100, E	VLDL → LDL
LDL	Low	Cholesterol	IDL	B-100	Chol delivery to tissues
HDL	Highest	Cholesterol	Liver/Intestine	A-I, A-II	Reverse chol transport

14.5 Lipoprotein Metabolism Pathways

14.5.1 Exogenous Pathway (Dietary Lipids)

1. **Chylomicron assembly** in intestinal enterocytes
   - Dietary TG and cholesterol packaged with apoB-48
   - Secreted into lymphatics → thoracic duct → bloodstream
2. **Acquisition of apolipoproteins**
   - HDL donates apoC-II and apoE to nascent chylomicrons
3. **Triglyceride hydrolysis**
   - **Lipoprotein lipase (LPL)** on capillary endothelium
   - Activated by apoC-II
   - Releases fatty acids for tissue uptake
   - Chylomicron shrinks to **chylomicron remnant**
4. **Remnant uptake by liver**
   - ApoE recognized by hepatic remnant receptors
   - Receptor-mediated endocytosis

14.5.2 Endogenous Pathway (Hepatic Lipids)

1. **VLDL assembly** in liver
   - Hepatic TG and cholesterol packaged with apoB-100
   - Requires microsomal triglyceride transfer protein (MTP)
2. **VLDL secretion and modification**
   - Acquires apoC-II and apoE from HDL
   - LPL hydrolyzes triglycerides
   - VLDL → IDL (intermediate-density lipoprotein)
3. **IDL fate**
   - 50% taken up by liver via apoE
   - 50% converted to LDL by hepatic lipase
4. **LDL metabolism**
   - Main carrier of cholesterol to peripheral tissues
   - Taken up via LDL receptor (recognizes apoB-100)
   - Receptor-mediated endocytosis

14.5.3 Reverse Cholesterol Transport (HDL Pathway)

1. **Nascent HDL formation**
   - Liver and intestine secrete apoA-I
   - Forms discoidal particles with phospholipids
2. **Cholesterol acquisition**
   - **ABCA1 transporter** on peripheral cells effluxes cholesterol
   - Free cholesterol transferred to nascent HDL
3. **Cholesterol esterification**
   - **LCAT** (lecithin-cholesterol acyltransferase) on HDL
   - Activated by apoA-I
   - Esterifies cholesterol → moves to hydrophobic core
   - Discoidal HDL → spherical HDL_3 → HDL_2
4. **Cholesterol delivery to liver**
   - Direct: SR-B1 receptor on liver (selective uptake)
   - Indirect: CETP transfers cholesteryl esters to VLDL/LDL

Clinical Significance of Lipoproteins

Atherogenic:

- LDL ("bad cholesterol"): Deposits cholesterol in arterial walls
- Oxidized LDL: Taken up by macrophages → foam cells → atherosclerosis

Atheroprotective:

- HDL ("good cholesterol"): Removes cholesterol from arteries
- Higher HDL associated with lower cardiovascular risk

14.6 Bile Acid Synthesis

Definition

Bile acids are amphipathic molecules synthesized from cholesterol in the liver. They are essential for dietary lipid digestion and absorption, and represent the major route of cholesterol elimination.

Functions of Bile Acids:

- Emulsification of dietary fats
- Solubilization of cholesterol in bile
- Facilitation of fat-soluble vitamin absorption
- Elimination of excess cholesterol

Primary Bile Acids (synthesized in liver):

- **Cholic acid** (3α, 7α, 12α-trihydroxy)
- **Chenodeoxycholic acid** (3α, 7α-dihydroxy)

Rate-Limiting Step:

$$\text{Cholesterol} \xrightarrow{\textbf{Cholesterol 7}\alpha\textbf{-hydroxylase (CYP7A1)}} 7\alpha\text{-Hydroxycholesterol}$$

- Cytochrome P450 enzyme in liver ER
- Inhibited by bile acids (negative feedback)
- Induced by cholesterol

Conjugation:

- Bile acids conjugated with glycine or taurine
- Forms glycocholic acid, taurocholic acid, etc.
- Conjugation increases water solubility and decreases pKa
- Conjugated bile acids = bile salts

Secondary Bile Acids (formed by intestinal bacteria):

- Cholic acid $\rightarrow$ **Deoxycholic acid**
- Chenodeoxycholic acid $\rightarrow$ **Lithocholic acid**
- Formed by 7α-dehydroxylation

Enterohepatic Circulation:

- 95% of bile acids reabsorbed in terminal ileum
- Return to liver via portal circulation
- Re-secreted into bile
- Pool cycles 6-10 times per day
- Only 5% lost in feces (major route of cholesterol elimination)

Clinical Application: Bile Acid Sequestrants

Drugs: Cholestyramine, colestipol, colesevelam
Mechanism:

- Bind bile acids in intestine
- Prevent reabsorption
- Increase fecal bile acid excretion
- Liver upregulates bile acid synthesis from cholesterol
- Upregulates LDL receptors to obtain more cholesterol
- Net effect: Decreased plasma LDL cholesterol

14.7 Steroid Hormone Synthesis

Cholesterol serves as the precursor for all steroid hormones, synthesized in the adrenal cortex, gonads, and placenta.

Rate-Limiting Step:

$$\text{Cholesterol} \xrightarrow{\textbf{Cholesterol desmolase (CYP11A1)}} \text{Pregnenolone}$$

- Located in inner mitochondrial membrane
- Also called side-chain cleavage enzyme (SCC)
- Removes 6-carbon side chain
- Stimulated by ACTH (adrenal) and LH (gonads)
- Requires StAR protein for cholesterol transport into mitochondria

Classes of Steroid Hormones:

Class	Examples	Site of Synthesis
Glucocorticoids	Cortisol	Adrenal cortex (zona fasciculata)
Mineralocorticoids	Aldosterone	Adrenal cortex (zona glomerulosa)
Androgens	Testosterone, DHEA	Adrenal cortex, testes
Estrogens	Estradiol	Ovaries, placenta
Progestogens	Progesterone	Ovaries, placenta

Clinical Correlation: Congenital Adrenal Hyperplasia (CAH)

Most common cause: 21-Hydroxylase deficiency (CYP21A2)
Pathophysiology:

- Cannot synthesize cortisol or aldosterone
- Precursors shunted to androgen pathway
- Elevated ACTH (no cortisol feedback)
- Adrenal hyperplasia

Clinical features:

- Virilization of female infants (ambiguous genitalia)
- Salt-wasting crisis in severe cases
- Precocious puberty in males
- Elevated 17-hydroxyprogesterone (diagnostic)

Treatment: Glucocorticoid and mineralocorticoid replacement

15 Amino Acid Metabolism

15.1 Overview of Amino Acid Metabolism

Definition

Amino acid metabolism encompasses the pathways for synthesis, degradation, and interconversion of amino acids, as well as the disposal of their nitrogen component.

Fates of Amino Acids:

1. **Protein synthesis**: Primary use
2. **Precursors for specialized compounds**: Neurotransmitters, hormones, heme, nucleotides
3. **Energy production**: Carbon skeletons enter TCA cycle
4. **Gluconeogenesis**: Glucogenic amino acids
5. **Ketogenesis**: Ketogenic amino acids

Sources of Amino Acids:

- Dietary protein digestion

- Intracellular protein turnover
- De novo synthesis (nonessential amino acids only)

15.2 Essential vs. Nonessential Amino Acids

Definition

Essential amino acids cannot be synthesized by the body and must be obtained from the diet. **Nonessential amino acids** can be synthesized endogenously.

Essential (9)	Nonessential (11)
Histidine	Alanine
Isoleucine	Arginine*
Leucine	Asparagine
Lysine	Aspartate
Methionine	Cysteine*
Phenylalanine	Glutamate
Threonine	Glutamine
Tryptophan	Glycine
Valine	Proline
	Serine
	Tyrosine*

*Conditionally essential: Required in diet under certain conditions (growth, illness)

Mnemonic for Essential Amino Acids

"PVT TIM HALL"

- **P**henylalanine
- **V**aline
- **T**hreonine
- **T**ryptophan
- **I**soleucine
- **M**ethionine
- **H**istidine
- **A**rginine (conditionally essential)
- **L**eucine
- **L**ysine

15.3 Nitrogen Removal from Amino Acids

Before amino acid carbon skeletons can be used for energy or biosynthesis, the amino group must be removed. This occurs through two main processes:

15.3.1 Transamination

Definition

Transamination is the transfer of an amino group from an amino acid to an α-keto acid, producing a new amino acid and a new α-keto acid.

General Reaction:

$$\text{Amino acid}_1 + \alpha\text{-Keto acid}_2 \xleftrightarrow{\text{Aminotransferase}} \alpha\text{-Keto acid}_1 + \text{Amino acid}_2$$

Key Features:

- **Reversible** reactions
- Require **pyridoxal phosphate (PLP)** — active form of vitamin B_6
- Do not result in net loss of nitrogen
- Collect amino groups onto glutamate

Important Aminotransferases:

1. **Alanine Aminotransferase (ALT/GPT):**

$$\text{Alanine} + \alpha\text{-Ketoglutarate} \leftrightarrow \text{Pyruvate} + \text{Glutamate}$$

   - Primarily in liver (cytosol)
   - More specific for liver damage
   - Important in glucose-alanine cycle

2. **Aspartate Aminotransferase (AST/GOT):**

$$\text{Aspartate} + \alpha\text{-Ketoglutarate} \leftrightarrow \text{Oxaloacetate} + \text{Glutamate}$$

   - Found in liver, heart, muscle, kidney
   - Present in both cytosol and mitochondria
   - Elevated in liver disease and myocardial infarction

Clinical Correlation: Liver Function Tests

ALT and AST are measured as markers of liver damage:

- **ALT ¿ AST**: Suggests hepatocellular injury (hepatitis)
- **AST ¿ ALT (ratio ¿ 2:1)**: Suggests alcoholic liver disease
- **Both elevated**: General liver damage

Note: AST is also elevated in muscle damage and MI

15.3.2 Oxidative Deamination

Definition

Oxidative deamination removes the amino group as free ammonia (NH_3), primarily from glutamate.

Key Enzyme: Glutamate Dehydrogenase (GDH)

$$\text{Glutamate} + \text{NAD}^+(\text{or NADP}^+) + \text{H}_2\text{O} \leftrightarrow \alpha\text{-Ketoglutarate} + \text{NH}_4^+ + \text{NADH}$$

Characteristics:

- Located in **mitochondrial matrix**
- Can use either NAD^+ or $NADP^+$
- **Reversible**: Direction depends on energy state
- Releases free ammonia for urea synthesis

Regulation of Glutamate Dehydrogenase:

- **Activated by:** ADP, GDP (low energy signals)
- **Inhibited by:** ATP, GTP (high energy signals)

15.4 Ammonia Transport

Free ammonia is toxic, especially to the brain. Two main mechanisms safely transport nitrogen to the liver for urea synthesis:

1. **Glutamine Transport (Most Tissues):**

$$\text{Glutamate} + \text{NH}_4^+ + \text{ATP} \xrightarrow{\text{Glutamine synthetase}} \text{Glutamine} + \text{ADP} + \text{P}_i$$

- Glutamine is non-toxic and can cross membranes
- In liver: Glutaminase releases NH_4^+ for urea synthesis

- In kidney: Glutaminase releases NH_4^+ for acid-base regulation

2. Glucose-Alanine Cycle (Muscle):

Glucose-Alanine Cycle

In Muscle:

1. Amino acids undergo transamination with pyruvate
2. Alanine is formed and released into blood

In Liver:

1. Alanine undergoes transamination to pyruvate
2. Pyruvate used for gluconeogenesis
3. Amino group enters urea cycle
4. Glucose released to blood and returns to muscle

Purpose: Transports nitrogen safely while recycling carbon

15.5 The Urea Cycle

Definition

The **urea cycle** (Krebs-Henseleit cycle) converts toxic ammonia to non-toxic urea for excretion by the kidneys. It is the major pathway for nitrogen disposal in humans.

Location: Liver (primarily periportal hepatocytes)

- First two reactions: Mitochondrial matrix
- Remaining reactions: Cytosol

Overall Reaction:

$$2NH_3 + CO_2 + 3ATP + \text{Aspartate} + 2H_2O \rightarrow \text{Urea} + 2ADP + 2P_i + AMP + PP_i + \text{Fumarate}$$

15.5.1 Reactions of the Urea Cycle

Reaction 1: Carbamoyl Phosphate Synthesis (Mitochondria)

$$NH_4^+ + CO_2 + 2ATP \xrightarrow{\textbf{CPS I}} \text{Carbamoyl phosphate} + 2ADP + P_i$$

Carbamoyl Phosphate Synthetase I (CPS I)

- **Rate-limiting enzyme** of urea cycle
- Located in mitochondrial matrix
- Requires **N-acetylglutamate (NAG)** as obligate activator
- Uses 2 ATP (one for activation, one for phosphorylation)
- Different from CPS II (cytosolic, pyrimidine synthesis)

Reaction 2: Citrulline Formation (Mitochondria)

$$\text{Carbamoyl phosphate} + \text{Ornithine} \xrightarrow{\text{Ornithine transcarbamoylase (OTC)}} \text{Citrulline} + \text{P}_i$$

- Citrulline is transported to cytosol via ornithine/citrulline antiporter

Reaction 3: Argininosuccinate Formation (Cytosol)

$$\text{Citrulline} + \text{Aspartate} + \text{ATP} \xrightarrow{\text{Argininosuccinate synthetase}} \text{Argininosuccinate} + \text{AMP} + \text{PP}_i$$

- Second nitrogen enters from aspartate
- ATP $\rightarrow$ AMP (equivalent to 2 ATP)

Reaction 4: Argininosuccinate Cleavage (Cytosol)

$$\text{Argininosuccinate} \xrightarrow{\text{Argininosuccinate lyase}} \text{Arginine} + \text{Fumarate}$$

- Fumarate can enter TCA cycle (links urea cycle to TCA cycle)
- This connection is called the "Krebs bicycle"

Reaction 5: Urea Formation (Cytosol)

$$\text{Arginine} + \text{H}_2\text{O} \xrightarrow{\text{Arginase}} \text{Urea} + \text{Ornithine}$$

- Ornithine returns to mitochondria to continue cycle
- Urea diffuses into blood, excreted by kidneys
- Arginase is found almost exclusively in liver

15.5.2 Regulation of the Urea Cycle

Short-term Regulation:

- **N-Acetylglutamate (NAG)**: Allosteric activator of CPS I

$$\text{Glutamate} + \text{Acetyl-CoA} \xrightarrow{\text{NAG synthase}} \text{N-Acetylglutamate}$$

- NAG synthase is activated by arginine
- High protein diet → more amino acids → more arginine → more NAG → increased urea synthesis

Long-term Regulation:

- High protein diet: Induces all urea cycle enzymes (10-20 fold)
- Starvation: Also induces enzymes (amino acids from protein breakdown)
- Low protein diet: Decreases enzyme levels

15.5.3 Energy Cost of Urea Synthesis

ATP Cost of Urea Synthesis

Direct cost:

- CPS I: 2 ATP
- Argininosuccinate synthetase: 1 ATP → AMP + PP_i (= 2 ATP equivalents)

Total: 4 ATP equivalents per urea molecule
However: Fumarate produced can generate ∼2.5 ATP via malate → OAA → NADH
Net cost: ∼1.5 ATP per urea

15.5.4 Clinical Correlations: Urea Cycle Disorders

Urea Cycle Disorders

General Features:

- All cause **hyperammonemia**
- Present in neonatal period or with protein load
- Symptoms: Vomiting, lethargy, seizures, coma
- Treatment: Low protein diet, nitrogen scavengers, dialysis

Specific Disorders:

Enzyme Deficiency	Accumulated Metabolite	Notes
CPS I deficiency	NH_4^+	Decreased citrulline
OTC deficiency	Orotic acid, NH_4^+	X-linked; most common
Argininosuccinate synthetase	Citrulline, NH_4^+	
Argininosuccinate lyase	Argininosuccinate	
Arginase	Arginine	Least severe
NAG synthase	NH_4^+	Treat with N-carbamylglutamate

Why Orotic Acid in OTC Deficiency?

In OTC deficiency:

1. Carbamoyl phosphate accumulates in mitochondria
2. Leaks into cytosol
3. Enters pyrimidine synthesis pathway
4. Excess orotic acid is produced and excreted

Distinguishing feature: OTC deficiency has elevated orotic acid; CPS I deficiency does not

15.6 Amino Acid Carbon Skeleton Metabolism

After nitrogen removal, amino acid carbon skeletons are converted to common metabolic intermediates.

15.6.1 Classification by Metabolic Fate

Definition

Glucogenic amino acids yield intermediates that can be converted to glucose via gluconeogenesis. **Ketogenic amino acids** yield acetyl-CoA or acetoacetate, which can form ketone bodies but not glucose.

Category	Amino Acids
Purely Ketogenic	Leucine, Lysine
Both Glucogenic & Ketogenic	Isoleucine, Phenylalanine, Threonine, Tryptophan, Tyrosine
Purely Glucogenic	All others (13 amino acids)

Mnemonic for Ketogenic Amino Acids

"The Leu-Lys are purely ketogenic"
For both ketogenic and glucogenic, remember: **"PITTT"**

- **P**henylalanine
- **I**soleucine
- **T**hreonine
- **T**ryptophan
- **T**yrosine

15.6.2 Entry Points into Central Metabolism

Amino acid carbon skeletons enter metabolism at seven points:

1. **Pyruvate:** Alanine, Serine, Glycine, Cysteine, Threonine, Tryptophan
2. **Acetyl-CoA:** Leucine, Isoleucine, Lysine, Threonine, Tryptophan
3. **Acetoacetyl-CoA:** Leucine, Lysine, Phenylalanine, Tyrosine, Tryptophan
4. **α-Ketoglutarate:** Glutamate, Glutamine, Pr oline, Histidine, Arginine
5. **Succinyl-CoA:** Isoleucine, Valine, Methionine, Threonine
6. **Fumarate:** Phenylalanine, Tyrosine, Aspartate
7. **Oxaloacetate:** Aspartate, Asparagine

15.7 Metabolism of Specific Amino Acids

15.7.1 Branched-Chain Amino Acids (BCAAs)

The branched-chain amino acids (leucine, isoleucine, valine) share common initial catabolic steps.

Location: Primarily in **skeletal muscle** (not liver)

Step 1: Transamination

$$\text{BCAA} + \alpha\text{-ketoglutarate} \xrightarrow{\text{BCAA aminotransferase}} \alpha\text{-keto acid} + \text{Glutamate}$$

Step 2: Oxidative Decarboxylation (Rate-Limiting)

$$\alpha\text{-Keto acid} + \text{NAD}^+ + \text{CoA} \xrightarrow{\textbf{BCKDH}} \text{Acyl-CoA} + \text{CO}_2 + \text{NADH}$$

Branched-Chain α-Ketoacid Dehydrogenase (BCKDH)

Structure: Similar to pyruvate dehydrogenase and α-ketoglutarate dehydrogenase

Cofactors required:

- Thiamine pyrophosphate (TPP) — B_1
- Lipoic acid
- CoA (contains pantothenic acid — B_5)
- FAD — B_2
- NAD^+ — B_3

Regulation:

- Activated by: BCAA excess, exercise
- Inhibited by: Phosphorylation (kinase), products (NADH, acyl-CoA)

Clinical Correlation: Maple Syrup Urine Disease (MSUD)

Defect: Branched-chain α-ketoacid dehydrogenase (BCKDH) deficiency
Inheritance: Autosomal recessive
Biochemistry:

- Accumulation of BCAAs and their α-keto acids
- Elevated leucine is most toxic (neurotoxic)

Clinical features:

- Sweet, maple syrup odor of urine
- Poor feeding, vomiting
- Neurological deterioration, seizures
- Intellectual disability if untreated
- Onset: First week of life

Treatment:

- Dietary restriction of BCAAs
- Thiamine supplementation (some variants are thiamine-responsive)
- Liver transplantation in severe cases

15.7.2 Phenylalanine and Tyrosine Metabolism

Phenylalanine Hydroxylation:

$$\text{Phenylalanine} + O_2 + BH_4 \xrightarrow{\textbf{Phenylalanine hydroxylase}} \text{Tyrosine} + H_2O + BH_2$$

- $\mathbf{BH_4}$ = tetrahydrobiopterin (cofactor)
- BH_2 is regenerated to BH_4 by dihydrobiopterin reductase
- This reaction makes tyrosine nonessential (if phenylalanine is adequate)

Clinical Correlation: Phenylketonuria (PKU)

Defect: Phenylalanine hydroxylase deficiency (most common) or BH_4 synthesis/regeneration defects
Inheritance: Autosomal recessive
Biochemistry:

- Elevated phenylalanine in blood
- Decreased tyrosine (becomes essential)
- Alternative pathway produces phenylketones:
  - Phenylpyruvate
  - Phenyllactate
  - Phenylacetate (musty odor)

Clinical features:

- Normal at birth
- Intellectual disability if untreated
- Fair skin, light hair, blue eyes (decreased melanin from low tyrosine)
- Musty/mousy body odor
- Eczema
- Seizures

Screening: Newborn screening (Guthrie test or tandem mass spectrometry)
Treatment:

- Dietary phenylalanine restriction
- Tyrosine supplementation
- Avoid aspartame (contains phenylalanine)
- BH_4 supplementation for responsive variants

Maternal PKU: Pregnant women with PKU must maintain strict diet to prevent fetal damage (microcephaly, heart defects, intellectual disability)

Tyrosine Degradation Products:

- **Catecholamines**: Dopamine, norepinephrine, epinephrine
- **Thyroid hormones**: T_3, T_4

- **Melanin**: Skin and hair pigment
- **Fumarate and acetoacetate**: Energy metabolism

Clinical Correlation: Alkaptonuria

Defect: Homogentisate oxidase deficiency
Inheritance: Autosomal recessive
Biochemistry:

- Accumulation of homogentisic acid (intermediate in Tyr degradation)
- Homogentisic acid oxidizes to dark pigment

Clinical features:

- Dark urine (turns black on standing/exposure to air)
- Ochronosis (blue-black pigmentation of cartilage, connective tissue)
- Arthritis (especially spine, large joints)
- Dark earwax
- Generally benign; arthritis develops in adulthood

Treatment: Symptomatic; vitamin C may slow pigment deposition

15.7.3 Homocysteine Metabolism

Homocysteine is a key intermediate in methionine metabolism with important clinical implications.

Sources of Homocysteine:

$$\text{Methionine} \rightarrow \text{SAM} \rightarrow \text{SAH} \rightarrow \text{Homocysteine}$$

Fates of Homocysteine:

1. Remethylation to Methionine:

$$\text{Homocysteine} + \text{N}^5\text{-methyl-THF} \xrightarrow{\text{Methionine synthase}} \text{Methionine} + \text{THF}$$

- Requires **vitamin $\mathbf{B}_{12}$** (methylcobalamin)
- Requires **folate** (as N^5-methyl-THF)

2. Transsulfuration to Cysteine:

$$\text{Homocysteine} + \text{Serine} \xrightarrow{\text{Cystathionine } \beta\text{-synthase}} \text{Cystathionine}$$

$$\text{Cystathionine} \xrightarrow{\text{Cystathionase}} \text{Cysteine} + \alpha\text{-ketobutyrate}$$

- Both enzymes require **vitamin B_6** (PLP)
- Irreversible pathway

Clinical Correlation: Homocystinuria

Causes of elevated homocysteine:

1. **Cystathionine β-synthase (CBS) deficiency** (classic homocystinuria)
   - Most common cause
   - Autosomal recessive
   - Elevated: Homocysteine, methionine
   - Decreased: Cysteine
2. **Vitamin B_{12} or folate deficiency**
   - Impaired remethylation
   - Elevated homocysteine, normal/low methionine
3. **MTHFR deficiency**
   - Cannot produce N^5-methyl-THF
   - Impaired remethylation

Clinical features of CBS deficiency:

- Marfanoid habitus (tall, long limbs)
- Downward lens dislocation (vs. upward in Marfan)
- Intellectual disability
- Thromboembolism (major cause of death)
- Osteoporosis

Treatment:

- High-dose vitamin B_6 (some patients are responsive)
- Dietary methionine restriction
- Cysteine supplementation
- Folate and B_{12} supplementation
- Betaine (alternative methyl donor)

15.8 Amino Acids as Precursors

15.8.1 Neurotransmitter Synthesis

Amino Acid	Product	Key Enzyme/Cofactor
Tryptophan	Serotonin	Tryptophan hydroxylase, BH_4, B_6
Tyrosine	Dopamine	Tyrosine hydroxylase, BH_4, B_6
Tyrosine	Norepinephrine	Dopamine β-hydroxylase, vitamin C
Tyrosine	Epinephrine	PNMT, SAM
Glutamate	GABA	Glutamate decarboxylase, B_6
Histidine	Histamine	Histidine decarboxylase, B_6

15.8.2 Other Important Products

- **Glycine** → Heme, purines, glutathione, creatine
- **Glutamate** → Glutathione, GABA, glutamine
- **Arginine** → Nitric oxide (via NOS), creatine, urea
- **Tryptophan** → $NAD^+/NADP^+$ (niacin synthesis), melatonin
- **Methionine** → SAM (universal methyl donor), cysteine
- **Histidine** → Histamine

16 Nucleotide Metabolism

16.1 Overview of Nucleotide Structure and Function

Definition

Nucleotides are the building blocks of DNA and RNA, consisting of three components: a nitrogenous base, a pentose sugar, and one or more phosphate groups.

Components:

1. **Nitrogenous base:**
   - Purines: Adenine (A), Guanine (G) — two rings
   - Pyrimidines: Cytosine (C), Thymine (T), Uracil (U) — one ring
2. **Pentose sugar:**
   - Ribose (in RNA, ribonucleotides)
   - Deoxyribose (in DNA, deoxyribonucleotides)
3. **Phosphate groups:** 1-3 phosphates (mono-, di-, triphosphate)

Mnemonic: Purines vs. Pyrimidines

"PURe As Gold" — **PUR**ines are **A**denine and **G**uanine
"CUT the PY" — Pyrimidines are **C**ytosine, **U**racil, **T**hymine
"PURines are Pure and double-ringed" — Purines have 2 rings

Functions of Nucleotides:

- Building blocks for DNA and RNA synthesis
- Energy currency (ATP, GTP)
- Coenzymes (NAD^+, FAD, CoA)
- Second messengers (cAMP, cGMP)
- Activated intermediates (UDP-glucose, CDP-choline)
- Allosteric regulators

16.2 Purine Synthesis

16.2.1 De Novo Purine Synthesis

Location: Cytosol
Key feature: The purine ring is built directly on ribose-5-phosphate (unlike pyrimidines)
Starting material: Ribose-5-phosphate (from pentose phosphate pathway)
Sources of atoms in the purine ring:

- **N1**: Aspartate
- **C2, C8**: N^{10}-formyl-THF (folate)
- **N3, N9**: Glutamine (amide nitrogen)
- **C4, C5, N7**: Glycine
- **C6**: CO_2

Step 1: Formation of PRPP (Rate-Limiting for Substrate)

$$\text{Ribose-5-P} + \text{ATP} \xrightarrow{\textbf{PRPP synthetase}} \text{PRPP} + \text{AMP}$$

- PRPP = 5-phosphoribosyl-1-pyrophosphate
- Activated by: Inorganic phosphate
- Inhibited by: Purine nucleotides (feedback inhibition)

Step 2: Committed Step

$$\text{PRPP}+\text{Glutamine} \xrightarrow{\textbf{Glutamine-PRPP amidotransferase}} \text{5-Phosphoribosylamine}+\text{Glutamate}+\text{PP}_i$$

- **Rate-limiting step** of purine synthesis
- Inhibited by: AMP, GMP, IMP (feedback inhibition)
- Activated by: PRPP

Subsequent Steps:

1. Nine more reactions build the purine ring
2. First complete purine nucleotide formed: **IMP** (inosine monophosphate)
3. Requires: Glycine, glutamine, aspartate, CO_2, N^{10}-formyl-THF
4. Consumes: 6 ATP equivalents

Conversion of IMP to AMP and GMP:
IMP → AMP:

1. IMP + Aspartate + GTP → Adenylosuccinate
2. Adenylosuccinate → AMP + Fumarate

- Requires **GTP** (reciprocal regulation)

IMP → GMP:

1. IMP + NAD^+ → XMP (xanthosine monophosphate)
2. XMP + Glutamine + ATP → GMP

- Requires **ATP** (reciprocal regulation)

Reciprocal Regulation of AMP and GMP Synthesis

- AMP synthesis requires **GTP**
- GMP synthesis requires **ATP**
- This ensures balanced production of both purines
- Excess of one stimulates synthesis of the other

16.2.2 Pu rine Salvage Pathway

Definition

Salvage pathways recycle free purine bases (adenine, guanine, hypoxanthine) back to nucleotides, conserving energy compared to de novo synthesis.

Key Enzymes:

1. **Adenine Phosphoribosyltransferase (APRT):**

$$\text{Adenine} + \text{PRPP} \xrightarrow{\text{APRT}} \text{AMP} + \text{PP}_i$$

2. **Hypoxanthine-Guanine Phosphoribosyltransferase (HGPRT):**

$$\text{Hypoxanthine} + \text{PRPP} \xrightarrow{\text{HGPRT}} \text{IMP} + \text{PP}_i$$

$$\text{Guanine} + \text{PRPP} \xrightarrow{\text{HGPRT}} \text{GMP} + \text{PP}_i$$

Clinical Correlation: Lesch-Nyhan Syndrome

Cause: X-linked recessive deficiency of **HGPRT**
Pathophysiology:

- Cannot salvage hypoxanthine or guanine
- PRPP accumulates → increased de novo purine synthesis
- Excess purines degraded to uric acid
- Severe hyperuricemia

Clinical Features:

- Hyperuricemia and gout (early childhood)
- Intellectual disability
- **Self-mutilation** (lip and finger biting) — pathognomonic
- Choreoathetosis and spasticity
- Aggressive behavior

Treatment: Allopurinol (reduces uric acid but does not improve neurological symptoms)

16.2.3 Purine Degradation

Purines are degraded to **uric acid**, which is excreted in urine.

Pathway:

1. AMP → IMP → Inosine → Hypoxanthine
2. GMP → Guanosine → Guanine → Xanthine
3. Hypoxanthine $\xrightarrow{\text{Xanthine oxidase}}$ Xanthine $\xrightarrow{\text{Xanthine oxidase}}$ **Uric acid**

Xanthine Oxidase: Key Enzyme in Purine Degradation

- Converts hypoxanthine → xanthine → uric acid
- Contains molybdenum, FAD, iron-sulfur centers
- Produces hydrogen peroxide (H_2O_2) as byproduct
- **Target of allopurinol** (gout treatment)

Clinical Correlation: Gout

Definition: Inflammatory arthritis caused by deposition of monosodium urate crystals in joints

Causes of Hyperuricemia:

- **Overproduction:** Lesch-Nyhan syndrome, PRPP synthetase overactivity, tumor lysis syndrome
- **Underexcretion:** Renal insufficiency, thiazide diuretics, alcohol

Clinical Features:

- Acute monoarticular arthritis (classically 1st MTP joint — podagra)
- Tophi (urate deposits in soft tissue)
- Uric acid kidney stones

Treatment:

- **Acute:** NSAIDs, colchicine, corticosteroids
- **Chronic:** Allopurinol/febuxostat (xanthine oxidase inhibitors), probenecid (uricosuric)

16.3 Pyrimidine Metabolism

16.3.1 De Novo Pyrimidine Synthesis

Unlike purines, the pyrimidine ring is synthesized **first as a free base**, then attached to ribose-5-phosphate.

Location: Cytosol

Sources of Pyrimidine Ring Atoms:

- **Aspartate**: Contributes N1, C4, C5, C6
- **Carbamoyl phosphate**: Contributes N3, C2 (from glutamine and CO_2)

Step 1: Formation of Carbamoyl Phosphate

$$\text{Glutamine} + CO_2 + 2\text{ATP} \xrightarrow{\textbf{CPS II}} \text{Carbamoyl phosphate} + 2\text{ADP} + P_i$$

Feature	CPS I	CPS II
Location	Mitochondria	Cytosol
Nitrogen source	Free NH_4^+	Glutamine
Function	Urea cycle	Pyrimidine synthesis
Activator	N-acetylglutamate	ATP, PRPP
Inhibitor	—	UTP (feedback)

Step 2: Formation of Carbamoyl Aspartate

$$\text{Carbamoyl phosphate} + \text{Aspartate} \xrightarrow{\textbf{Aspartate transcarbamoylase (ATCase)}} \text{Carbamoyl aspartate}$$

- **Rate-limiting step** in bacteria
- Inhibited by CTP, activated by ATP

Step 3: Ring Closure

$$\text{Carbamoyl aspartate} \xrightarrow{\text{Dihydroorotase}} \text{Dihydroorotate}$$

Step 4: Oxidation

$$\text{Dihydroorotate} \xrightarrow{\text{Dihydroorotate dehydrogenase}} \text{Orotate}$$

- Only step occurring in **mitochondria** (inner membrane)
- Uses ubiquinone as electron acceptor
- Target of **leflunomide** (immunosuppressant for rheumatoid arthritis)

Step 5: Attachment to Ribose-5-Phosphate

$$\text{Orotate} + \text{PRPP} \xrightarrow{\text{Orotate phosphoribosyltransferase}} \text{OMP} + PP_i$$

Step 6: Decarboxylation

$$\text{OMP} \xrightarrow{\text{OMP decarboxylase}} \text{UMP}$$

UMP Synthase: Bifunctional Enzyme

In mammals, orotate phosphoribosyltransferase and OMP decarboxylase activities are on a single bifunctional enzyme called **UMP synthase**.
Deficiency: Hereditary orotic aciduria

Formation of Other Pyrimidine Nucleotides:
UMP → UTP:

$$\text{UMP} \xrightarrow{\text{kinases}} \text{UDP} \xrightarrow{\text{kinases}} \text{UTP}$$

UTP → CTP:

$$\text{UTP} + \text{Glutamine} + \text{ATP} \xrightarrow{\text{CTP synthetase}} \text{CTP} + \text{Glutamate} + \text{ADP}$$

Clinical Correlation: Hereditary Orotic Aciduria

Cause: Autosomal recessive deficiency of **UMP synthase**
Pathophysiology:

- Cannot convert orotate to UMP
- Orotic acid accumulates and is excreted
- Pyrimidine deficiency impairs DNA/RNA synthesis

Clinical Features:

- Megaloblastic anemia (does NOT respond to B12 or folate)
- Failure to thrive
- Orotic acid crystals in urine
- No hyperammonemia (distinguishes from OTC deficiency)

Treatment: Oral **uridine** supplementation (bypasses the block)

16.3.2 Pyrimidine Salvage

Pyrimidine salvage is less important than purine salvage but does occur:

$$\text{Uracil} + \text{Ribose-1-P} \xrightarrow{\text{Uridine phosphorylase}} \text{Uridine}$$

$$\text{Uridine} + \text{ATP} \xrightarrow{\text{Uridine kinase}} \text{UMP} + \text{ADP}$$

16.3.3 Pyrimidine Degradation

Unlike purines (which produce uric acid), pyrimidines are degraded to **water-soluble products**:

Cytosine → Uracil (by deamination)
Uracil degradation:

$$\text{Uracil} \rightarrow \text{Dihydrouracil} \rightarrow \beta\text{-ureidopropionate} \rightarrow \beta\text{-alanine} + NH_3 + CO_2$$

Thymine degradation:

$$\text{Thymine} \rightarrow \text{Dihydrothymine} \rightarrow \beta\text{-ureidoisobutyrate} \rightarrow \beta\text{-aminoisobutyrate} + NH_3 + CO_2$$

- β-alanine: Component of CoA and carnosine
- β-aminoisobutyrate: Elevated in urine during chemotherapy (tumor lysis)

16.4 Deoxyribonucleotide Synthesis

Definition

Deoxyribonucleotides are synthesized by reduction of ribonucleotides at the C2' position of the ribose sugar.

16.4.1 Ribonucleotide Reductase

Reaction:

$$\text{NDP} \xrightarrow{\textbf{Ribonucleotide reductase}} \text{dNDP}$$

Substrates: ADP, GDP, CDP, UDP (diphosphate forms)
Products: dADP, dGDP, dCDP, dUDP
Mechanism:

- Requires **thioredoxin** (reduced form) as electron donor
- Thioredoxin is regenerated by **thioredoxin reductase** using NADPH
- Alternative: Glutaredoxin system (uses glutathione)

Regulation of Ribonucleotide Reductase:

1. **Activity site:** Controls overall enzyme activity
   - ATP: Activates enzyme
   - dATP: Inhibits enzyme (signals sufficient dNTPs)
2. **Specificity site:** Determines which substrate is reduced
   - ATP/dATP bound: Favors CDP and UDP reduction
   - dTTP bound: Favors GDP reduction
   - dGTP bound: Favors ADP reduction

Clinical Significance of Ribonucleotide Reductase

Drug target:

- **Hydroxyurea**: Inhibits ribonucleotide reductase; used in sickle cell disease and myeloproliferative disorders
- **Gemcitabine**: Nucleoside analog that inhibits the enzyme; used in cancer chemotherapy

16.4.2 Thymidylate Synthesis

dTMP (thymidylate) is synthesized from dUMP, not from reduction of TDP.

Reaction:

$$\text{dUMP} + \text{N}^5,\text{N}^{10}\text{-methylene-THF} \xrightarrow{\textbf{Thymidylate synthase}} \text{dTMP} + \text{DHF}$$

Key Points:

- Methyl group transferred from N^5,N^{10}-methylene-THF
- THF is oxidized to **DHF** (dihydrofolate)
- DHF must be reduced back to THF by **dihydrofolate reductase (DHFR)**

Regeneration of THF:

$$\text{DHF} + \text{NADPH} \xrightarrow{\textbf{DHFR}} \text{THF} + \text{NADP}^+$$

Clinical Correlation: Anticancer and Antimicrobial Drugs

Thymidylate Synthase Inhibitors:

- **5-Fluorouracil (5-FU)**: Converted to 5-FdUMP; suicide inhibitor of thymidylate synthase
- Used in colorectal, breast, and other cancers

DHFR Inhibitors:

- **Methotrexate**: Anticancer drug; inhibits human DHFR
- **Trimethoprim**: Antibiotic; selectively inhibits bacterial DHFR
- **Pyrimethamine**: Antiprotozoal; inhibits protozoal DHFR

Mechanism of Selectivity:

- Trimethoprim has 100,000× higher affinity for bacterial vs. human DHFR
- Allows selective killing of bacteria

16.5 Regulation of Nucleotide Metabolism

Purine Synthesis Regulation:

1. **PRPP synthetase:**
   - Inhibited by: ADP, GDP (feedback)
   - Activated by: Inorganic phosphate

2. **Glutamine-PRPP amidotransferase** (committed step):
   - Inhibited by: AMP, GMP, IMP
   - Activated by: PRPP
3. **Reciprocal regulation:** AMP inhibits its own synthesis; GMP inhibits its own synthesis

Pyrimidine Synthesis Regulation:

1. **CPS II:**
   - Inhibited by: UTP
   - Activated by: ATP, PRPP
2. **CTP synthetase:**
   - Inhibited by: CTP (product inhibition)

17 NCLEX-Style Practice Problems

Practice Problem

A 45-year-old man presents with severe pain in his right great toe that began suddenly during the night. The joint is red, swollen, and extremely tender. Serum uric acid is elevated. Which enzyme is the target of the medication that would be most appropriate to reduce uric acid production in this patient?

A. Glutamine-PRPP amidotransferase

B. Xanthine oxidase

C. HGPRT

D. Adenosine deaminase

Solution

Answer: B. Xanthine oxidase
Rationale: This patient has acute gouty arthritis caused by hyperuricemia. **Allopurinol** and **febuxostat** are xanthine oxidase inhibitors that reduce uric acid production by blocking the conversion of hypoxanthine to xanthine and xanthine to uric acid. This is the most appropriate target for reducing uric acid synthesis.

- **A. Glutamine-PRPP amidotransferase**: Rate-limiting enzyme of de novo purine synthesis; not a current drug target
- **C. HGPRT**: Salvage pathway enzyme; deficiency causes Lesch-Nyhan syndrome
- **D. Adenosine deaminase**: Deficiency causes SCID; not a target for gout treatment

Practice Problem

A newborn presents with lethargy, poor feeding, and seizures on day 3 of life. Laboratory studies reveal hyperammonemia (850 µmol/L) and elevated orotic acid in the urine. Which enzyme is most likely deficient?

A. Carbamoyl phosphate synthetase I

B. Ornithine transcarbamylase

C. Argininosuccinate synthetase

D. Arginase

Solution

Answer: B. Ornithine transcarbamylase
Rationale: The combination of hyperammonemia with elevated urinary orotic acid is diagnostic of **ornithine transcarbamylase (OTC) deficiency**. In OTC deficiency, carbamoyl phosphate accumulates in mitochondria, leaks into the cytosol, and enters the pyrimidine synthesis pathway, leading to excess orotic acid production.

- **A. CPS I deficiency**: Causes hyperammonemia but **without** orotic aciduria (carbamoyl phosphate is not made)
- **C. Argininosuccinate synthetase**: Would show elevated citrulline
- **D. Arginase**: Would show elevated arginine

Key Point: Orotic aciduria distinguishes OTC deficiency from CPS I deficiency.

Practice Problem

A 6-month-old infant presents with failure to thrive, musty body odor, fair skin, and developmental delay. Newborn screening reveals elevated phenylalanine levels. The parents are counseled about dietary restrictions. Which amino acid becomes essential in this patient's diet?

A. Tryptophan

B. Methionine

C. Tyrosine

D. Cysteine

Solution

Answer: C. Tyrosine
Rationale: This infant has **phenylketonuria (PKU)**, caused by deficiency of phenylalanine hydroxylase. Normally, tyrosine is nonessential because it can be synthesized from phenylalanine. However, in PKU, this conversion is blocked, making **tyrosine conditionally essential** and requiring dietary supplementation.

- **A. Tryptophan**: Essential amino acid, unrelated to PKU
- **B. Methionine**: Essential amino acid, unrelated to PKU
- **D. Cysteine**: Synthesized from methionine, not phenylalanine

Key Point: PKU patients require tyrosine supplementation and phenylalanine restriction.

Practice Problem

A 35-year-old woman with type 1 diabetes presents with nausea, vomiting, and abdominal pain. She has fruity breath odor. Laboratory studies show glucose 450 mg/dL, pH 7.15, and elevated serum ketones. Which of the following best explains the metabolic basis for ketone body production in this patient?

A. Increased insulin promotes lipolysis

B. Excess acetyl-CoA from fatty acid oxidation exceeds TCA cycle capacity

C. Decreased glucagon inhibits ketogenesis

D. Malonyl-CoA activates CPT-I

Solution

Answer: B. Excess acetyl-CoA from fatty acid oxidation exceeds TCA cycle capacity
Rationale: In diabetic ketoacidosis (DKA), insulin deficiency leads to:

1. Increased lipolysis → elevated free fatty acids
2. Increased β-oxidation in liver → excess acetyl-CoA
3. Oxaloacetate is diverted to gluconeogenesis
4. TCA cycle capacity is overwhelmed
5. Excess acetyl-CoA is channeled to ketogenesis

- **A**: Insulin **inhibits** lipolysis; its absence promotes lipolysis
- **C**: Glucagon is **elevated** in DKA and **stimulates** ketogenesis
- **D**: Malonyl-CoA **inhibits** CPT-I; in DKA, malonyl-CoA is low

Practice Problem

A 50-year-old man with hypercholesterolemia is started on atorvastatin. This medication works by inhibiting which enzyme?

A. Acetyl-CoA carboxylase

B. HMG-CoA reductase

C. Squalene synthase

D. Lipoprotein lipase

Solution

Answer: B. HMG-CoA reductase
Rationale: Statins (atorvastatin, simvastatin, etc.) are competitive inhibitors of **HMG-CoA reductase**, the rate-limiting enzyme of cholesterol synthesis. This enzyme catalyzes the conversion of HMG-CoA to mevalonate.
Effects of statin therapy:

- Decreased hepatic cholesterol synthesis
- Upregulation of LDL receptors (via SREBP-2)
- Increased LDL clearance from blood
- Reduced serum LDL cholesterol
- **A. Acetyl-CoA carboxylase**: Rate-limiting enzyme of fatty acid synthesis
- **C. Squalene synthase**: Later step in cholesterol synthesis
- **D. Lipoprotein lipase**: Releases fatty acids from VLDL/chylomicrons

Practice Problem

A 2-year-old child presents with developmental regression, hepatomegaly, and cherry-red spot on fundoscopic examination. Enzyme assay reveals deficiency of hexosaminidase A. Which of the following substances accumulates in this condition?

A. Glucocerebroside

B. Sphingomyelin

C. GM2 ganglioside

D. Galactocerebroside

Solution

Answer: C. GM2 ganglioside
Rationale: This child has **Tay-Sachs disease**, caused by deficiency of hexosaminidase A. This enzyme normally degrades GM2 ganglioside. Deficiency leads to accumulation of GM2 ganglioside in neurons, causing progressive neurodegeneration.

- **A. Glucocerebroside**: Accumulates in Gaucher disease (glucocerebrosidase deficiency)
- **B. Sphingomyelin**: Accumulates in Niemann-Pick disease (sphingomyelinase deficiency)
- **D. Galactocerebroside**: Accumulates in Krabbe disease (galactocerebrosidase deficiency)

Key Features of Tay-Sachs:

- Cherry-red spot on macula
- Progressive neurodegeneration
- NO hepatosplenomegaly (distinguishes from Niemann-Pick)
- Common in Ashkenazi Jewish population

Practice Problem

During prolonged fasting, which of the following metabolic changes occurs in the liver?

A. Increased malonyl-CoA concentration

B. Activation of acetyl-CoA carboxylase

C. Increased CPT-I activity

D. Decreased gluconeogenesis

Solution

Answer: C. Increased CPT-I activity
Rationale: During prolonged fasting:

- Glucagon is elevated, insulin is low
- Acetyl-CoA carboxylase is **inhibited** (phosphorylated)
- Malonyl-CoA levels **decrease**
- CPT-I is **relieved from inhibition** and becomes more active
- Fatty acid β-oxidation increases
- Ketogenesis increases
- **A**: Malonyl-CoA **decreases** during fasting
- **B**: ACC is **inhibited** during fasting
- **D**: Gluconeogenesis **increases** during fasting

Practice Problem

A patient with severe combined immunodeficiency (SCID) is found to have adenosine deaminase deficiency. Which of the following best explains the mechanism of immunodeficiency in this condition?

A. Accumulation of uric acid is toxic to lymphocytes

B. Elevated dATP inhibits ribonucleotide reductase

C. Decreased purine synthesis impairs DNA replication

D. Excess adenosine activates T-cell apoptosis receptors

Solution

Answer: B. Elevated dATP inhibits ribonucleotide reductase
Rationale: In adenosine deaminase (ADA) deficiency:

1. Adenosine and deoxyadenosine accumulate
2. Deoxyadenosine is phosphorylated to dATP
3. dATP is a potent **inhibitor of ribonucleotide reductase**
4. This blocks synthesis of all deoxyribonucleotides
5. DNA synthesis is impaired
6. Rapidly dividing lymphocytes are most affected

- **A**: Uric acid is not the toxic metabolite in ADA deficiency
- **C**: Purine synthesis is not decreased; degradation is blocked
- **D**: While adenosine has signaling effects, the primary mechanism is dATP toxicity

Practice Problem

A 25-year-old woman is being treated for acute lymphoblastic leukemia with methotrexate. Which of the following enzymes is directly inhibited by this drug?

A. Thymidylate synthase

B. Dihydrofolate reductase

C. Ribonucleotide reductase

D. CTP synthetase

Solution

Answer: B. Dihydrofolate reductase
Rationale: Methotrexate is a structural analog of folate that competitively inhibits **dihydrofolate reductase (DHFR)**. This enzyme is required to regenerate tetrahydrofolate (THF) from dihydrofolate.
Effects of DHFR inhibition:

- Decreased THF and its derivatives
- Impaired one-carbon transfer reactions
- Blocked thymidylate synthesis (requires N^5,N^{10}-methylene-THF)
- Impaired purine synthesis (requires N^{10}-formyl-THF)
- Inhibition of DNA synthesis
- **A. Thymidylate synthase**: Inhibited by 5-fluorouracil (5-FU)
- **C. Ribonucleotide reductase**: Inhibited by hydroxyurea
- **D. CTP synthetase**: Not a major chemotherapy target

Practice Problem

A 40-year-old man with a history of alcohol abuse presents with peripheral neuropathy. Laboratory studies reveal megaloblastic anemia with elevated homocysteine and methylmalonic acid levels. Which vitamin deficiency is most likely?

A. Folate (B9)

B. Cobalamin (B12)

C. Pyridoxine (B6)

D. Thiamine (B1)

Solution

Answer: B. Cobalamin (B12)
Rationale: The combination of megaloblastic anemia, elevated homocysteine, AND elevated methylmalonic acid (MMA) indicates **vitamin B12 deficiency**. B12 is a cofactor for two enzymes:

1. **Methionine synthase**: Converts homocysteine to methionine
   - Deficiency → elevated homocysteine
2. **Methylmalonyl-CoA mutase**: Converts methylmalonyl-CoA to succinyl-CoA
   - Deficiency → elevated methylmalonic acid

Distinguishing B12 from Folate Deficiency:

Finding	B12 Deficiency	Folate Deficiency
Homocysteine	Elevated	Elevated
Methylmalonic acid	**Elevated**	Normal
Neurologic symptoms	Present	Absent

- **A. Folate**: Would have elevated homocysteine but **normal** MMA
- **C. Pyridoxine**: Causes peripheral neuropathy but not megaloblastic anemia
- **D. Thiamine**: Causes Wernicke-Korsakoff syndrome, not megaloblastic anemia

18 Summary Tables and Quick Reference

18.1 Key Enzymes and Their Regulation

Enzyme	Pathway	Activators	Inhibitors
HMG-CoA reductase	Cholesterol synthesis	Insulin, SREBP-2	Statins, cholesterol, glucagon
Acetyl-CoA carboxylase	Fatty acid synthesis	Insulin, citrate	Glucagon, palmitoyl-CoA, AMPK
Fatty acid synthase	Fatty acid synthesis	Insulin, fed state	Fasting
CPT-I	β-oxidation	Fasting, glucagon	Malonyl-CoA
HMG-CoA synthase (mito)	Ketogenesis	Fasting, glucagon	Fed state

18.2 Lysosomal Storage Diseases Summary

Disease	Enzyme Deficient	Accumulated Substrate	Key Features
Tay-Sachs	Hexosaminidase A	GM2 ganglioside	Cherry-red spot, no HSM
Niemann-Pick	Sphingomyelinase	Sphingomyelin	Cherry-red spot, HSM
Gaucher	Glucocerebrosidase	Glucocerebroside	HSM, bone crises, Gaucher cells
Fabry	α-galactosidase A	Globotriaosylceramide	Peripheral neuropathy, angiokeratomas
Krabbe	Galactocerebrosidase	Galactocerebroside	Globoid cells, peripheral neuropathy
Metachromatic leukodystrophy	Arylsulfatase A	Sulfatides	Demyelination, ataxia
Hurler syndrome	α-L-iduronidase	Heparan/dermatan sulfate	Corneal clouding, HSM, dysostosis
Hunter syndrome	Iduronate sulfatase	Heparan/dermatan sulfate	No corneal clouding, X-linked

18.3 Amino Acid Disorders Summary

Disease	Enzyme Deficient	Accumulated	Key Features
PKU	Phenylalanine hydroxylase	Phenylalanine	Intellectual disability, musty odor
Maple syrup urine disease	BCKDH complex	BCAAs, α-keto acids	Sweet urine odor, neurologic damage
Homocystinuria	Cystathionine β-synthase	Homocysteine	Marfanoid, lens dislocation (down), thrombosis
Alkaptonuria	Homogentisate oxidase	Homogentisic acid	Dark urine, ochronosis, arthritis
Cystinuria	Dibasic AA transporter	Cystine in urine	Hexagonal kidney stones
Hartnup disease	Neutral AA transporter	Neutral AAs in urine	Pellagra-like symptoms

18.4 Urea Cycle Disorders Summary

Disease	Enzyme Deficient	Orotic Acid	Key Features
CPS I deficiency	CPS I	Normal/Low	Hyperammonemia, autosomal recessive
OTC deficiency	OTC	**Elevated**	Hyperammonemia, **X-linked**
Citrullinemia	ASS	Normal	Elevated citrulline
Argininosuccinic aciduria	ASL	Normal	Elevated argininosuccinate
Arginase deficiency	Arginase	Normal	Elevated arginine, spastic diplegia

18.5 Nucleotide Metabolism Disorders Summary

Disease	Enzyme Deficient	Key Features
Lesch-Nyhan syndrome	HGPRT	Self-mutilation, hyperuricemia, gout, choreoathetosis, X-linked
ADA deficiency	Adenosine deaminase	SCID (severe combined immunodeficiency), lymphopenia
Gout	Overproduction or underexcretion of uric acid	Painful arthritis, tophi, negatively birefringent crystals
Orotic aciduria	UMP synthase	Megaloblastic anemia (not responsive to B12/folate), orotic acid in urine

18.6 Key Vitamins in Lipid and Amino Acid Metabolism

Vitamin	Active Form	Metabolic Roles
B1 (Thiamine)	TPP	PDH, α-KGDH, BCKDH, transketolase
B2 (Riboflavin)	FAD, FMN	Acyl-CoA dehydrogenases, succinate DH
B3 (Niacin)	NAD^+, $NADP^+$	Oxidation-reduction reactions
B5 (Pantothenate)	CoA	Acyl group carrier
B6 (Pyridoxine)	PLP	Transamination, decarboxylation
B7 (Biotin)	Biotin-enzyme	Carboxylation reactions (ACC, PC, PCC)
B9 (Folate)	THF	One-carbon transfers, nucleotide synthesis
B12 (Cobalamin)	Methylcobalamin, Adenosylcobalamin	Methionine synthase, methylmalonyl-CoA mutase

19 Metabolic Integration and Clinical Applications

19.1 Fed State Metabolism

Fed State: Anabolic Predominance

Hormonal Signal: High insulin/glucagon ratio
Liver:

- Glycogen synthesis ↑
- Fatty acid synthesis ↑
- VLDL secretion ↑
- Cholesterol synthesis ↑
- Protein synthesis ↑

Adipose Tissue:

- Lipoprotein lipase ↑ (uptake of FA from VLDL/chylomicrons)
- Fatty acid esterification ↑
- Lipolysis ↓

Muscle:

- Glucose uptake ↑ (GLUT4)
- Glycogen synthesis ↑
- Protein synthesis ↑
- BCAA uptake and oxidation ↑

19.2 Fasting State Metabolism

Fasting State: Catabolic Predominance

Hormonal Signal: Low insulin/glucagon ratio, elevated cortisol
Liver:

- Glycogenolysis ↑ (first 12-18 hours)
- Gluconeogenesis ↑ (sustained fasting)
- Fatty acid oxidation ↑
- Ketogenesis ↑
- Urea synthesis ↑ (from amino acid catabolism)

Adipose Tissue:

- Lipolysis ↑ (hormone-sensitive lipase activated)
- Release of free fatty acids and glycerol

Muscle:

- Proteolysis ↑ (releases amino acids)
- BCAA oxidation for energy
- Alanine and glutamine release to liver

Brain:

- Glucose utilization (early fasting)
- Ketone body utilization (prolonged fasting)

19.3 Starvation Adaptation

Timeline of Metabolic Changes:

1. **0-4 hours (Absorptive state):**
   - Dietary nutrients being absorbed
   - Insulin dominant
   - Anabolic processes active
2. **4-12 hours (Early fasting):**
   - Glycogenolysis maintains blood glucose

- Glucagon rising
- Lipolysis begins

3. **12-24 hours (Intermediate fasting):**
   - Liver glycogen depleted
   - Gluconeogenesis becomes primary glucose source
   - Ketogenesis begins
   - Muscle proteolysis provides amino acids
4. **1-3 days (Early starvation):**
   - Ketone body production increases
   - Brain begins using ketones
   - Protein breakdown continues
5. **¿3 days (Prolonged starvation):**
   - Ketones supply 60-70% of brain energy
   - Protein breakdown slows (protein sparing)
   - Decreased gluconeogenesis from amino acids
   - Metabolic rate decreases

19.4 Diabetes Mellitus: Metabolic Derangements

Type 1 Diabetes: Absolute Insulin Deficiency

Metabolic Consequences:
Carbohydrate:

- Decreased glucose uptake by muscle and adipose
- Hyperglycemia despite elevated blood glucose
- Increased gluconeogenesis (unopposed glucagon)

Lipid:

- Uncontrolled lipolysis → elevated FFAs
- Increased hepatic β-oxidation
- **Diabetic ketoacidosis (DKA):** Excessive ketone production
- Hypertriglyceridemia (decreased LPL activity)

Protein:

- Increased muscle proteolysis
- Negative nitrogen balance
- Weight loss despite adequate caloric intake

DKA Features:

- Metabolic acidosis (ketoacids)
- Fruity breath odor (acetone)
- Kussmaul respirations (compensation)
- Dehydration (osmotic diuresis)

Type 2 Diabetes: Insulin Resistance

Metabolic Consequences:

- Hyperglycemia (impaired glucose disposal)
- Hyperinsulinemia (compensatory, early disease)
- Dyslipidemia:
  - Elevated triglycerides
  - Low HDL
  - Small, dense LDL particles
- **Hyperosmolar hyperglycemic state (HHS):**
  - Severe hyperglycemia (¿600 mg/dL)
  - Enough insulin to prevent ketosis
  - Severe dehydration
  - Altered mental status

19.5 Alcohol Metabolism and Its Effects

Ethanol Metabolism:

$$\text{Ethanol} \xrightarrow{\text{Alcohol DH}} \text{Acetaldehyde} \xrightarrow{\text{Aldehyde DH}} \text{Acetate} \rightarrow \text{Acetyl-CoA}$$

Metabolic Consequences of Alcohol:

1. **Elevated NADH/NAD$^+$ ratio:**
   - Inhibits gluconeogenesis → **hypoglycemia**
   - Inhibits fatty acid oxidation → **fatty liver**
   - Inhibits TCA cycle → acetyl-CoA shunted to fatty acid synthesis
   - Converts pyruvate to lactate → **lactic acidosis**
2. **Acetaldehyde toxicity:**
   - Protein adducts
   - Oxidative stress
   - Hepatocyte damage
3. **Nutritional deficiencies:**
   - Thiamine (B1) → Wernicke-Korsakoff syndrome

- Folate → megaloblastic anemia
- Pyridoxine (B6) → peripheral neuropathy

Clinical Pearl: Disulfiram Reaction

Disulfiram inhibits aldehyde dehydrogenase, causing acetaldehyde accumulation when alcohol is consumed.
Symptoms:

- Flushing
- Nausea, vomiting
- Headache
- Hypotension
- Tachycardia

Similar reactions occur with:

- Metronidazole
- Certain cephalosporins (cefotetan, cefoperazone)
- Sulfonylureas (chlorpropamide)

20 Additional NCLEX-Style Practice Problems

Practice Problem

A 3-month-old infant presents with poor feeding, vomiting, and lethargy. Laboratory studies show hyperammonemia, elevated orotic acid in urine, and low BUN. The infant is male. What is the most likely inheritance pattern of this disorder?

A. Autosomal dominant

B. Autosomal recessive

C. X-linked recessive

D. Mitochondrial

Solution

Answer: C. X-linked recessive
Rationale: The combination of hyperammonemia with **elevated orotic acid** is diagnostic of **ornithine transcarbamylase (OTC) deficiency**. OTC deficiency is the **only X-linked** urea cycle disorder.
Why orotic acid is elevated:

- Carbamoyl phosphate accumulates in mitochondria
- Leaks into cytosol
- Enters pyrimidine synthesis pathway
- Excess orotic acid produced

Why other answers are incorrect:

- **A & B**: All other urea cycle defects are autosomal recessive and do NOT have elevated orotic acid
- **D**: Mitochondrial inheritance would show maternal transmission pattern

Practice Problem

A 55-year-old woman with breast cancer is being treated with methotrexate. Which of the following best describes the mechanism by which this drug inhibits cancer cell proliferation?

A. Inhibits purine ring synthesis by blocking PRPP synthetase

B. Inhibits thymidylate synthesis by blocking dihydrofolate reductase

C. Inhibits DNA polymerase directly

D. Inhibits ribonucleotide reductase

Solution

Answer: B. Inhibits thymidylate synthesis by blocking dihydrofolate reductase
Rationale: Methotrexate is a structural analog of folate that competitively inhibits **dihydrofolate reductase (DHFR)**. This enzyme is required to regenerate tetrahydrofolate (THF) from dihydrofolate after the thymidylate synthase reaction.
Mechanism:

1. Thymidylate synthase converts dUMP to dTMP
2. This reaction requires N^5,N^{10}-methylene-THF as a one-carbon donor
3. THF is oxidized to DHF in this reaction
4. DHFR normally regenerates THF from DHF
5. Methotrexate blocks DHFR → THF depletion → no dTMP synthesis
6. Without dTMP, DNA synthesis stops

Why other answers are incorrect:

- **A**: Methotrexate does not directly inhibit PRPP synthetase
- **C**: Methotrexate does not directly inhibit DNA polymerase
- **D**: Hydroxyurea inhibits ribonucleotide reductase, not methotrexate

Practice Problem

A 25-year-old man presents with self-mutilating behavior, intellectual disability, and gouty arthritis. His serum uric acid level is markedly elevated. Which enzyme deficiency is responsible for this condition?

A. Adenosine deaminase

B. Xanthine oxidase

C. Hypoxanthine-guanine phosphoribosyltransferase

D. Purine nucleoside phosphorylase

Solution

Answer: C. Hypoxanthine-guanine phosphoribosyltransferase
Rationale: This patient has **Lesch-Nyhan syndrome**, caused by complete deficiency of **HGPRT** (hypoxanthine-guanine phosphoribosyltransferase). This is an X-linked recessive disorder.
Classic Triad:

1. **Hyperuricemia with gout** (due to increased de novo purine synthesis)
2. **Intellectual disability**
3. **Self-mutilating behavior** (lip and finger biting)

Pathophysiology:

- HGPRT normally salvages hypoxanthine and guanine back to IMP and GMP
- Without salvage, PRPP accumulates
- Excess PRPP drives de novo purine synthesis
- Increased purines → increased degradation → hyperuricemia

Why other answers are incorrect:

- **A**: Adenosine deaminase deficiency causes SCID, not gout
- **B**: Xanthine oxidase deficiency would cause **low** uric acid
- **D**: PNP deficiency causes T-cell immunodeficiency

Practice Problem

A newborn is found to have elevated phenylalanine on newborn screening. Further testing reveals normal dihydrobiopterin reductase activity. The infant is started on a phenylalanine-restricted diet. Which of the following foods must be avoided?

A. Fruits and vegetables only

B. Foods containing aspartame

C. Foods high in tyrosine

D. Foods containing sucrose

Solution

Answer: B. Foods containing aspartame
Rationale: This infant has classic **phenylketonuria (PKU)** due to phenylalanine hydroxylase deficiency (confirmed by normal BH4 metabolism). **Aspartame** is an artificial sweetener that contains phenylalanine.
Aspartame structure:

- Dipeptide of **aspartic acid** and **phenylalanine**
- When metabolized, releases free phenylalanine
- Products containing aspartame must carry warning labels for PKU patients

PKU Dietary Management:

- Restrict phenylalanine intake (found in high-protein foods)
- Avoid: meat, fish, eggs, dairy, nuts, aspartame
- Supplement with tyrosine (becomes essential)
- Special medical formulas provide protein without phenylalanine

Why other answers are incorrect:

- **A**: Fruits and vegetables are generally low in phenylalanine and allowed
- **C**: Tyrosine should be **supplemented**, not avoided
- **D**: Sucrose does not contain amino acids

Practice Problem

A 40-year-old man with a history of alcohol use disorder presents with peripheral neuropathy and heart failure. Which vitamin deficiency is most likely responsible?

A. Thiamine (B1)

B. Riboflavin (B2)

C. Niacin (B3)

D. Pyridoxine (B6)

Solution

Answer: A. Thiamine (B1)
Rationale: This patient has **wet beriberi**, caused by thiamine (vitamin B1) deficiency. Alcoholics are at high risk due to poor nutrition and impaired thiamine absorption.
Thiamine Deficiency Syndromes:

- **Wet beriberi**: High-output cardiac failure, peripheral edema
- **Dry beriberi**: Peripheral neuropathy, muscle wasting
- **Wernicke encephalopathy**: Confusion, ataxia, ophthalmoplegia
- **Korsakoff syndrome**: Anterograde amnesia, confabulation

Thiamine Function:

- Cofactor (as TPP) for:
  - Pyruvate dehydrogenase
  - α-ketoglutarate dehydrogenase
  - Branched-chain α-ketoacid dehydrogenase
  - Transketolase

Why other answers are incorrect:

- **B**: Riboflavin deficiency causes cheilosis, glossitis, corneal vascularization
- **C**: Niacin deficiency causes pellagra (dermatitis, diarrhea, dementia)
- **D**: Pyridoxine deficiency causes peripheral neuropathy but not cardiomyopathy

Practice Problem

A pharmaceutical company is developing a new cholesterol-lowering drug. The drug is designed to inhibit the rate-limiting enzyme of cholesterol synthesis. Which enzyme is the target?

A. Thiolase

B. HMG-CoA synthase

C. HMG-CoA reductase

D. Squalene synthase

Solution

Answer: C. HMG-CoA reductase
Rationale: HMG-CoA reductase catalyzes the rate-limiting step of cholesterol synthesis: the conversion of HMG-CoA to mevalonate. This is the target of **statins**, the most widely prescribed cholesterol-lowering drugs.
Reaction:

$$\text{HMG-CoA} + 2\text{NADPH} \xrightarrow{\text{HMG-CoA reductase}} \text{Mevalonate} + 2\text{NADP}^+ + \text{CoA}$$

Statin Mechanism:

- Competitive inhibitors of HMG-CoA reductase
- Structural analogs of HMG-CoA
- Decrease hepatic cholesterol synthesis
- Upregulate LDL receptors → increased LDL clearance

Why other answers are incorrect:

- **A**: Thiolase is involved in ketogenesis and β-oxidation
- **B**: HMG-CoA synthase is upstream of the rate-limiting step
- **D**: Squalene synthase is downstream; inhibition would still allow mevalonate pathway intermediates to accumulate

Practice Problem

A 6-month-old infant presents with hepatomegaly, hypoglycemia, lactic acidosis, and hyperuricemia. Liver biopsy shows glycogen with normal structure but increased quantity. Which enzyme is most likely deficient?

A. Glucose-6-phosphatase

B. Lysosomal acid maltase

C. Debranching enzyme

D. Muscle phosphorylase

Solution

Answer: A. Glucose-6-phosphatase
Rationale: This infant has **Von Gierke disease (GSD Type I)**, caused by glucose-6-phosphatase deficiency. The glycogen has **normal structure** but is present in **increased amounts**.
Classic Features of Von Gierke Disease:

- Severe fasting hypoglycemia
- Hepatomegaly (massive)
- Lactic acidosis (glucose-6-P shunted to glycolysis)
- Hyperuricemia (increased purine degradation)
- Hyperlipidemia
- "Doll-like" facies

Why glycogen is normal structure:

- Glycogen synthesis is normal
- Glycogen breakdown to glucose-6-P is normal
- Only the final step (G6P $\rightarrow$ glucose) is blocked

Why other answers are incorrect:

- **B**: Pompe disease (acid maltase deficiency) causes cardiomegaly, not hepatomegaly
- **C**: Cori disease (debranching enzyme) produces **abnormal** glycogen (short outer branches)
- **D**: McArdle disease affects muscle, not liver

21 Final Summary: High-Yield Integration Points

21.1 Metabolic Pathway Integration

The Fed State (Insulin Dominant)

Active Pathways:

- Glycolysis (glucose → pyruvate)
- Glycogenesis (glucose → glycogen)
- Fatty acid synthesis (acetyl-CoA → palmitate)
- Triglyceride synthesis
- Cholesterol synthesis
- Protein synthesis
- Pentose phosphate pathway

Key Regulatory Events:

- Insulin activates phosphatases (dephosphorylation)
- Glycogen synthase: dephosphorylated = ACTIVE
- Acetyl-CoA carboxylase: dephosphorylated = ACTIVE
- PFK-2: dephosphorylated = kinase active → F-2,6-BP ↑

The Fasted State (Glucagon Dominant)

Active Pathways:

- Glycogenolysis (glycogen → glucose)
- Gluconeogenesis (pyruvate → glucose)
- β-oxidation (fatty acids → acetyl-CoA)
- Ketogenesis (acetyl-CoA → ketone bodies)
- Proteolysis (muscle protein → amino acids)

Key Regulatory Events:

- Glucagon activates PKA (phosphorylation via cAMP)
- Glycogen phosphorylase: phosphorylated = ACTIVE
- Glycogen synthase: phosphorylated = INACTIVE
- Acetyl-CoA carboxylase: phosphorylated = INACTIVE
- Low malonyl-CoA allows CPT-I activity

21.2 Master Mnemonic Summary

Essential Mnemonics for Board Exams

Essential Amino Acids: "PVT TIM HALL"

- Phenylalanine, Valine, Threonine, Tryptophan, Isoleucine, Methionine, Histidine, Arginine, Leucine, Lysine

Glucogenic and Ketogenic: "The Leu-Lys are purely ketogenic"
B Vitamin Deficiencies:

- B1 (Thiamine): "BeriBeri and Wernicke"
- B2 (Riboflavin): "2 C's" - Cheilosis, Corneal vascularization
- B3 (Niacin): "3 D's" - Dermatitis, Diarrhea, Dementia
- B6 (Pyridoxine): Peripheral neuropathy, Sideroblastic anemia
- B12 (Cobalamin): "Megaloblastic anemia with Neuro symptoms"

Sphingolipidoses with Cherry-Red Spot:

- Tay-Sachs: "Tay-Sachs lacks heXosaminidase" (no HSM)
- Niemann-Pick: "No man picks his nose with his sphinger" (has HSM)

Urea Cycle: "Ordinarily Careless Crappers Are Also Frivolous About Urination"

- Ornithine, Carbamoyl phosphate, Citrulline, Aspartate, Argininosuccinate, Fumarate, Arginine, Urea

21.3 Clinical Correlation Summary

Lab Finding	Consider	Mechanism
Hyperammonemia + elevated orotic acid	OTC deficiency	Carbamoyl-P enters pyrimidine pathway
Hyperammonemia + normal orotic acid	CPS I deficiency	No excess carbamoyl-P
Elevated uric acid + self-mutilation	Lesch-Nyhan syndrome	HGPRT deficiency
Hypoglycemia + lactic acidosis + hepatomegaly	Von Gierke disease	G6Pase deficiency
Hypoglycemia + cardiomegaly	Pompe disease	Acid maltase deficiency
Elevated phenylalanine + musty odor	PKU	PAH deficiency
Maple syrup urine odor	MSUD	BCKDH deficiency

22 Appendix: Metabolic Pathway Integration Maps

22.1 Central Metabolic Hub: Acetyl-CoA

Acetyl-CoA: The Metabolic Crossroads

Sources of Acetyl-CoA:

1. Pyruvate (via pyruvate dehydrogenase) - from glucose
2. β-oxidation of fatty acids
3. Ketogenic amino acids (Leu, Lys, Ile, Phe, Trp, Tyr)
4. Ketone body utilization (in peripheral tissues)
5. Ethanol metabolism

Fates of Acetyl-CoA:

1. TCA cycle $\rightarrow$ ATP production
2. Fatty acid synthesis (when energy is abundant)
3. Cholesterol synthesis $\rightarrow$ steroids, bile acids
4. Ketone body synthesis (in liver during fasting)
5. Acetylcholine synthesis (in neurons)

Key Point: Acetyl-CoA **cannot** be converted to glucose because:

- Pyruvate dehydrogenase is irreversible
- No net conversion of acetyl-CoA to oxaloacetate in TCA cycle

22.2 Integration of Carbohydrate, Lipid, and Protein Metabolism

Fed State (High Insulin/Glucagon Ratio)

Tissue	Metabolic Activities
Liver	Glycogenesis, glycolysis, fatty acid synthesis, VLDL secretion, protein synthesis
Adipose	Glucose uptake (GLUT4), lipogenesis, TAG storage, LPL activation
Muscle	Glucose uptake (GLUT4), glycogenesis, protein synthesis
Brain	Glucose oxidation (obligate glucose user in fed state)

Fasted State (Low Insulin/Glucagon Ratio)

Tissue	Metabolic Activities
Liver	Glycogenolysis, gluconeogenesis, β-oxidation, ketogenesis
Adipose	Lipolysis, release of fatty acids and glycerol
Muscle	β-oxidation, protein degradation (supplies amino acids), ketone utilization
Brain	Glucose oxidation; ketone utilization after prolonged fasting

22.3 Tissue-Specific Enzyme Distribution

Enzyme/Pathway	Tissue Location
Glucokinase	Liver, pancreatic β-cells
Hexokinase	All tissues
Glucose-6-phosphatase	Liver, kidney, intestine (NOT muscle)
Fructose-1,6-bisphosphatase	Liver, kidney (gluconeogenic tissues)
Glycogen phosphorylase	Liver and muscle (different isozymes)
Pyruvate carboxylase	Liver, kidney (mitochondria)
HMG-CoA synthase (cytosolic)	Liver (cholesterol synthesis)
HMG-CoA synthase (mitochondrial)	Liver (ketogenesis)
HMG-CoA lyase	Liver only (ketogenesis)
Branched-chain aminotransferase	Primarily skeletal muscle
Urea cycle enzymes	Liver only
Carnitine palmitoyltransferase I	Outer mitochondrial membrane (all tissues)

22.4 Hormonal Regulation Summary

Insulin vs. Glucagon: Metabolic Effects

Process	Insulin	Glucagon
Glycolysis	↑	↓
Gluconeogenesis	↓	↑
Glycogenesis	↑	↓
Glycogenolysis	↓	↑
Fatty acid synthesis	↑	↓
β-oxidation	↓	↑
Ketogenesis	↓	↑
Lipolysis (adipose)	↓	↑
Protein synthesis	↑	↓
Cholesterol synthesis	↑	↓

Mechanism of Regulation:

- **Insulin**: Activates phosphatases → dephosphorylation
- **Glucagon**: Activates protein kinase A → phosphorylation

23 Final Comprehensive Review

23.1 High-Yield Facts for Board Examinations

1. **Rate-limiting enzymes** are the most important regulatory points and common drug targets
2. **Irreversible reactions** in pathways often have separate enzymes for forward and reverse directions (e.g., glucokinase vs. glucose-6-phosphatase)
3. **Compartmentalization** is crucial:
   - Fatty acid synthesis: Cytosol
   - β-oxidation: Mitochondrial matrix
   - Urea cycle: Both mitochondria and cytosol
   - Cholesterol synthesis: Cytosol/ER
   - Ketogenesis: Mitochondria (liver only)
4. **Vitamin cofactors** are essential:
   - B1 (Thiamine/TPP): Dehydrogenase complexes, transketolase
   - B2 (Riboflavin/FAD): Oxidation reactions

- B3 (Niacin/NAD^+): Redox reactions
- B5 (Pantothenate/CoA): Acyl transfers
- B6 (Pyridoxine/PLP): Transamination, decarboxylation
- B7 (Biotin): Carboxylation reactions
- B9 (Folate/THF): One-carbon transfers
- B12 (Cobalamin): Methylation, isomerization

5. **Energy yields**:
   - Glucose (aerobic): ∼30-32 ATP
   - Palmitate: ∼106 ATP
   - Glucose (anaerobic): 2 ATP

23.2 Common Board Question Patterns

Recognizing Question Patterns

Pattern 1: Enzyme Deficiency

- Presents with: Accumulated substrate, deficient product
- Look for: Characteristic clinical features, inheritance pattern
- Example: "Infant with musty odor, intellectual disability, fair skin" → PKU

Pattern 2: Drug Mechanism

- Know: Target enzyme, pathway affected, clinical use
- Example: "Statin mechanism" → HMG-CoA reductase inhibition

Pattern 3: Metabolic State

- Identify: Fed vs. fasted, insulin vs. glucagon dominant
- Predict: Which pathways are active
- Example: "After 24-hour fast" → gluconeogenesis, ketogenesis active

Pattern 4: Vitamin Deficiency

- Match: Clinical features to specific vitamin
- Know: Biochemical function of each vitamin
- Example: "Dermatitis, diarrhea, dementia" → Niacin (B3) deficiency

Pattern 5: Lab Value Interpretation

- Correlate: Abnormal values with metabolic block
- Example: "Hyperammonemia + elevated orotic acid" → OTC deficiency

23.3 Critical Concepts Checklist

☐ Understand the fed vs. fasted metabolic states

☐ Know all rate-limiting enzymes and their regulators

☐ Memorize vitamin cofactors and their associated enzymes

☐ Understand the urea cycle and causes of hyperammonemia

☐ Know the lysosomal storage diseases and their enzyme deficiencies

- ☐ Understand amino acid metabolism disorders (PKU, MSUD, homocystinuria)
- ☐ Know purine and pyrimidine synthesis, salvage, and degradation
- ☐ Understand lipid transport (lipoproteins) and associated disorders
- ☐ Know cholesterol synthesis regulation and statin mechanism
- ☐ Understand ketone body metabolism and diabetic ketoacidosis

24 Glossary of Key Terms

Anaplerotic reaction A reaction that replenishes TCA cycle intermediates

Apoenzyme The protein portion of an enzyme without its cofactor

Coenzyme An organic molecule required for enzyme activity

Committed step The first irreversible step unique to a pathway

Deamination Removal of an amino group from a molecule

Feedback inhibition Inhibition of an enzyme by the end product of its pathway

Glucogenic Capable of being converted to glucose

Holoenzyme Complete enzyme with all cofactors bound

Ketogenic Capable of being converted to ketone bodies

Lipolysis Breakdown of triglycerides to fatty acids and glycerol

Lipogenesis Synthesis of fatty acids and triglycerides

Oxidative phosphorylation ATP synthesis coupled to electron transport

Proteolysis Breakdown of proteins to amino acids

Rate-limiting step The slowest step in a pathway that determines overall flux

Salvage pathway Recycling of bases or nucleosides to form nucleotides

Substrate-level phosphorylation Direct transfer of phosphate to ADP

Transamination Transfer of an amino group between molecules

25 Recommended Resources

25.1 Primary Textbooks

1. Ferrier, D.R. *Lippincott Illustrated Reviews: Biochemistry* (8th Edition)
2. Nelson, D.L. & Cox, M.M. *Lehninger Principles of Biochemistry* (8th Edition)
3. Rodwell, V.W. et al. *Harper's Illustrated Biochemistry* (31st Edition)

25.2 Review Resources

1. First Aid for the USMLE Step 1
2. Kaplan Biochemistry Lecture Notes
3. Boards and Beyond Video Series
4. Pixorize (Visual mnemonics for biochemistry)

25.3 Practice Questions

1. UWorld USMLE Question Bank
2. Amboss Question Bank
3. Kaplan Qbank

Final Note

Success in biochemistry requires understanding the **logic** of metabolic pathways rather than rote memorization. Focus on:

- **Why** pathways exist (physiological purpose)
- **When** pathways are active (metabolic state)
- **Where** pathways occur (tissue and subcellular location)
- **How** pathways are regulated (hormones, allosteric effectors)
- **What happens** when pathways fail (clinical correlations)

This integrated understanding will serve you well on board examinations and in clinical practice.